

Magnetic correlations and neutron scattering

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Magnetic correlations and neutron scattering†

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Contents

	Page
1. Introduction	706
2. Formal relationships	708
2.1. The neutron-scattering cross section	708
2.2. Scattering and fluctuations	708
2.3. The wavelength-dependent susceptibility	709
2.4. Susceptibility and fluctuations	711
2.5. The power spectrum in terms of the relaxation function	711
2.6. Time-dependent susceptibilities	713
2.7. Relaxation and susceptibility	715
2.8. Relaxation of the system after a disturbance	715
2.9. Frequency-dependent susceptibility	716
2.10. Properties of the functions	717
2.11. The power spectrum and the susceptibility	718
2.12. Significance of the generalized susceptibility	718
2.13. Green functions	719
2.14. Expressions for the inelastic neutron scattering	720
3. Significance of the wavelength-dependent susceptibility	720
3.1. $\chi^{zz}(\mathbf{K})$ above T_C	720
3.2. Just below T_C ; $\chi^{zz}(\mathbf{K})$	722
3.3. Above T_C ; ferromagnet and antiferromagnet, including anisotropy	725
3.4. Below T_C ; $\chi^{xx}(\mathbf{K})$	727
3.5. The instantaneous critical correlation function	728
3.6. Computation of the critical parameters	729
3.7. Field dependence of the correlations	731
3.8. Other cases	731
4. Properties of the spectral-shape function	732
4.1. The moments of $F(\mathbf{K}, \omega)$	732
4.2. The second moment	733
4.3. High temperatures	736
4.4. Experimentally determined fluctuation spectra	740
4.5. $F(\mathbf{K}, \omega)$ at high T and large K	742
4.6. Moderately high temperatures	745
5. Spin diffusion	747
5.1. Diffusion at high temperatures	747
5.2. Diffusion at low temperatures	756
6. Critical phenomena	758
6.1. The question of critical diffusion	758
6.2. Thermodynamic slowing-down	759
6.3. The instantaneous spin- and spin-velocity-correlations at small K as a function of temperature above T_C	762
6.4. Facts and fallacies about critical scattering	763
6.5. Interpretation of the observed critical relaxation	767
7. Concluding remarks	771
References	772

† An article by the latter author which began with five lectures of the former.

Abstract. An introduction is given to the methods and results of some recent researches into statistical thermodynamics bearing upon the correlation functions of magnetic moments in Heisenberg-coupled spin-only magnets, and their intimate connection with neutron-scattering theory and practice is brought out. The interrelationships between the correlation function, the relaxation function, the generalized susceptibility, the power spectrum of the fluctuations and the neutron scattering are explained, and it is shown that insights into any one of these aspects can serve to illuminate the others. Different forms of approximate theory are seen to be suitable in approaching the topic through these different avenues; the method of moments for analysing the power spectrum and the connection with Green function theory are described in particular. Some examples are calculated in molecular-field theory and with various other simple approximations; expressions for the frequency spectrum of the susceptibility are obtained in various temperature ranges. The extreme forms of the frequency spectrum in appropriate conditions as a set of delta functions, as a pseudo-Gaussian and as a pseudo-Lorentzian are derived, and the transitions between these cases are considered. The approximation of spin diffusion and its limitations are analysed; the problem of the dynamical slowing-down of magnetic fluctuations near the phase-transition point is examined and the current position in this enquiry is set out.

1. Introduction

Powerful new methods of analysing the behaviour of condensed matter have recently become available as the result of discoveries in statistical mechanics. They spring from theoretical advances that have greatly improved our understanding of strongly-interacting systems, and which in so doing have made it possible to interpret experiments in a more penetrating fashion than before. For our purposes here, in discussing the application of these ideas to magnetism, their essence can be expressed by saying that the frequency spectrum of the spontaneous fluctuations in magnetization, the relaxation function controlling the decay of a macroscopic perturbation, the susceptibility, the spin correlation function and the scattering cross section of a magnet are all related.

From the theoretical standpoint, these interconnections between different physical properties have proved extremely fruitful. On the one hand, approximate mathematical techniques appropriate to some special field—such as the Green function method for calculating correlations—are extended in scope to the other domains; on the other hand, any direct physical intuition one may gain into phenomena of one kind can be taken over to give insight in another connection. In this way, several doors have been opened at once onto the problem of explaining co-operative effects in spin systems.

On the experimental side, the frequency distribution of the scattered neutron spectrum can now be related immediately to the most fundamental properties of the scattering magnet. The comparison of data with theories can accordingly now be made with a clear indication of what are the model-dependent, what the thermodynamic and what the merely technological limitations. The situation here is analogous to that which has arisen in the same fashion in the parallel study of density fluctuations in liquids.

In this article we attempt to review the position thus created in the study of spin dynamics. We have given prominence to the special place of neutron-scattering experiments in the subject, at the expense of other techniques such as nuclear

magnetic resonance for investigating what is essentially the same problem. The reason is that in solid state physics magnetic coupling energies between atoms of a few times 10^{-3} eV are typical, and this order of magnitude is just such as to render their effects conveniently observable in thermal neutron experiments. Over essentially the whole compass of interest in electron, rather than nuclear, magnetism a uniquely versatile technique is provided by neutron beam research, and quite detailed investigations into spin dynamics are practicable by this means.

We have put the emphasis throughout on simple arguments, so that the reader will be equipped with the main ideas and prepared for further reading. For simplicity again, the whole discussion is centred around insulating magnetic substances subject to Heisenberg exchange coupling only. Although the restriction to 'Heisenberg substances' does less than justice to the available theories, it does nevertheless direct attention to a Hamiltonian on which a great deal of mathematics has been done and which unarguably applies to a vast class of real materials. We do not claim to quote every paper on the subject, nor even all the good ones; our aim has been rather to develop a thread of argument, extending from what we hope are the well-known properties of the pair-distribution function through to the present-day perplexities of the problem of phase transitions.

An analytical framework having been set up, the central task is still that of comprehending the motions of strongly interacting particles, and it is of course notorious that this is extremely difficult. In fact the problem is one that has consistently defeated intuition at most levels of thinking. Thus spin-wave interactions, critical 'exponents', dynamical slowing-down and the validity of spin-diffusion theory are all topics where the observed phenomena have been surprising. In this difficult situation, mathematical experiments are increasingly being employed as well as physical ones. The technique of simulating many-particle aggregates with computers is still at an early stage, but applications to magnetism are beginning to appear, and we shall have occasion to quote two extremely interesting projects of this kind.

Our discussion could be summed up by saying that we examine the influence and characteristics of the generalized susceptibility $\chi_T(\mathbf{K}, \omega)$ of the Heisenberg magnet as a function of wave vector $\mathbf{K}$, frequency ω and specimen temperature T . After outlining in §2 the widely ranging connections of this quantity, we go on in the remaining sections to rough out its general functional form. The argument essentially revolves around the fluctuation-dissipation theorem, which in this field relates the generalized susceptibility, as the function governing the dissipation of an impressed disturbance, to the spectrum of spontaneous fluctuations of the magnetic moments. The evolution of the ideas here follows a well-known path explored originally by Nyquist (1928), Callen and Welton (1951), Van Hove (1954) and Kubo (1959). The principles were first extensively applied to the problem of Heisenberg spins by Van Vleck (1939) and de Gennes (1958 a); more recently the whole study was carried a great stride forward in a famous investigation by Mori and Kawasaki (1962 a), whose article is still required reading in the subject. Special mention must also be made of a dozen subsequent papers by the last-mentioned authors, and of the classical paper by Izuyama *et al.* (1963) showing how to apply the ideas to conductors. Valuable reviews of some of this material, overlapping the province of the present article, have been written by de Gennes (1963),

Kadanoff and Martin (1963) and Mori (1966 a, b). We draw extensively on the arguments of these various authors in what follows.

2. Formal relationships

2.1. The neutron-scattering cross section

We start at the neutron-scattering end of the discussion, on the ground that the physical connection between scattering cross sections and pair distributions is the most likely to be familiar to the reader. Defining the neutron energy and momentum loss respectively as

$$\hbar\omega = E - E' = \frac{\hbar^2}{2m_0}(k^2 - k'^2), \quad \hbar\mathbf{K} = \hbar(\mathbf{k} - \mathbf{k}'), \quad (1)$$

the cross section to unpolarized neutrons of a system of N localized spins is

$$\frac{\partial^2 \sigma}{\partial \Omega' \partial E'} = \mathcal{A}(\mathbf{k}, \mathbf{k}') \sum_{\alpha\beta} (\delta_{\alpha\beta} - \hat{K}_\alpha \hat{K}_\beta) \mathcal{S}^{\alpha\beta}(\mathbf{K}, \omega), \quad (2)$$

where

$$\mathcal{A}(\mathbf{k}, \mathbf{k}') = \frac{N}{\hbar} (r_0 \gamma)^2 \frac{k'}{k} |f(\mathbf{K})|^2, \quad (3)$$

$r_0 = e^2/mc^2$, f in $\mathcal{A}$ is the neutron-scattering form factor, and

$$\mathcal{S}^{\alpha\beta}(\mathbf{K}, \omega) = \frac{1}{2\pi} \sum_{\mathbf{r}} \int_{-\infty}^{\infty} e^{i(\mathbf{K} \cdot \mathbf{r} - \omega t)} \langle S_0^\alpha(0) S_{\mathbf{r}}^\beta(t) \rangle dt. \quad (4)$$

Actually, for systems with only exchange coupling, $\alpha\beta$ terms like

$$\langle S_0^y S_{\mathbf{r}}^z \rangle = \left\langle \frac{1}{2i} (S_0^+ - S_0^-) S_{\mathbf{r}}^z \right\rangle = 0, \quad (5)$$

while

$$\begin{aligned} \langle S_0^x S_{\mathbf{r}}^y \rangle &= \left\langle \frac{1}{2} (S_0^+ + S_0^-) \frac{1}{2i} (S_{\mathbf{r}}^+ - S_{\mathbf{r}}^-) \right\rangle \\ &= \frac{1}{4i} \langle S_0^- S_{\mathbf{r}}^+ - S_0^+ S_{\mathbf{r}}^- \rangle \\ &= -\langle S_0^y S_{\mathbf{r}}^x \rangle. \end{aligned} \quad (6)$$

In such cases equation (2) reduces to

$$\frac{\partial^2 \sigma}{\partial \Omega' \partial E'} = \mathcal{A}(\mathbf{k}, \mathbf{k}') \sum_{\alpha} (1 - \hat{K}_\alpha^2) \mathcal{S}^{\alpha\alpha}(\mathbf{K}, \omega). \quad (7)$$

2.2. Scattering and fluctuations

In terms of the transform

$$S^z(\mathbf{K}, t) = \sum_{\mathbf{n}} e^{i\mathbf{K} \cdot \mathbf{n}} S_{\mathbf{n}}^z(t), \quad (8)$$

$\mathcal{S}^{\alpha\beta}$ can be written as

$$\mathcal{S}^{\alpha\beta}(\mathbf{K}, \omega) = \frac{1}{2\pi N} \int_{-\infty}^{\infty} e^{-i\omega t} \langle S^z(-\mathbf{K}, 0) S^{\beta}(\mathbf{K}, t) \rangle dt. \quad (9)$$

This is the Fourier transform of the evolution in time of a spin correlation; one may therefore say that neutron measurements give directly the power spectrum of the spontaneous fluctuations of the $S(\mathbf{K})$.

It is clear from (9) that

$$\int_{-\infty}^{\infty} \mathcal{S}^{\alpha\beta}(\mathbf{K}, \omega) d\omega = \frac{1}{N} \langle S^{\alpha}(-\mathbf{K}, 0) S^{\beta}(\mathbf{K}, 0) \rangle. \quad (10)$$

Now

$$S^{\alpha}(0, t) = S_{\text{tot}}^{\alpha}, \quad (11)$$

so that

$$\int_{-\infty}^{\infty} \mathcal{S}^{\alpha\alpha}(0, \omega) d\omega = \frac{1}{N} \langle S_{\text{tot}}^{\alpha 2} \rangle = \frac{1}{Ng^2 \mu^2} \langle M^{\alpha 2} \rangle$$

and this, according to a well-known thermodynamic result that we shall confirm directly, is

$$= \frac{1}{Ng^2 \mu^2} (\langle M^{\alpha} \rangle^2 + NkT\chi^{\alpha\alpha}). \quad (12)$$

We shall see later that an identical formula applies at all reciprocal lattice vectors $\mathbf{K} \equiv \boldsymbol{\tau}$. The term in $\langle M^{\alpha} \rangle^2$ gives a measure of the Bragg scattering, and the second term shows that the remainder of the scattering is related to the susceptibility.

Note, in passing, that the total scattering in any one direction does not correspond to an integral such as (10). On the one hand, for a fixed direction of scattering $\mathbf{K}$ is not constant; on the other, the neutron cannot lose more energy than it possesses to begin with, so that the upper limit of the ω integral in an estimate of total scattering is ω_{inc} . Nevertheless if the incident neutron energy is high enough, and the range of energy transfer to the neutron sufficiently restricted, (10) is a limitingly good approximation to the total scattering. These conditions are matters into which we enquire in the later pages. We shall also derive various functions from $\mathcal{S}$ that are even in ω , and the employment of these in a practical case will always allow the fluctuation spectrum to be estimated on whichever side of zero is desired.

2.3. The wavelength-dependent susceptibility

We now go on to generalize (12) for non-zero $\mathbf{K}$. This will require us to introduce a wavelength-dependent susceptibility $\chi(\mathbf{K})$ with the following significance: Let the magnetic response at $\mathbf{r}$ due to fields $H(\mathbf{r}')$ at $\mathbf{r}'$ be

$$M(\mathbf{r}) = \sum_{\mathbf{r}'} \chi(\mathbf{r} - \mathbf{r}') H(\mathbf{r}'). \quad (13)$$

Then in terms of the Fourier transforms

$$M(\mathbf{K}) = \sum_{\mathbf{r}} M(\mathbf{r}) e^{i\mathbf{K} \cdot \mathbf{r}}, \quad \chi(\mathbf{K}) = \sum_{\mathbf{r}} \chi(\mathbf{r}) e^{i\mathbf{K} \cdot \mathbf{r}}, \quad H(\mathbf{K}) = \sum_{\mathbf{r}} H(\mathbf{r}) e^{i\mathbf{K} \cdot \mathbf{r}}, \quad (14)$$

we have

$$M(\mathbf{K}) = \chi(\mathbf{K}) H(\mathbf{K}). \quad (15)$$

The generalization is important, in part because, of course, the experimental information is at non-zero $\mathbf{K}$, and in part because $\chi(\mathbf{K})$ is a quantity of clear theoretical significance which can be compared with model calculations.

Writing for generality

$$\langle M^\beta(\mathbf{K}) \rangle = \langle M^\beta(\mathbf{K}) \rangle_{H=0} + H^\alpha(\mathbf{K}) \chi^{\alpha\beta}(\mathbf{K}), \quad (16)$$

we shall show that

$$\chi^{\alpha\beta}(\mathbf{K}) = \frac{g^2 \mu^2}{N} \{S^\alpha(-\mathbf{K}), S^\beta(\mathbf{K})\}, \quad (17)$$

in which

$$\{A, B\} = \int_0^\beta \langle e^{\lambda \mathcal{H}} A e^{-\lambda \mathcal{H}} B \rangle d\lambda - \beta \langle A \rangle \langle B \rangle \quad (18)$$

is known as a *relaxation function* for reasons that will become apparent. We shall then show that the cross section can be expressed naturally in terms of the relaxation function.

Call the Hamiltonian

$$\mathcal{H} = \mathcal{H}_0 + g\mu \sum_{\alpha, \mathbf{n}} H_{\mathbf{n}}^\alpha S_{\mathbf{n}}^\alpha \quad (19)$$

$$= \mathcal{H}_0 + \frac{g\mu}{N} \sum_{\alpha, \mathbf{K}} H^\alpha(\mathbf{K}) S^\alpha(-\mathbf{K}) = \mathcal{H}_0 + V, \text{ say.} \quad (20)$$

The response of the system will be given by

$$\langle M^\alpha(\mathbf{K}) \rangle = -g\mu \langle S^\alpha(\mathbf{K}) \rangle = -g\mu \text{Tr} \{ \rho S^\alpha(\mathbf{K}) \}, \quad (21)$$

and we have to calculate the density matrix

$$\rho = Z^{-1} e^{-\beta \mathcal{H}}. \quad (22)$$

Write

$$\rho = Z^{-1} e^{-\beta(\mathcal{H}_0 + V)} = Z^{-1} e^{-\beta \mathcal{H}_0} (1 + \alpha). \quad (23)$$

Then

$$\alpha = e^{\beta \mathcal{H}_0} e^{-\beta(\mathcal{H}_0 + V)} - 1.$$

$$\frac{d\alpha}{d\beta} = e^{\beta \mathcal{H}_0} \mathcal{H}_0 e^{-\beta(\mathcal{H}_0 + V)} - e^{\beta \mathcal{H}_0} (\mathcal{H}_0 + V) e^{-\beta(\mathcal{H}_0 + V)}$$

$$= -e^{\beta \mathcal{H}_0} V e^{-\beta(\mathcal{H}_0 + V)},$$

i.e.

$$= -e^{\beta \mathcal{H}_0} V e^{-\beta \mathcal{H}_0} + \text{higher order terms.}$$

Integrating with respect to β , we obtain

$$\alpha = - \int_0^\beta e^{\lambda \mathcal{H}_0} V e^{-\lambda \mathcal{H}_0} d\lambda; \quad (24)$$

the lower limit is seen to be 0 from the fact that $\alpha = 0$ at $T = \infty$. Therefore

$$\rho = Z^{-1} e^{-\beta \mathcal{H}_0} \left(1 - \int_0^\beta e^{\lambda \mathcal{H}_0} V e^{-\lambda \mathcal{H}_0} d\lambda \right). \quad (25)$$

We deduce Z^{-1} from

$$\text{Tr} \rho = 1 = Z^{-1} \left\{ Z_0 - \int_0^\beta \text{Tr} (e^{-\beta \mathcal{H}_0} e^{\lambda \mathcal{H}_0} V e^{-\lambda \mathcal{H}_0}) d\lambda \right\} \quad (Z_0 = \text{Tr} e^{-\beta \mathcal{H}_0}).$$

In the trace, $e^{\lambda \mathcal{H}_0}$ can be transferred to the left and thence to the right-hand end, where it cancels; thus

$$1 = Z^{-1} Z_0 (1 - \beta \langle V \rangle);$$

$$Z^{-1} \simeq Z_0^{-1} (1 + \beta \langle V \rangle). \quad (26)$$

Finally

$$\rho = Z_0^{-1} e^{-\beta \mathcal{H}_0} \left(1 - \int_0^\beta e^{\lambda \mathcal{H}_0} (V - \langle V \rangle) e^{-\lambda \mathcal{H}_0} d\lambda \right). \quad (27)$$

Then from (21),

$$\langle M^\beta(\mathbf{K}) \rangle = \langle M^\beta(\mathbf{K}) \rangle_{H=0} + g\mu \int_0^\beta \langle e^{\lambda \mathcal{H}_0} (V - \langle V \rangle) e^{-\lambda \mathcal{H}_0} S^\beta(\mathbf{K}) \rangle d\lambda; \quad (28)$$

using (20),

$$\langle M^\beta(\mathbf{K}) \rangle$$

$$= \langle M^\beta(\mathbf{K}) \rangle_{H=0} + \frac{g^2 \mu^2}{N} H^\alpha(\mathbf{K}) \int_0^\beta \langle e^{\lambda \mathcal{H}_0} \{ S^\alpha(-\mathbf{K}) - \langle S^\alpha(-\mathbf{K}) \rangle \} e^{-\lambda \mathcal{H}_0} S^\beta(\mathbf{K}) \rangle d\lambda. \quad (29)$$

On comparison with (16) we see we have deduced

$$\chi^{\alpha\beta}(\mathbf{K}) = \frac{g^2 \mu^2}{N} \int_0^\beta \langle e^{\lambda \mathcal{H}_0} \{ S^\alpha(-\mathbf{K}) - \langle S^\alpha(-\mathbf{K}) \rangle \} e^{-\lambda \mathcal{H}_0} S^\beta(\mathbf{K}) \rangle d\lambda,$$

$$= \frac{g^2 \mu^2}{N} \int_0^\beta \{ \langle e^{\lambda \mathcal{H}_0} S^\alpha(-\mathbf{K}) e^{-\lambda \mathcal{H}_0} S^\beta(\mathbf{K}) \rangle - \langle S^\alpha(-\mathbf{K}) \rangle \langle S^\beta(\mathbf{K}) \rangle \} d\lambda, \quad (30)$$

i.e. equation (17), as required.

2.4. Susceptibility and fluctuations

Notice that at $K = 0$, (30) reduces to

$$\chi^{\alpha\alpha}(0) = \frac{g^2 \mu^2}{N k_B T} \{ \langle S_{\text{tot}}^{\alpha 2} \rangle - \langle S_{\text{tot}}^\alpha \rangle^2 \}, \quad (31)$$

which in the light of (10) reproduces (12). And writing (31) as

$$\chi^{\alpha\alpha}(0) = N \frac{\langle (M_{\mathbf{r}}^\alpha - \langle M_{\mathbf{r}}^\alpha \rangle)^2 \rangle}{k_B T}, \quad (32)$$

we see that the classical susceptibility of an assemblage of a large number N of spins is proportional to the mean square fluctuation of the magnetization per atom. The more general case (30) can be seen from this point of view to involve a more complicated average over the fluctuations.

2.5. The power spectrum in terms of the relaxation function

Consider the function

$$\frac{\hbar\omega}{1 - e^{-\hbar\omega\beta}} \frac{1}{2\pi N} \int_{-\infty}^{\infty} e^{-i\omega t} \{ S^\alpha(-\mathbf{K}, 0), S^\beta(\mathbf{K}, t) \} dt. \quad (33)$$

The second term of (18) contributes

$$-\frac{\hbar\omega}{1-e^{-\hbar\omega\beta}} \frac{1}{2\pi N} \int_{-\infty}^{\infty} e^{-i\omega t} \beta \langle S^{\alpha}(-\mathbf{K}, 0) \rangle \langle S^{\beta}(\mathbf{K}, t) \rangle dt;$$

but $\langle S^{\beta}(\mathbf{K}, t) \rangle$ is independent of t , so that this comes to

$$= -\frac{1}{N} \delta(\omega) \langle S^{\alpha}(\mathbf{K}, 0) \rangle \langle S^{\beta}(\mathbf{K}, 0) \rangle \quad (34)$$

and is just $-\mathcal{S}_{\text{Bragg}}^{\alpha\beta}$. The first term of (18) contributes

$$\begin{aligned} & \frac{\hbar\omega}{1-e^{-\hbar\omega\beta}} \frac{1}{2\pi N} \int_{-\infty}^{\infty} dt e^{-i\omega t} \int_0^{\beta} d\lambda \langle e^{\lambda\mathcal{H}} S^{\alpha}(-\mathbf{K}, 0) e^{-\lambda\mathcal{H}} e^{i\mathcal{H}t/\hbar} S^{\beta}(\mathbf{K}, 0) e^{-i\mathcal{H}t/\hbar} \rangle \\ &= \frac{\hbar\omega}{1-e^{-\hbar\omega\beta}} \frac{1}{2\pi N} \int_{-\infty}^{\infty} dt e^{-i\omega t} \int_0^{\beta} d\lambda \sum_{ij} Z^{-1} e^{(-\beta+\lambda)E_i} S^{\alpha}(-\mathbf{K})_{ij} e^{-\lambda E_j} e^{iE_j t/\hbar} S^{\beta}(\mathbf{K})_{ji} e^{-iE_i t/\hbar} \\ &= \frac{\hbar\omega}{1-e^{-\hbar\omega\beta}} \frac{1}{2\pi N} \sum_{ij} \int_{-\infty}^{\infty} dt e^{-i\{\omega-(E_j-E_i)/\hbar\}t} \int_0^{\beta} d\lambda Z^{-1} e^{-\beta E_i - \lambda(E_j-E_i)} S^{\alpha}(-\mathbf{K})_{ij} S^{\beta}(\mathbf{K})_{ji} \\ &= \frac{\hbar\omega}{1-e^{-\hbar\omega\beta}} \frac{1}{N} \sum_{ij} \delta\left(\omega - \frac{E_j-E_i}{\hbar}\right) Z^{-1} e^{-\beta E_i} S^{\alpha}(-\mathbf{K})_{ij} S^{\beta}(\mathbf{K})_{ji} \int_0^{\beta} e^{-\lambda\hbar\omega} d\lambda \\ &= \frac{1}{N} \sum_{ij} \delta\left(\omega - \frac{E_j-E_i}{\hbar}\right) Z^{-1} e^{-\beta E_i} S^{\alpha}(-\mathbf{K})_{ij} S^{\beta}(\mathbf{K})_{ji} \\ &= \frac{1}{2\pi N} \int_{-\infty}^{\infty} e^{-i\omega t} Z^{-1} \sum_{ij} e^{-\beta E_i} S^{\alpha}(-\mathbf{K})_{ij} e^{iE_j t/\hbar} S^{\beta}(\mathbf{K})_{ji} e^{-iE_i t/\hbar} dt \\ &= \frac{1}{2\pi N} \int_{-\infty}^{\infty} e^{-i\omega t} \langle S^{\alpha}(-\mathbf{K}, 0) S^{\beta}(\mathbf{K}, t) \rangle dt \\ &= \mathcal{S}^{\alpha\beta}(\mathbf{K}, \omega). \end{aligned} \quad (35)$$

Thus

$$\mathcal{S}^{\alpha\beta}(\mathbf{K}, \omega) - \mathcal{S}_{\text{Bragg}}^{\alpha\beta}(\mathbf{K}, \omega) = \frac{\hbar\omega}{1-e^{-\hbar\omega\beta}} \frac{1}{2\pi N} \int_{-\infty}^{\infty} e^{-i\omega t} \{S^{\alpha}(-\mathbf{K}, 0), S^{\beta}(\mathbf{K}, t)\} dt. \quad (36)$$

From (17) and (36) we at once get

$$\int_{-\infty}^{\infty} \frac{1-e^{-\hbar\omega\beta}}{\hbar\omega\beta} \{ \mathcal{S}^{\alpha\beta}(\mathbf{K}, \omega) - \mathcal{S}_{\text{Bragg}}^{\alpha\beta}(\mathbf{K}, \omega) \} d\omega = \frac{1}{3} S(S+1) \frac{\chi^{\alpha\beta}(\mathbf{K})}{\chi_0}, \quad (37)$$

where $\chi_0 = g^2 \mu^2 S(S+1)/3k_B T$. This is our desired generalization of (12); it shows that in the limit of high incident neutron energies, or in practice when the material energy transfers are much smaller than the incident neutron energy, the total scattering at general $\mathbf{K}$ is proportional to $\chi(\mathbf{K})$.

Let us call the relaxation function

$$\{S^{\alpha}(-\mathbf{K}, 0), S^{\beta}(\mathbf{K}, t)\} = R^{\alpha\beta}(\mathbf{K}, t), \quad (38)$$

and define

$$F^{\alpha\beta}(\mathbf{K}, t) = \frac{R^{\alpha\beta}(\mathbf{K}, t)}{R^{\alpha\beta}(\mathbf{K}, 0)} = \frac{\{S^{\alpha}(-\mathbf{K}, 0), S^{\beta}(\mathbf{K}, t)\}}{\{S^{\alpha}(-\mathbf{K}, 0), S^{\beta}(\mathbf{K}, 0)\}}. \quad (39)$$

We could introduce Fourier transforms

$$\begin{aligned} R^{\alpha\beta}(\mathbf{K}, \omega) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega t} R^{\alpha\beta}(\mathbf{K}, t) dt; \\ F^{\alpha\beta}(\mathbf{K}, \omega) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega t} F^{\alpha\beta}(\mathbf{K}, t) dt. \end{aligned} \quad (40)$$

The relationship is simply

$$R(\mathbf{K}, \omega) = \frac{N}{g^2 \mu^2} \chi(\mathbf{K}) F(\mathbf{K}, \omega); \quad (41)$$

however,

$$\int_{-\infty}^{\infty} F^{\alpha\beta} d\omega = 1, \quad (42)$$

and this property of unit area renders the function F rather more convenient to think about when one is concentrating on the *shape* of the frequency distribution of the response function, as we shall find ourselves doing in later sections.

Then

$$\mathcal{S}^{\alpha\beta} - \mathcal{S}_{\text{Bragg}}^{\alpha\beta} = N^{-1} \frac{\hbar\omega}{1 - e^{-\hbar\omega\beta}} R^{\alpha\beta}(\mathbf{K}, \omega); \quad (43)$$

an alternative form is

$$= \frac{k_B T}{g^2 \mu^2} \chi^{\alpha\beta}(\mathbf{K}) \frac{\hbar\omega\beta}{1 - e^{-\hbar\omega\beta}} F^{\alpha\beta}(\mathbf{K}, \omega). \quad (44)$$

If, for instance, we were to assume that

$$F^{\alpha\beta}(\mathbf{K}, t) = e^{-\Gamma_K^{\alpha\beta} |t|}, \quad (45)$$

we would get

$$\mathcal{S}^{\alpha\beta}(\mathbf{K}, \omega) - \mathcal{S}_{\text{Bragg}}^{\alpha\beta}(\mathbf{K}, \omega) = \frac{\hbar\omega\beta}{1 - e^{-\hbar\omega\beta}} \frac{k_B T}{g^2 \mu^2} \chi^{\alpha\beta}(\mathbf{K}) \frac{1}{\pi} \frac{\Gamma_K^{\alpha\beta}}{\omega^2 + \Gamma_K^{\alpha\beta/2}}. \quad (46)$$

2.6. Time-dependent susceptibilities

We now introduce the more general concepts $\chi^{\alpha\beta}(\mathbf{r}, t)$ and $\chi^{\alpha\beta}(\mathbf{K}, t)$. Their relationships are less elementary than those we have met of $\chi^{\alpha\beta}(\mathbf{K})$ because of an essential asymmetry; an atom is not susceptible to effects that have not yet taken place, and consequently $\chi^{\alpha\beta}(\mathbf{r}, t)$ is zero for negative t . In fact,

$$\chi^{\alpha\beta}(\mathbf{K}) \neq \chi^{\alpha\beta}(\mathbf{K}, t = 0) \quad (47)$$

although

$$\chi^{\alpha\beta}(\mathbf{K}) = g^2 \mu^2 N^{-1} R^{\alpha\beta}(\mathbf{K}, t = 0). \quad (48)$$

One need not be too surprised at this difference, because $\chi^{\alpha\beta}(\mathbf{K})$ describes the response to a steady field, whereas $\chi^{\alpha\beta}(\mathbf{K}, t = 0)$ essentially mediates an instantaneous response.

$\chi^{\alpha\beta}(\mathbf{r}, t)$ is known as a response function, and is defined by

$$\langle M^\beta(\mathbf{n}, t) \rangle = \langle M^\beta(\mathbf{n}, t) \rangle_{H=0} + \int_{-\infty}^t \sum_{\mathbf{m}} H^\alpha(\mathbf{m}, t') \chi^{\alpha\beta}(\mathbf{n} - \mathbf{m}, t - t') e^{\alpha t'} dt'. \quad (49)$$

The factor $e^{\epsilon t'}$ is inserted here to demonstrate that convergence can always be guaranteed at the $-\infty$ end of the integration; it can be put equal to unity at will. On operating through with $\sum_{\mathbf{n}} e^{i\mathbf{K} \cdot \mathbf{n}}$ we obtain

$$\langle M^\beta(\mathbf{K}, t) \rangle = \langle M^\beta(\mathbf{K}, t) \rangle_{H=0} + \int_{-\infty}^t \sum_{\mathbf{m}, \mathbf{n}} e^{i\mathbf{K} \cdot \mathbf{m}} H^\alpha(\mathbf{m}, t') e^{i\mathbf{K} \cdot (\mathbf{n} - \mathbf{m})} \chi^{\alpha\beta}(\mathbf{n} - \mathbf{m}, t - t') e^{\epsilon t'} dt' \quad (50)$$

$$= \langle M^\beta(\mathbf{K}, t) \rangle_{H=0} + \int_{-\infty}^t H^\alpha(\mathbf{K}, t') \chi^{\alpha\beta}(\mathbf{K}, t - t') e^{\epsilon t'} dt', \quad (51)$$

with definitions similar to (14), in particular

$$\chi^{\alpha\beta}(\mathbf{K}, t - t') = \sum_{\mathbf{n}} e^{i\mathbf{K} \cdot (\mathbf{n} - \mathbf{m})} \chi^{\alpha\beta}(\mathbf{n} - \mathbf{m}, t - t'). \quad (52)$$

An explicit formula for $\chi^{\alpha\beta}(\mathbf{K}, t)$ may be obtained as follows. We have

$$i\hbar \frac{\partial \psi}{\partial t} = \mathcal{H} \psi = (\mathcal{H}_0 + V) \psi = \left\{ \mathcal{H}_0 + H^\alpha(\mathbf{K}, t) \frac{g\mu}{N} S^\alpha(-\mathbf{K}, t) \right\} \psi, \quad (53)$$

say (referring back to (20)). Now let

$$\psi(t) = e^{-iE_0 t/\hbar} |i\rangle + \sum_j a_j(t) e^{-iE_j t/\hbar} |j\rangle. \quad (54)$$

Then by (53),

$$\dot{a}_j(t) = \frac{1}{i\hbar} e^{i(E_j - E_0)t/\hbar} \langle j | V(t) | i \rangle + \text{higher order terms.} \quad (55)$$

Therefore

$$a_j(t) = \frac{1}{i\hbar} \int_{-\infty}^t e^{i(E_j - E_0)t'/\hbar} \langle j | V(t') | i \rangle dt' \quad (56)$$

$$= \frac{g\mu}{i\hbar N} \int_{-\infty}^t H^\alpha(\mathbf{K}, t') \langle j | S^\alpha(-\mathbf{K}, t') | i \rangle dt'. \quad (57)$$

Now

$$\langle \psi | M^\beta(\mathbf{K}, t) | \psi \rangle = \langle i | M^\beta(\mathbf{K}, t) | i \rangle - \sum_j a_j(t) \langle i | g\mu S^\beta(\mathbf{K}, t) | j \rangle - \sum_j a_j(t)^* \langle j | g\mu S^\beta(\mathbf{K}, t) | i \rangle, \quad (58)$$

$$= \langle i | M^\beta(\mathbf{K}, t) | i \rangle + \frac{i}{\hbar} \frac{g^2 \mu^2}{N} \int_{-\infty}^t H^\alpha(\mathbf{K}, t') \langle i | [S^\beta(\mathbf{K}, t), S^\alpha(-\mathbf{K}, t')] | i \rangle dt'; \quad (59)$$

hence

$$\langle M^\beta(\mathbf{K}, t) \rangle = \langle M^\beta(\mathbf{K}, t) \rangle_{H=0} + \frac{i}{\hbar} \frac{g^2 \mu^2}{N} \int_{-\infty}^t H^\alpha(\mathbf{K}, t') \langle [S^\beta(\mathbf{K}, t), S^\alpha(-\mathbf{K}, t')] \rangle dt', \quad (60)$$

and so by (51)

$$\chi^{\alpha\beta}(\mathbf{K}, t - t') = \frac{1}{i\hbar} \frac{g^2 \mu^2}{N} \langle [S^\alpha(-\mathbf{K}, t'), S^\beta(\mathbf{K}, t)] \rangle \quad t > t'. \quad (61)$$

2.7. Relaxation and susceptibility

An important property of the relaxation function is that

$$\begin{aligned}
 \frac{\partial}{\partial t} \{A, B(t)\} &= \frac{1}{i\hbar} \{A, [B(t), \mathcal{H}]\} \\
 &= \frac{1}{i\hbar Z} \int_0^\beta \sum_i (\langle i | e^{-\beta \mathcal{H}} e^{\lambda \mathcal{H}} A e^{-\lambda \mathcal{H}} e^{i\mathcal{H}t/\hbar} B e^{-i\mathcal{H}t/\hbar} \mathcal{H} | i \rangle \\
 &\quad - \langle i | e^{-\beta \mathcal{H}} e^{\lambda \mathcal{H}} A e^{-\lambda \mathcal{H}} \mathcal{H} e^{i\mathcal{H}t/\hbar} B e^{-i\mathcal{H}t/\hbar} | i \rangle) d\lambda \\
 &= \frac{1}{i\hbar Z} \sum_{ij} A_{ij} B_{ji} e^{-\beta E_i} e^{i(E_j - E_i)t/\hbar} (E_i - E_j) \int_0^\beta e^{\lambda(E_i - E_j)} d\lambda \\
 &= \frac{i}{\hbar Z} \sum_{ij} A_{ij} B_{ji} e^{i(E_j - E_i)t/\hbar} (e^{-\beta E_i} - e^{-\beta E_j}) \\
 &= \frac{i}{\hbar} \langle AB(t) - B(t)A \rangle \\
 &= \frac{i}{\hbar} \langle [A, B(t)] \rangle.
 \end{aligned} \tag{62}$$

Thus

$$\frac{g^2 \mu^2}{N} \frac{\partial}{\partial t} R^{\alpha\beta}(\mathbf{K}, t) = -\chi^{\alpha\beta}(\mathbf{K}, t). \tag{63}$$

2.8. Relaxation of the system after a disturbance

Consider how the spin system relaxes if an imposed field is suddenly switched off. Suppose that the system be exposed from $t = -\infty$ to a steady field $H^\alpha(\mathbf{m}, t)$ such that

$$\begin{aligned}
 H^\alpha(\mathbf{m}, t) &= H^\alpha(\mathbf{m}), \quad t < 0; \\
 &= 0, \quad t > 0.
 \end{aligned} \tag{64}$$

Then for $t > 0$, we have (49) but with the upper limit of the integral zero:

$$\langle M^\beta(\mathbf{n}, t) \rangle = \langle M^\beta(\mathbf{n}, t) \rangle_{H=0} + \int_{-\infty}^0 \sum_{\mathbf{m}} H^\alpha(\mathbf{m}) \chi^{\alpha\beta}(\mathbf{n} - \mathbf{m}, t - t') e^{\epsilon t'} dt'. \tag{65}$$

Alternatively, by (51),

$$\langle M^\beta(\mathbf{K}, t) \rangle = \langle M^\beta(\mathbf{K}, t) \rangle_{H=0} + H^\alpha(\mathbf{K}) \int_t^\infty \chi^{\alpha\beta}(\mathbf{K}, x) e^{-\epsilon x} dx. \tag{66}$$

But according to (62),

$$\int_t^\infty \chi^{\alpha\beta}(\mathbf{K}, x) dx = -\frac{g^2 \mu^2}{N} \int_t^\infty \frac{\partial}{\partial x} R^{\alpha\beta}(\mathbf{K}, x) dx \tag{67}$$

$$\begin{aligned}
 &= \frac{g^2 \mu^2}{N} \{R^{\alpha\beta}(\mathbf{K}, t) - R^{\alpha\beta}(\mathbf{K}, \infty)\} \\
 &= \frac{g^2 \mu^2}{N} R^{\alpha\beta}(\mathbf{K}, t).
 \end{aligned} \tag{68}$$

Thus

$$\langle M^\beta(\mathbf{K}, t) \rangle = \langle M^\beta(\mathbf{K}, t) \rangle_{H=0} + H^\alpha(\mathbf{K}) \frac{g^2 \mu^2}{N} R^{\alpha\beta}(\mathbf{K}, t). \quad (69)$$

We now see why our R is called a relaxation function. Its appearance in (69) is at first sight surprising; it is in the place where we have been accustomed to find a susceptibility, whereas we have just shown that R is the integral of a susceptibility. $g^2 \mu^2 N^{-1} R^{\alpha\beta}(\mathbf{K}, t)$ is, of course, to be clearly distinguished from $\chi^{\alpha\beta}(\mathbf{K}, t)$, a symbol that has been reserved for (52). Actually $\chi^{\alpha\beta}(\mathbf{K}, t)$ is given by (63) while $\chi(\mathbf{K})$ is given by (48).

2.9. Frequency-dependent susceptibility

A quantity $\chi^{\alpha\beta}(\mathbf{K}, \omega)$ can be introduced as the response function corresponding to an impressed field

$$H^\alpha(\mathbf{K}, t) = H^\alpha(\mathbf{K}) e^{i\omega t}. \quad (70)$$

Then following on (51),

$$\langle M^\beta(\mathbf{K}, t) \rangle = \langle M^\beta(\mathbf{K}, t) \rangle_{H=0} + H^\alpha(\mathbf{K}) e^{i\omega t} \int_0^\infty e^{-i\omega x} \chi^{\alpha\beta}(\mathbf{K}, x) e^{-ex} dx; \quad (71)$$

we make the definition

$$\langle M^\beta(\mathbf{K}, t) \rangle = \langle M^\beta(\mathbf{K}, t) \rangle_{H=0} + \{e^{i\omega t} H^\alpha(\mathbf{K})\} \chi^{\alpha\beta}(\mathbf{K}, \omega), \quad (72)$$

and this leads to

$$\chi^{\alpha\beta}(\mathbf{K}, \omega) = \int_0^\infty e^{-i\omega t} \chi^{\alpha\beta}(\mathbf{K}, t) e^{-et} dt. \quad (73)$$

This form is slightly different from what would have been written for a symmetric function, having a $\int_0^\infty$ as χ is zero for negative t . In fact we could replace $\int_0^\infty$ with $\int_{-\infty}^\infty$, and then invert; the correct result is

$$\chi^{\alpha\beta}(\mathbf{K}, t) = \frac{1}{2\pi} \int_C e^{i\omega t} \chi^{\alpha\beta}(\mathbf{K}, \omega) d\omega, \quad (74)$$

where C is a contour running below all singularities in the plane of complex ω .

Now from the definition of $\chi^{\alpha\beta}(\mathbf{K}, \omega)$ at equation (73) together with (63),

$$\chi^{\alpha\beta}(\mathbf{K}, \omega) = -\frac{g^2 \mu^2}{N} \int_0^\infty e^{-i\omega t} \frac{\partial}{\partial t} R^{\alpha\beta}(\mathbf{K}, t) dt; \quad (75)$$

integrating by parts,

$$\begin{aligned} \chi^{\alpha\beta}(\mathbf{K}, \omega) &= \frac{g^2 \mu^2}{N} R^{\alpha\beta}(\mathbf{K}, 0) - \frac{g^2 \mu^2}{N} i\omega \int_0^\infty e^{-i\omega t} R^{\alpha\beta}(\mathbf{K}, t) dt \\ &= \chi^{\alpha\beta}(\mathbf{K}) \left\{ 1 - i\omega \int_0^\infty e^{-i\omega t} F^{\alpha\beta}(\mathbf{K}, t) dt \right\}. \end{aligned} \quad (76)$$

We have related $\chi^{\alpha\beta}(\mathbf{K}, \omega)$ to $\chi^{\alpha\beta}(\mathbf{K})$, and shown that

$$\chi^{\alpha\beta}(\mathbf{K}) = \chi^{\alpha\beta}(\mathbf{K}, \omega = 0). \quad (77)$$

2.10. Properties of the functions

Now

$$\begin{aligned}
 \{A, B\} &= Z^{-1} \int_0^\beta \sum_i \langle i | e^{-\beta \mathcal{H}} e^{\lambda \mathcal{H}} A e^{-\lambda \mathcal{H}} B | i \rangle d\lambda - \beta \langle A \rangle \langle B \rangle \\
 &= Z^{-1} \sum_{ij} A_{ij} B_{ji} e^{-\beta E_i} \int_0^\beta e^{\lambda(E_i - E_j)} d\lambda - \beta \langle A \rangle \langle B \rangle \\
 &= Z^{-1} \sum_{ij} A_{ij} B_{ji} \frac{e^{-\beta E_j} - e^{-\beta E_i}}{E_i - E_j} - \beta \langle A \rangle \langle B \rangle,
 \end{aligned}$$

which by interchanging i and j

$$= \{B, A\}. \quad (78)$$

It follows that

$$R^{\alpha\beta}(\mathbf{K}, t) = R^{\beta\alpha}(-\mathbf{K}, -t). \quad (79)$$

Also, by simple inspection,

$$R^{\alpha\beta}(\mathbf{K}, t) = R^{\alpha\beta}(-\mathbf{K}, t)^*. \quad (80)$$

Thus in the most general case $R^{\alpha\beta}$ is complex and is neither even nor odd in $\mathbf{K}$ or t , except that at $K = 0$ it is real by (80). And

$$\text{Re } R^{\alpha\alpha} \text{ is even in } \mathbf{K} \text{ and } t; \quad (81)$$

$$\text{Im } R^{\alpha\alpha} \text{ is odd in } \mathbf{K} \text{ and } t. \quad (82)$$

Rules exactly similar to (79)–(82) apply to $F^{\alpha\beta}$:

$$R^{\alpha\beta}(\mathbf{K}, \omega) = R^{\beta\alpha}(-\mathbf{K}, -\omega) = R^{\beta\alpha}(\mathbf{K}, \omega)^*; \quad (83)$$

$R^{\alpha\alpha}(\mathbf{K}, \omega)$ is real, without being necessarily even or odd in $\mathbf{K}$ or ω .

From (79) and (80),

$$\chi^{\alpha\beta}(\mathbf{K})^* = \chi^{\alpha\beta}(-\mathbf{K}) = \chi^{\beta\alpha}(\mathbf{K}); \quad (84)$$

thus $\chi^{\alpha, \beta \neq \alpha}(\mathbf{K})$ is in the most general case complex, but $\chi^{\alpha\alpha}(\mathbf{K})$ is always real and even in $\mathbf{K}$:

$$\chi^{\alpha\alpha}(\mathbf{K})^* = \chi^{\alpha\alpha}(\mathbf{K}) = \chi^{\alpha\alpha}(-\mathbf{K}). \quad (85)$$

From (75)–(76), $\chi^{\alpha\beta}(\mathbf{K}, \omega)$ is complex and not necessarily even in either $\mathbf{K}$ or ω , irrespective of whether $\beta = \alpha$; however,

$$\chi^{\alpha\beta}(\mathbf{K}, \omega) = \chi^{\alpha\beta}(-\mathbf{K}, -\omega)^*. \quad (86)$$

$F^{\alpha, \beta \neq \alpha}$ is in the most general case complex, and from (79):

$$F^{\alpha\beta}(\mathbf{K}, \omega) = F^{\beta\alpha}(-\mathbf{K}, -\omega) = F^{\beta\alpha}(\mathbf{K}, \omega)^*; \quad (87)$$

$F^{\alpha\alpha}(\mathbf{K}, \omega)$ is consequently real, without being necessarily even or odd in $\mathbf{K}$ or ω . Finally, in the most general case $\mathcal{S}^{\alpha\beta} - \mathcal{S}_{\text{Bragg}}^{\alpha\beta}$ is not necessarily even in either $\mathbf{K}$ or ω , even when $\alpha = \beta$.

The position is particularly simple in structures whose magnetic symmetry is such that $\mathbf{K}$ and $-\mathbf{K}$ are equivalent in (8). In this situation everything becomes even in $\mathbf{K}$, and by (80) and (52)

$$R^{\alpha\beta}(\mathbf{K}, t), F^{\alpha\beta}(\mathbf{K}, t), \chi^{\alpha\beta}(\mathbf{K}, t) \text{ and } \chi^{\alpha\beta}(\mathbf{K}) \text{ become real, } (\mathbf{K} \equiv -\mathbf{K}) \quad (88)$$

implying by (81) that $R^{\alpha\alpha}(\mathbf{K}, t)$ becomes even in t . $R^{\alpha, \beta \neq \alpha}(\mathbf{K}, \omega)$ and $F^{\alpha, \beta \neq \alpha}(\mathbf{K}, \omega)$ are still in general complex, but

$$\left. \begin{aligned} R^{\alpha\alpha}(\mathbf{K}, \omega) &= R^{\alpha\alpha}(\mathbf{K}, \omega)^* = R^{\alpha\alpha}(\mathbf{K}, -\omega) \\ F^{\alpha\alpha}(\mathbf{K}, \omega) &= F^{\alpha\alpha}(\mathbf{K}, \omega)^* = F^{\alpha\alpha}(\mathbf{K}, -\omega) \end{aligned} \right\} (\mathbf{K} \equiv -\mathbf{K}) \quad (89)$$

so that $R^{\alpha\alpha}(\mathbf{K}, \omega)$ and $F^{\alpha\alpha}(\mathbf{K}, \omega)$ are real and even in ω . Equation (86) shows that

$$\left. \begin{aligned} \text{Re } \chi^{\alpha\beta}(\mathbf{K}, \omega) &\text{ is even in } \omega \\ \text{Im } \chi^{\alpha\beta}(\mathbf{K}, \omega) &\text{ is odd in } \omega \end{aligned} \right\} (\mathbf{K} \equiv -\mathbf{K}). \quad (90)$$

From (36), (38) and (79) we see that on defining an

$$\mathcal{S}'^{\alpha\beta}(\mathbf{K}, \omega) = e^{-\frac{1}{2}\hbar\omega\beta} \mathcal{S}^{\alpha\beta}(\mathbf{K}, \omega), \quad (91)$$

we obtain

$$\mathcal{S}'^{\alpha\beta}(\mathbf{K}, \omega) = \mathcal{S}'^{\beta\alpha}(-\mathbf{K}, -\omega); \quad (92)$$

and where $\mathbf{K} \equiv -\mathbf{K}$,

$$\mathcal{S}'^{\alpha\alpha}(\mathbf{K}, \omega) \text{ is even in } \omega \quad (\mathbf{K} \equiv -\mathbf{K}). \quad (93)$$

Going back to equation (2), we notice that where α and β are different, $\mathcal{S}^{\alpha\beta}$ and $\mathcal{S}^{\beta\alpha}$ occur in observable quantities always together in the combination

$$\hat{K}_\alpha \hat{K}_\beta (\mathcal{S}^{\alpha\beta} + \mathcal{S}^{\beta\alpha}).$$

The statement that

$$e^{-\frac{1}{2}\hbar\omega\beta} \frac{1}{2} (\mathcal{S}^{\alpha\beta} + \mathcal{S}^{\beta\alpha}) \text{ is real and even in } \omega \quad (\mathbf{K} \equiv -\mathbf{K}) \quad (94)$$

expresses as simply as possible the so-called condition of detailed balance for magnetic inelastic cross sections. As in crystal-vibration theory, it has been possible to obtain an elementary form of the condition only in the case where $\mathbf{K}$ and $-\mathbf{K}$ are interchangeable in the lattice sums.

2.11. The power spectrum and the susceptibility

From (76), in the light of the foregoing,

$$\begin{aligned} \chi^{\alpha\beta}(\mathbf{K}, \omega) - \chi^{\beta\alpha}(\mathbf{K}, \omega)^* &= -i\chi^{\alpha\beta}(\mathbf{K}) \omega \int_{-\infty}^{\infty} e^{-i\omega t} F^{\alpha\beta}(\mathbf{K}, t) dt \\ &= -\frac{g^2 \mu^2}{N} 2\pi i \omega R^{\alpha\beta}(\mathbf{K}, \omega) \end{aligned} \quad (95)$$

leading to a relationship between the $\mathcal{S}^{\alpha\beta}$'s and the susceptibilities. In the simple case of $\alpha\alpha$,

$$\mathcal{S}^{\alpha\alpha}(\mathbf{K}, \omega) - \mathcal{S}_{\text{Bragg}}^{\alpha\alpha}(\mathbf{K}, \omega) = \frac{\hbar}{\pi} \frac{1}{g^2 \mu^2} \frac{1}{1 - e^{-\hbar\omega\beta}} \text{Im } \chi^{\alpha\alpha}(\mathbf{K}, \omega). \quad (96)$$

2.12. Significance of the generalized susceptibility

We have seen that the generalized susceptibility is simply $\Delta M(\mathbf{K}, \omega)/H(\mathbf{K}, \omega)$. Now any disturbance affecting the spin system, however elaborate—such as the passage of a neutron through it—can be described as a sum of $H(\mathbf{K}, \omega)$'s. Thus if $\chi(\mathbf{K}, \omega)$ is known, the linear response of the system to any perturbation follows immediately from writing χH .

For instance, if a field $H(\mathbf{r}, t) = H_0 \cos(\mathbf{K} \cdot \mathbf{r} - \omega t)$ be applied to a centrosymmetric substance, it is easily shown that the magnetic response will be

$$M(\mathbf{r}, t) = [\text{Re} \chi(\mathbf{K}, \omega)] H_0 \cos(\mathbf{K} \cdot \mathbf{r} - \omega t) + [\text{Im} \chi(\mathbf{K}, \omega)] H_0 \sin(\mathbf{K} \cdot \mathbf{r} - \omega t). \quad (97)$$

The real part of χ therefore gives the direct, in-phase response of the spin system, while the imaginary part gives the out-of-phase response, exactly as in the standard analogous case in classical electricity and magnetism where the existence of a non-zero imaginary part to the dielectric constant produces a complex refractive index and all the consequent phenomena of dispersion, absorption and scattering.

$\text{Im} \chi$ is the absorptive part of the susceptibility, governing the dissipation of energy from the source of the disturbance. If the disturbance is a beam of neutrons, its energy can only be 'dissipated', in Born approximation, by processes which remove neutrons physically from the scene; hence $\text{Im} \chi$ appears as a factor multiplying the scattering cross section.

Equation (96) is an expression of the fluctuation-dissipation theorem, relating as it does the dissipation function on the right (which refers to small macroscopic disturbances) to the spectrum of spontaneous microscopic fluctuations on the left (Callen 1961, Kubo 1966 a, b). Equation (96) can in fact be written

$$\mathcal{S}^{\alpha\alpha}(\mathbf{K}, \omega) - \mathcal{S}_{\text{Bragg}}^{\alpha\alpha}(\mathbf{K}, \omega) = -\frac{\hbar}{\pi} \frac{1}{g^2 \mu^2} (n_{|\omega|} + \tfrac{1}{2} \pm \tfrac{1}{2}) \text{Im} \chi(\mathbf{K}, |\omega|), \quad \omega = \pm |\omega| \quad (98)$$

in which form it exhibits the power spectrum of spontaneous fluctuations as the product of a Bose oscillator-occupation number and an admittance $\text{Im} \chi$. Thus $\text{Im} \chi$ as a function of temperature describes the further weighting of the oscillator spectrum, over and above the Bose probability, that according to statistical mechanics the particular system enjoys in the ensemble average of its spontaneous fluctuations. The generalized $\text{Im} \chi$ therefore embodies the tendency of the system to prefer any $(\mathbf{K}, \omega)$ region by its natural motion. For instance, at low temperatures its poles give the spin-wave spectrum. We shall return to this matter of the poles at the end of the article.

2.13. Green functions

Introducing χ^{+-} , with an obvious notation and the property that

$$\chi^{xx}(\mathbf{K}, t) + \chi^{yy}(\mathbf{K}, t) = \tfrac{1}{2} \{ \chi^{+-}(\mathbf{K}, t) + \chi^{-+}(\mathbf{K}, t) \}, \quad (99)$$

the connection with Green functions can be seen most directly from the fact that $\chi^{+-}(\mathbf{K}, t)$ is just $-g^2 \mu^2 / \hbar N$ times the retarded Green function G_{R} , where this and the advanced Green function G_{A} are defined by

$$G_{\text{R}}(\mathbf{K}, t) = \mp i \theta(\pm t) \langle [S^-(\mathbf{K}, t), S^+(-\mathbf{K}, 0)] \rangle \quad (100)$$

($\theta(t) = 1$ for $t > 0$, 0 for $t < 0$). It follows from the standard results of Green function theory that the spectral density $g(\mathbf{K}, \omega)$ of Zubarev is given by

$$g(\mathbf{K}, \omega) = G_{\text{R}}(\mathbf{K}, \omega) - G_{\text{A}}(\mathbf{K}, \omega) \quad (101)$$

evaluated along the real axis of the complex ω plane. Then the transverse

$$R^{+-}(\mathbf{K}, \omega) = \frac{N}{i\hbar\omega} g(\mathbf{K}, \omega). \quad (102)$$

2.14. Expressions for the inelastic neutron scattering

When a system has the unembellished Heisenberg Hamiltonian (106), so that the special conditions (5)–(6) apply, the inelastic cross section can be written in a variety of simple ways, useful examples being:

$$\frac{\partial^2 \sigma}{\partial \Omega' \partial E'} = \mathcal{A}(\mathbf{k}, \mathbf{k}') \left\{ (1 - \hat{K}_z^2) \mathcal{S}^{zz} + (1 + \hat{K}_z^2) \frac{1}{4} (\mathcal{S}^{+-} + \mathcal{S}^{-+}) \right\} \quad (103)$$

$$= N^{-1} \mathcal{A}(\mathbf{k}, \mathbf{k}') \frac{\hbar \omega}{1 - e^{-\hbar \omega \beta}} \left\{ (1 - \hat{K}_z^2) R^{zz} + (1 + \hat{K}_z^2) \frac{1}{4} (R^{+-} + R^{-+}) \right\} \quad (104)$$

$$= \frac{\hbar}{\pi g^2 \mu^2} \mathcal{A}(\mathbf{k}, \mathbf{k}') \frac{-1}{1 - e^{-\hbar \omega \beta}} \left\{ (1 - \hat{K}_z^2) \text{Im} \chi^{zz} + (1 + \hat{K}_z^2) \frac{1}{4} (\chi^{+-} + \chi^{-+}) \right\} \quad (105)$$

where the functions with superfixes have argument $(\mathbf{K}, \omega)$.

3. Significance of the wavelength-dependent susceptibility

At the beginning of this article we pointed out that the treatment we have just given opens the way for various approximate discussions to be brought to bear on the problem of magnetic correlations. We shall now proceed to sketch in the general behaviour of our fundamental response function, so that the broad outlines of its dependence on $\mathbf{K}$, ω and T become apparent. $\chi(\mathbf{K})$ has a marked and highly characteristic temperature dependence—more pronounced in some respects, as we shall see, than that of $F(\mathbf{K}, \omega)$ —and it will be interesting to examine it first. We accordingly calculate $\chi(\mathbf{K})$ below for a number of situations.

We use molecular field theory. At the level of quantitative detail this approach would be altogether unsatisfactory; but it gives very quickly a comprehensive picture of the main qualitative phenomena in a way that no alternative approach can. Taking the Hamiltonian to be (equation (19))

$$\mathcal{H} = g\mu \sum_{\mathbf{r}} \mathbf{S}(\mathbf{r}) \cdot \mathbf{H}(\mathbf{r}) - 2 \sum_{\mathbf{r} \neq \mathbf{r}'} J(\mathbf{r} - \mathbf{r}') \mathbf{S}(\mathbf{r}) \cdot \mathbf{S}(\mathbf{r}') \quad (106)$$

(so that in the ferromagnetic ground state the magnetic moments point upwards, the spins downwards), the molecular field approximation consists in writing (106) as

$$\mathcal{H} = g\mu \sum_{\mathbf{r}} \mathbf{S}(\mathbf{r}) \cdot \mathbf{H}_{\text{eff}}(\mathbf{r}) = g\mu \sum_{\mathbf{r}} \mathbf{S}(\mathbf{r}) \cdot \left\{ \mathbf{H}(\mathbf{r}) - \frac{2}{g\mu} \sum_{\mathbf{r} \neq \mathbf{r}'} J(\mathbf{r} - \mathbf{r}') \langle \mathbf{S}(\mathbf{r}') \rangle \right\}. \quad (107)$$

3.1. $\chi^{zz}(\mathbf{K})$ above T_C

Above T_C , $\langle S^z(\mathbf{r}) \rangle$ is due only to the small applied $H(\mathbf{r})$ and is small. We may thus write

$$M(\mathbf{r}) = \chi_0 H_{\text{eff}} = \chi_0 \left\{ H(\mathbf{r}) + \frac{2}{g^2 \mu^2} \sum_{\mathbf{r} \neq \mathbf{r}'} J(\mathbf{r} - \mathbf{r}') M(\mathbf{r}') \right\}, \quad (108)$$

with χ_0 as before $g^2 \mu^2 S(S+1)/3k_B T$. Operating through with $\sum_{\mathbf{r}} e^{i\mathbf{K} \cdot \mathbf{r}}$, it results that

$$M(\mathbf{K}) = \chi_0 H(\mathbf{K}) + \frac{2\chi_0}{g^2 \mu^2} J(\mathbf{K}) M(\mathbf{K}) \quad (109)$$

with definitions like (14);

$$\chi^{xx}(\mathbf{K}) = \chi^{yy}(\mathbf{K}) = \chi^{zz}(\mathbf{K}) = \frac{M(\mathbf{K})}{H(\mathbf{K})} = \frac{g^2 \mu^2 S(S+1)}{3k_B T - 2S(S+1)J(\mathbf{K})}, \quad (110)$$

a generalized form of the relation

$$\chi = \frac{C}{T - \theta}. \quad (111)$$

As the temperature is lowered from infinity, $\chi^{zz}(\mathbf{K})$ diverges at that value of $\mathbf{K}$ for which $J(\mathbf{K})$ has its highest maximum. Then, at lower temperatures, $\langle S^z \rangle$ is no longer small, the foregoing analysis is inapplicable and the substance goes over into an ordered phase. If the highest maximum is at $K = 0$ the material becomes a ferromagnet; if not, it becomes an antiferromagnet of some kind. Taking the case of the ferromagnet, we deduce that

$$k_B T_C = \frac{2}{3} S(S+1) J(0). \quad (112)$$

(The general case is taken up later.)

It follows from (32) that on approaching the critical point, the fluctuations of the magnetization increase without limit. Actually it is a general feature of critical points that they are associated with the so-called anomalous increases in various temperature-dependent parameters. These increases are physically traceable to the fact that with approach to T_C , less and less free energy is required to pass from one phase to the other. As another example, it would be possible to show in our own magnetic system that there is also a divergence of the specific heat, due to fluctuations in the energy of the ensembles. (For an introduction to critical fluctuation theory the reader is referred to an article by Münster (1965).)

Proceeding for the moment with the ferromagnetic case, we have in terms of T_C that

$$\chi^{zz}(\mathbf{K}) = \frac{\chi_C}{T/T_C - J(\mathbf{K})/J(0)} = \frac{\chi_C}{(T - T_C)/T_C + \{J(0) - J(\mathbf{K})\}/J(0)}, \quad (113)$$

where χ_C is written for

$$\chi_C = \frac{g^2 \mu^2 S(S+1)}{3k_B T_C}. \quad (114)$$

Now for small K ,

$$J(\mathbf{K}) = \sum_{\mathbf{r}} J(\mathbf{r}) \left[1 - \frac{(\mathbf{K} \cdot \mathbf{r})^2}{2!} + \frac{(\mathbf{K} \cdot \mathbf{r})^4}{4!} - \dots \right]; \quad (115)$$

and taking a spherical average, which for the quadratic term will also be correct for a cubic single crystal, this becomes

$$J(\mathbf{K}) = J^{(0)} - \frac{1}{6} K^2 J^{(2)} + \frac{1}{120} K^4 J^{(4)} - \dots, \quad (116)$$

where the n th moment

$$J^{(n)} = \sum_{\mathbf{r}} J(\mathbf{r}) \cdot r^n. \quad (117)$$

Then since $J(0) = J^{(0)}$, we obtain for small K

$$\chi^{zz}(\mathbf{K}) = \frac{\chi_C}{r_1^{-2} \{ \kappa_1^{-2} + K^2 - \frac{1}{20} K^4 (J^{(4)}/J^{(2)}) + \dots \}}. \quad (118)$$

Writing $\mathbf{K} = \boldsymbol{\tau} + \mathbf{q}$ it is apparent that $J(\mathbf{K}) = J(\mathbf{q})$ if $\boldsymbol{\tau}$ is a reciprocal lattice vector. Thus for $\mathbf{K}$ near reciprocal lattice vectors $\boldsymbol{\tau}$,

$$\chi^{zz}(\mathbf{K}) = \frac{\chi_C}{r_1^2 \{ \kappa_1^2 + q^2 - \frac{1}{20} q^4 (J^{(4)}/J^{(2)}) + \dots \}}. \quad (119)$$

In these equations

$$\left. \begin{aligned} r_1^2 &= \frac{1}{6} \frac{J^{(2)}}{J^{(0)}} \quad (\text{with only nearest-neighbour interactions } r_n^2/6); \\ \kappa_1^2 &= 6 \frac{J^{(0)}}{J^{(2)}} \frac{T - T_C}{T_C} = 6 \frac{J^{(0)}}{J^{(2)}} \frac{\chi_C}{\chi^{zz}(0)}; \\ \kappa_1^2 r_1^2 &= \frac{\chi_C}{\chi^{zz}(0)}. \end{aligned} \right\} \quad (120)$$

The formula is often abbreviated to the quadratic term and written in an elementary form as follows:

$$\chi_{\parallel}(\mathbf{K}) \simeq \frac{\chi_C}{r_1^2 (\kappa_1^2 + q^2)}. \quad (121)$$

If we substitute (121) into (46) and (2), we obtain Van Hove's (1954) oft-quoted

$$\frac{\partial^2 \sigma}{\partial \Omega' \partial E'} = \frac{1}{3\pi} \mathcal{A}(\mathbf{k}, \mathbf{k}') \frac{S(S+1)}{k_B T_C} \sum_{\alpha, \beta} (\delta_{\alpha\beta} - \hat{K}^\alpha \hat{K}^\beta) \frac{\hbar \omega e^{i\hbar \omega \beta}}{e^{i\hbar \omega \beta} - e^{-i\hbar \omega \beta}} \frac{1}{r_1^2 (\kappa_1^2 + q^2)} \frac{\Gamma}{\omega^2 + \Gamma^2}. \quad (122)$$

One should bear in mind the assumptions underlying equation (119), particularly (i) that the Hamiltonian is Heisenberg's; (ii) that $\chi_0 H_{\text{eff}}$ is small; (iii) that q is small; and (iv) that J has finite moments to all orders. Subject to these the theory suggests, at least qualitatively, that the neutron cross section diverges as $T \rightarrow T_C$ and as $q \rightarrow 0$, and this is confirmed by experiment. (The idea that this type of critical divergence occurs was first conceived by Landau (1937) in the analogous situation of x-ray scattering from an ordering alloy.)

We shall see later that the conditions of small energy transfer under which $\chi(\mathbf{K})$ is simply related to the total neutron scattering can easily be satisfied in practice at low K 's near the critical temperature. Several authors have used this fact to make tests of equation (119) (figure 1). As predicted by the more realistic theories (Fisher 1967), κ_1^2 is found to depend on $(T - T_C)^{4/3}$ in Heisenberg magnets that are suitably free from complications (figure 2). In other materials, other dependences are found (see Hamaguchi 1966 for an assessment).

3.2. Just below T_C ; $\chi^{zz}(\mathbf{K})$

We cannot use (108), as $\langle S(\mathbf{r}') \rangle$ is no longer of the same order of small quantities as it is above T_C . We have to say

$$\langle S^z \rangle = \frac{\sum_{S^z} S^z e^{-g\mu S^z H_{\text{eff}} \beta}}{\sum_{S^z} e^{-g\mu S^z H_{\text{eff}} \beta}}. \quad (123)$$

Just below T_C it will be sufficient to expand this expression in powers of $H_{\text{eff}}\beta$, which let us do as far as terms in β^3 . Using the formulae:

$$\sum_{m=-S}^S m^2 = \frac{1}{3}S(S+1)(2S+1);$$

$$\sum_{m=-S}^S m^4 = \frac{1}{5}S(S+1)(2S+1)(S^2+S+\frac{1}{2});$$
(124)

we find that

$$\langle S^z \rangle = -(g\mu H_{\text{eff}}\beta) \frac{1}{3}S(S+1) + (g\mu H_{\text{eff}}\beta)^3 \frac{1}{45}S(S+1)(S^2+S+\frac{1}{2}) - \dots \quad (125)$$

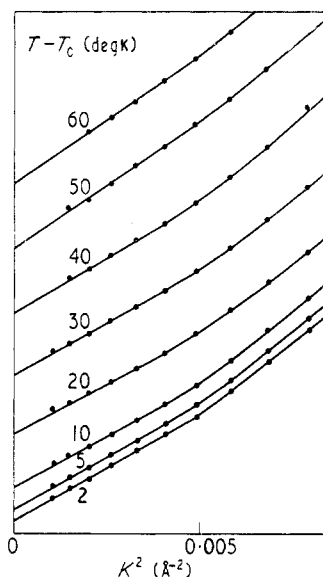


Figure 1. The inverse of the observed intensity scattered at small angles (ranging from 1.3° to 3.5° of arc) by an iron polycrystal near the critical temperature. The data are plotted against the square of the scattering angle, here expressed in terms of the corresponding K in the (roughly adequate) 'static approximation' whereby the intensity would be proportional to $\chi(\mathbf{K})$. A susceptibility like equation (118) is indicated. (After Passell *et al.* 1965.)

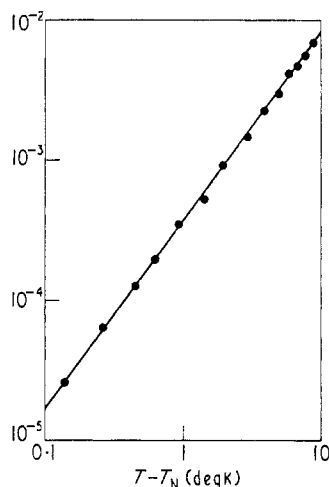


Figure 2. A log-log plot against temperature of a quantity proportional to the inverse susceptibility $\chi(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})^{-1}$ for KMnF_3 , obtained from neutron scattering. The straight line is drawn for a temperature dependence of $(T - T_C)^{1/4}$. (After Cooper and Nathans 1966.)

Notice in passing that this can be written

$$\langle M^z \rangle = \chi_0 H_{\text{eff}} - (\chi_0 H_{\text{eff}})^3 \frac{\frac{3}{5}(S^2+S+\frac{1}{2})}{g^2 \mu^2 S^2(S+1)^2} + \dots \quad (126)$$

We can determine both magnetization and susceptibility by writing

$$\langle S^z(\mathbf{r}) \rangle = \langle S^z \rangle_{H=0} + \delta S^z(\mathbf{r}), \quad (127)$$

which gives with (107)

$$g\mu H_{\text{eff}}(\mathbf{r}) = -2J^{(0)}\langle S^z \rangle_{H=0} + \left\{ g\mu H(\mathbf{r}) - 2 \sum_{\mathbf{r}'} J(\mathbf{r} - \mathbf{r}') \delta S(\mathbf{r}') \right\}. \quad (128)$$

Putting these into (125), we obtain to first order in δS :

$$\begin{aligned} \langle S^z \rangle_{H=0} + \delta S^z(\mathbf{r}) = \frac{1}{3}\beta S(S+1) & \left[2J^{(0)}\langle S^z \rangle_{H=0} - \left\{ g\mu H(\mathbf{r}) - 2 \sum_{\mathbf{r}'} J(\mathbf{r} - \mathbf{r}') \delta S(\mathbf{r}') \right\} \right] \\ & + \frac{1}{45}\beta^3 S(S+1)(S^2 + S + \frac{1}{2}) \left[-8J^{(0)3}\langle S^z \rangle_{H=0}^3 \right. \\ & \left. + 12J^{(0)2}\langle S^z \rangle_{H=0}^2 \left\{ g\mu H(\mathbf{r}) - 2 \sum_{\mathbf{r}'} J(\mathbf{r} - \mathbf{r}') \delta S(\mathbf{r}') \right\} \right]. \quad (129) \end{aligned}$$

The case $H(\mathbf{r}) \equiv 0$ gives

$$\langle S^z \rangle_{H=0} = \frac{2}{3}\beta J^{(0)} S(S+1) \langle S^z \rangle_{H=0} - \frac{8}{45}\beta^3 J^{(0)3} S(S+1)(S^2 + S + \frac{1}{2}) \langle S^z \rangle_{H=0}^3, \quad (130)$$

i.e.

$$\langle S^z \rangle_{H=0}^2 = 0 \quad \text{or} \quad \frac{\frac{2}{3}\beta J^{(0)} S(S+1) - 1}{\frac{8}{45}\beta^3 J^{(0)3} S(S+1)(S^2 + S + \frac{1}{2})}. \quad (131)$$

Specializing to a ferromagnet, this becomes

$$\langle S^z \rangle_{H=0}^2 = \left(\frac{T_C}{T} - 1 \right) \left(\frac{T}{T_C} \right)^3 \frac{5}{3} S^2(S+1)^2 (S^2 + S + \frac{1}{2})^{-1}; \quad (132)$$

$$\langle S^z \rangle_{H=0} \simeq \left(1 - \frac{T}{T_C} \right)^{1/2} \frac{S(S+1)}{\left\{ \frac{5}{3}(S^2 + S + \frac{1}{2}) \right\}^{1/2}} \quad (133)$$

(which, incidentally, predicts a linear decrease of the ferromagnetic Bragg intensity to zero).

Cancelling out now the terms in (129) that remain for $H = 0$, we get

$$\begin{aligned} \delta S^z(\mathbf{r}) = \frac{1}{3}\beta S(S+1) & \left\{ g\mu H(\mathbf{r}) - 2 \sum_{\mathbf{r}'} J(\mathbf{r} - \mathbf{r}') \delta S(\mathbf{r}') \right\} \\ & \times \left\{ -1 + \frac{4}{5}\beta^2 (S^2 + S + \frac{1}{2}) J^{(0)2} \langle S^z \rangle_{H=0}^2 \right\}. \quad (134) \end{aligned}$$

For a ferromagnet we use (112) and (132), and convert this to

$$\delta S^z(\mathbf{r}) = \frac{2T_C - 3T}{2J(0)T} \left\{ g\mu H(\mathbf{r}) - 2 \sum_{\mathbf{r}'} J(\mathbf{r} - \mathbf{r}') \delta S(\mathbf{r}') \right\}. \quad (135)$$

Then

$$\delta S^z(\mathbf{K}) = \frac{2T_C - 3T}{2J(0)T} \{ g\mu H(\mathbf{K}) - 2J(\mathbf{K}) \delta S(\mathbf{K}) \}; \quad (136)$$

$$\chi^{zz}(\mathbf{K}) = g^2 \mu^2 \frac{3T - 2T_C}{2J(0)T [1 + \{J(\mathbf{K})/J(0)\} \{(2T_C - 3T)/T\}]}; \quad (137)$$

and remembering that $T \simeq T_C$ this is approximately

$$= \frac{g^2 \mu^2}{2J(0) [2(T_C - T)/T_C + \{J(0) - J(\mathbf{K})\}/J(0)]}, \quad (138)$$

i.e. equation (119) again but with

$$\kappa_1^{zz} = 12 \frac{J^{(0)}}{J^{(2)}} \frac{T_C - T}{T_C} = 6 \frac{J^{(0)}}{J^{(2)}} \frac{\chi_C}{\chi^{zz}(0)}. \quad (139)$$

We thus have near T_C the general rule $\kappa_1^{zz} = 6(J^{(0)}/J^{(2)})\{\chi_C/\chi^{zz}(0)\}$; but $\chi^{zz}(0)$ diverges twice as rapidly below T_C as above (figure 3). With nearest-neighbour interactions only, the two lines are $(12/r_n^2)(1 - T/T_C)$ and $(6/r_n^2)(T/T_C - 1)$.

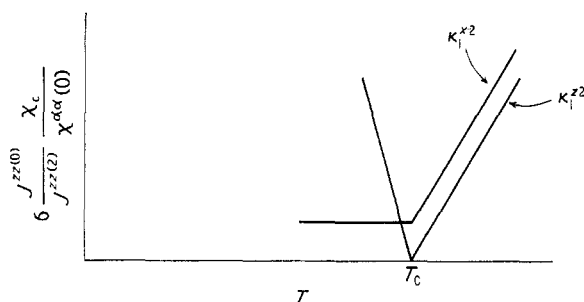


Figure 3. Temperature dependences of the different components of the Ornstein-Zernike range parameter κ_1 , according to molecular-field theory.

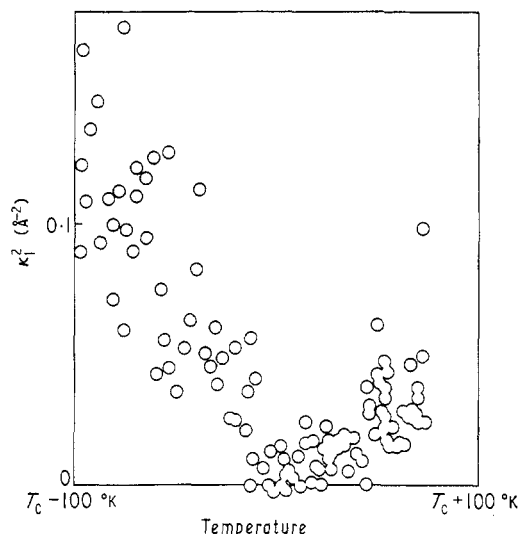


Figure 4. Different temperature dependences measured for κ_1 below and above the Curie temperature in iron. (After Lowde 1958.)

The difference of a factor 2 in slope can also be derived by thermodynamic arguments, and is not, therefore, restricted to 'Heisenberg substances'. About eleven years ago a claim was made that it had been confirmed in a neutron-scattering experiment on iron (figure 4). The analysis was based on ideas current at the time, according to which χ^{zz} and χ^{xx} just below T_C could be treated as similar. Such a view is in conflict with molecular-field theory, as will become apparent in §3.4; but in consequence of new developments discussed in §6 the idea has recently returned to favour. Although the true situation near the Curie point in iron (Shirane *et al.* 1968, Collins *et al.* 1968) is far from clear, it is of some interest to disinter this old experiment, performed at a reactor of neutron flux only $2 \times 10^{12} \text{ cm}^{-2} \text{ s}^{-1}$.

3.3. Above T_C ; ferromagnet and antiferromagnet, including anisotropy

The anisotropy arising from dipole-dipole and spin-orbit coupling effects can be treated by introducing a tensor J , such that

$$\mathcal{H} = -2 \sum_{\mathbf{r} \neq \mathbf{r}'} S^{\alpha}(\mathbf{r}) J^{\alpha\beta}(\mathbf{r} - \mathbf{r}') S^{\beta}(\mathbf{r}'); \quad (140)$$

in a cubic case

$$\mathcal{H} = -2 \sum_{\substack{\mathbf{r} \neq \mathbf{r}' \\ \alpha}} J^{\alpha\alpha}(\mathbf{r} - \mathbf{r}') S^{\alpha}(\mathbf{r}) S^{\alpha}(\mathbf{r}'). \quad (141)$$

We define

$$J^{\alpha\beta}(\mathbf{K}) = \sum_{\mathbf{r}} e^{i\mathbf{K} \cdot \mathbf{r}} J^{\alpha\beta}(\mathbf{r}), \quad (142)$$

for simplicity assuming that the principal axes x , y and z are independent of $\mathbf{K}$ (though strictly speaking the conditions for this should be studied more closely).

Above T_C we can see from (106)–(110) that

$$\chi^{\alpha\alpha}(\mathbf{K}) = \frac{g^2 \mu^2 S(S+1)}{3k_B T - 2S(S+1)J^{\alpha\alpha}(\mathbf{K})}. \quad (143)$$

Thus the critical temperature is determined by the largest value occurring among $J^{xx}(\mathbf{K})$, $J^{yy}(\mathbf{K})$, $J^{zz}(\mathbf{K})$. Let us define the z axis as that principal axis whose $J(\mathbf{K})$ attains the highest maximum. (The z axis is not necessarily an axis of the crystal.) Then in parallel with (112)

$$k_B T_C = \frac{2}{3} S(S+1) J^{zz}(\mathbf{K}_0), \quad (144)$$

$\mathbf{K}_0$ being the vector for which J^{zz} attains its highest maximum; and with (113),

$$\chi^{zz}(\mathbf{K}) = \frac{g^2 \mu^2 S(S+1)/3k_B T_C}{(T - T_C)/T_C + \{J^{zz}(\mathbf{K}_0) - J^{zz}(\mathbf{K})\}/J^{zz}(\mathbf{K}_0)}. \quad (145)$$

Let us now examine the case of a simple antiferromagnet in slightly more detail. If, again for simplicity, we exclude spiral structures we may write for a simple antiferromagnet $\mathbf{K}_0 = \boldsymbol{\tau}_0$, where $\boldsymbol{\tau}_0$ is a magnetic reciprocal lattice vector. For instance, $\boldsymbol{\tau}_0 = (0, 0, 1)$ in the body-centred tetragonal lattice generates the MnF_2 structure; $\boldsymbol{\tau}_0 = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ in the simple-cubic lattice a structure like that of KMnF_3 (with the spins along z), etc. Then just as at (115)–(117),

$$\begin{aligned} J^{zz}(\mathbf{K}) &= J^{zz}(\boldsymbol{\tau}_0 + \mathbf{q}) = \sum_{\mathbf{r}} e^{i\boldsymbol{\tau}_0 \cdot \mathbf{r} + i\mathbf{q} \cdot \mathbf{r}} J^{zz}(\mathbf{r}) \\ &= \sum_{\mathbf{r}} J^{zz}(\mathbf{r}) e^{i\boldsymbol{\tau}_0 \cdot \mathbf{r}} \left\{ 1 - \frac{(\mathbf{q} \cdot \mathbf{r})^2}{2!} + \frac{(\mathbf{q} \cdot \mathbf{r})^4}{4!} - \dots \right\}. \end{aligned} \quad (146)$$

The moments

$$\sum_{\mathbf{r}} J^{\alpha\beta}(\mathbf{r}) e^{i\boldsymbol{\tau}_0 \cdot \mathbf{r}} r^n \quad (147a)$$

to which this expression draws attention simplify on noticing that $e^{i\boldsymbol{\tau}_0 \cdot \mathbf{r}}$ equals 1 for sites with spin parallel to that at the origin and -1 for sites with spin antiparallel. Splitting the sum into parts for the lattices of same and opposite spin, the moments are

$$= \sum_{\text{same}} J^{\alpha\beta}(\mathbf{r}) r^n - \sum_{\text{opp}} J^{\alpha\beta}(\mathbf{r}) r^n \quad (147b)$$

and may be described as a difference moment

$$J_{\text{diff}}^{\alpha\beta(n)} = J_{\text{same}}^{\alpha\beta(n)} - J_{\text{opp}}^{\alpha\beta(n)}. \quad (148)$$

Then for small q , in the spherical average again, correct at the quadratic term for a single crystal:

$$J^{zz}(\mathbf{K}) = J^{zz}(\boldsymbol{\tau}_0) - \frac{1}{6} q^2 J_{\text{diff}}^{zz(2)} + \frac{1}{120} q^4 J_{\text{diff}}^{zz(4)} - \dots \quad (149)$$

and

$$\chi^{zz}(\mathbf{K}) = \frac{g_z^2 \mu^2 S(S+1)}{3k_B T_C} \frac{1}{r_1^2(\kappa_1^{z2} + q^2 - \dots)}, \quad (150)$$

with

$$r_1^2 = \frac{1}{6} \frac{J_{\text{diff}}^{zz(2)}}{J_{\text{diff}}^{zz(0)}}, \quad (151)$$

$$\kappa_1^{z2} = 6 \frac{J_{\text{diff}}^{zz(0)}}{J_{\text{diff}}^{zz(2)}} \frac{T - T_C}{T_C} = 6 \frac{J_{\text{diff}}^{zz(0)}}{J_{\text{diff}}^{zz(2)}} \frac{\chi_C^z}{\chi^{zz}(\tau_0)}. \quad (152)$$

By putting $\tau_0 = 0$ and striking out 'diff' these formulae become those for a ferromagnet (equation (120)).

To discuss the other $\chi^{\alpha\alpha}$'s let us assume for illustrative purposes that

$$J^{xx}(\mathbf{K}) = J^{yy}(\mathbf{K}) = J^{zz}(\mathbf{K}) \{1 - A(\mathbf{K})\}, \quad (153)$$

where $A(\mathbf{K})$ at $\mathbf{K} = \mathbf{K}_0$ is a positive quantity. Then according to equations (143) and (144),

$$\chi^{xx}(\mathbf{K}) = \frac{g_x^2 \mu^2 S(S+1)}{3k_B T_C} \frac{1}{(T - T_C)/T_C + [J^{zz}(\tau_0) - J^{zz}(\mathbf{K}) \{1 - A(\mathbf{K})\}]/J^{zz}(\tau_0)}; \quad (154)$$

at small q and A

$$\chi^{xx}(\mathbf{K}) = \frac{g_x^2 \mu^2 S(S+1)}{3k_B T_C} \frac{1}{r_1^2(\kappa_1^{x2} + q^2 - \dots)} \quad (155)$$

with (151) and

$$\kappa_1^{x2} = 6 \frac{J_{\text{diff}}^{zz(0)}}{J_{\text{diff}}^{zz(2)}} \left\{ \frac{T - T_C}{T_C} + A(\tau_0) \right\}. \quad (156)$$

Again by putting $\tau_0 = 0$ and striking out 'diff', these formulae become those for a ferromagnet. We have the situation of figure 3.

In the presence of anisotropy, critical effects are not isotropic in $\mathbf{q}$: in an anti-ferromagnet the critical divergence occurs only for magnetic reciprocal lattice points.

Cooper and Nathans (1966) have reported that the critical scattering from anti-ferromagnetic KMnF_3 is approximately the *same* below and above T_C ; this would appear to be in conflict with molecular-field theory.

3.4. Below T_C ; $\chi^{xx}(\mathbf{K})$

Imagine a situation in which the spins are in the z direction and the applied field is $H(\mathbf{K})$ along x . Then

$$\left. \begin{aligned} H_{\text{eff}}^z(\mathbf{r}) &= -\frac{2}{g\mu} \sum_{\mathbf{r}'} J(\mathbf{r} - \mathbf{r}') \langle S^z(\mathbf{r}') \rangle; \\ H_{\text{eff}}^x(\mathbf{r}) &= H^x(\mathbf{r}) - \frac{2}{g\mu} \sum_{\mathbf{r}'} J(\mathbf{r} - \mathbf{r}') \langle S^x(\mathbf{r}') \rangle. \end{aligned} \right\} \quad (157)$$

If both these are small, (126) shows that the expectation values of the spins are in proportion. In fact at any temperature the expectation value of $S(\mathbf{r})$ is along the

resultant of H_{eff}^z and H_{eff}^x , so that $\langle S^x(\mathbf{r}) \rangle / \langle S^z(\mathbf{r}) \rangle = H_{\text{eff}}^x(\mathbf{r}) / H_{\text{eff}}^z(\mathbf{r})$. Therefore

$$\langle S^x(\mathbf{r}) \rangle = \langle S^z(\mathbf{r}) \rangle \frac{g\mu H^x(\mathbf{r}) - 2 \sum_{\mathbf{r}'} J(\mathbf{r} - \mathbf{r}') \langle S^x(\mathbf{r}') \rangle}{-2 \sum_{\mathbf{r}'} J(\mathbf{r} - \mathbf{r}') \langle S^z(\mathbf{r}') \rangle}. \quad (158)$$

Proceeding as before, we have that for a ferromagnet

$$\begin{aligned} \langle S^x(\mathbf{K}) \rangle &= \langle S^z \rangle \frac{g\mu H^x(\mathbf{K}) - 2J(\mathbf{K}) \langle S^x(\mathbf{K}) \rangle}{-2 \langle S^z \rangle J^{(0)}}; \\ \chi^{xx}(\mathbf{K}) &= \frac{g^2 \mu^2}{2\{J^{zz}(0) - J^{xx}(\mathbf{K})\}}, \end{aligned} \quad (159)$$

at small q equivalent to equation (119) with

$$\kappa_1^{x2} = 6 \frac{J^{zz(0)}}{J^{zz(2)}} A(0). \quad (160)$$

Thus in contrast to κ_1^z , κ_1^x behaves as illustrated in figure 3. For the ferromagnet we can again represent the ordinate as $6(J^{zz(0)}/J^{zz(2)})(\chi_C/\chi^{xx}(0))$. In the case of an antiferromagnet, $\langle S^z(\mathbf{r}) \rangle$ is replaced by $-\langle S^z(\mathbf{r}) \rangle e^{i\tau_0 \cdot \mathbf{r}}$ in (158) and it is easily shown that

$$\chi^{xx}(\mathbf{K}) = \frac{g^2 \mu^2}{2\{J_{\text{diff}}^{zz}(0) - J^{xx}(\mathbf{K})\}}; \quad (161)$$

$$\kappa_1^{x2} = 6 \frac{J_{\text{diff}}^{zz(0)}}{J_{\text{diff}}^{zz(2)}} A(\mathbf{K}). \quad (162)$$

3.5. The instantaneous critical correlation function

$\chi(\mathbf{K})$ is not very simply related to the correlation functions, except when $\hbar\omega\beta$ is small over the important range of integration in (37) and the left-hand side of that expression simplifies to

$$\chi^{\alpha\beta}(\mathbf{K}) = \frac{3\chi_0}{NS(S+1)} \langle S^\alpha(-\mathbf{K}) S^\beta(\mathbf{K}) \rangle, \quad k_B T \gg \hbar\omega_{\mathcal{S}}. \quad (163)$$

This occurs if the energy range $\hbar\omega_{\mathcal{S}}$ over which $\mathcal{S}$ is significant is materially less than $k_B T$. We shall show in §6 that the energy width of $\mathcal{S}$ narrows drastically at low K as the critical temperature is approached; and therefore, certainly at sufficiently low K , the total critical scattering gives $\chi(\mathbf{K})$. This is interesting from two points of view: experimentally, in that we can determine $\chi(\mathbf{K})$ under critical conditions for comparison with theory; and theoretically, in that we can apply (8) and (150) to derive by Fourier inversion

$$\langle S_0^\alpha S_{\mathbf{r}}^\alpha \rangle = \frac{1}{3} S(S+1) \frac{T}{T_C} \frac{1}{r_1^2} \sum_{\mathbf{K}} \frac{e^{-i\mathbf{K} \cdot \mathbf{r}}}{\kappa_1^2 + K^2}, \quad K \rightarrow 0, T \rightarrow T_C^+ \quad (164)$$

e.g., for the temperature region just above the Curie point in a ferromagnet. (In an antiferromagnet, $\mathbf{q}$ would appear in (164) instead of $\mathbf{K}$.) Replacing $\sum_{\mathbf{K}}$ by

$$\frac{V}{(2\pi)^3} \int_{-\infty}^{\infty} d\mathbf{K},$$

with V the volume per atom, we obtain an integral whose integrand is spherically

symmetrical, apart from the factor $e^{-i\mathbf{K}\cdot\mathbf{r}}$ of which the spherical average is $(\sin Kr)/Kr$. This integral can therefore be written

$$\frac{\pi}{ir} \int_{-\infty}^{\infty} \frac{(e^{iKr} - e^{-iKr}) K dK}{\kappa_1^2 + K^2}, \quad (165)$$

and a simple contour integration leads to

$$\langle S_0^\alpha S_{\mathbf{r}}^\alpha \rangle = \frac{VS(S+1)}{12\pi} \frac{T}{T_C} \frac{1}{r_1^2} \frac{e^{-\kappa_1 r}}{r}, \quad r \rightarrow \infty, T \rightarrow T_C. \quad (166)$$

Equation (166) is the famous Ornstein–Zernike asymptotic critical correlation function. It shows that upon lowering the temperature of a spin assembly towards the phase transition point the spin fluctuations increase not only in magnitude, as we have already noted, but also in range. The susceptibility increase towards the critical point and towards $K = 0$ can now be ascribed to the fact that the alignment of any one spin increasingly tends to bring with it the alignment of a large surrounding neighbourhood. At the actual transition temperature, κ_1 vanishes and the spin pair correlation goes to r^{-1} ; this is a long-range form such that the sum of the spin expectation values correlated with S_0 goes to infinity with the size of the assembly.

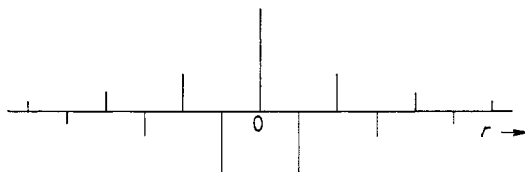


Figure 5. The spin correlation function $\langle S_0^\alpha S_{\mathbf{r}}^\alpha \rangle$ just above the Néel temperature in an antiferromagnet.

To carry through the same argument for an antiferromagnet, we must add a magnetic reciprocal lattice vector $\boldsymbol{\tau}_0$ at equation (164). A factor $e^{i\boldsymbol{\tau}_0 \cdot \mathbf{r}}$ thereupon appears in the final result, introducing a $+1$ at 'down' sites and a -1 at 'up' sites. The correlation function in this case is therefore as shown in figure 5.

Plots of the observed $\ln(r\langle S_0^\alpha S_{\mathbf{r}}^\alpha \rangle)$ for iron against r , obtained by Fourier analysis of the neutron-scattering pattern, appear in figure 6. The asymptotic Ornstein–Zernike form is seen to apply beyond about six nearest-neighbour separation distances. Spooner and Averbach (1966) have shown that between r_n and $4r_n$ in iron the critical correlation goes over to a quite accurate exponential shape $e^{-\kappa_2 r}$, which would give a susceptibility tending towards $(\kappa_2^2 + K^2)^{-2}$ in an appropriate range of K , and this has been verified by Bally *et al.* (1967).

More refined theories give asymptotes of $(\kappa_1^2 + K^2)^{\frac{1}{2}\eta-1}$, $e^{-\kappa_1 r}/r^{1+\eta}$, etc. (Fisher 1967; see also Kocinski 1968). Bally *et al.* (1968 a, b) have given an interesting demonstration that η is non-zero for both iron and cobalt.

3.6. Computation of the critical parameters

A number of attempts have been made to calculate κ_1 and r_1 by supposedly realistic methods, such as the 'cluster' method, that allow in greater or lesser degree for local order. The calculations may be summed up by saying that they are

all versions—more complicated versions—of the following argument, whereby we rederive the molecular-field theory κ_1 from an explicit examination of the magnetization at sites next to the origin.

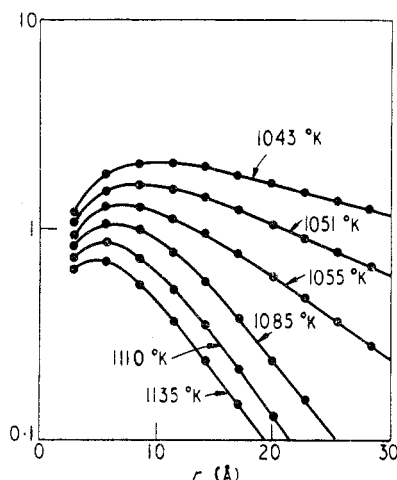


Figure 6. A logarithmic plot of $r\langle S_0^\alpha S_{\mathbf{r}}^\alpha \rangle / \frac{1}{3} S(S+1)$ obtained by Fourier analysis of the magnetic scattering from an iron crystal at and above the Curie temperature of 1043°K. An asymptotic approach to linear behaviour would conform with the Ornstein-Zernike type of correlation. (After Spooner and Averbach 1966.)

Apply the equations of classical molecular-field theory to a substance with nearest-neighbour coupling only:

$$M(\mathbf{r}) = \chi_0 H_{\text{eff}}(\mathbf{r}) = \chi_0 \{H(\mathbf{r}) + H_{\text{mol}}(\mathbf{r})\} = \chi_0 \{H(\mathbf{r}) + \lambda M_n(\mathbf{r})\}, \quad (167)$$

where $M_n(\mathbf{r})$ is the average moment

$$M_n(\mathbf{r}) = Z^{-1} \sum_{\mathbf{r}_n} M(\mathbf{r} + \mathbf{r}_n) \quad (168)$$

on the Z neighbours at distance $\mathbf{r}_n$ from $\mathbf{r}$. Then taking expectation values all through, according to molecular-field theory:

$$H_{\text{mol}}(\mathbf{r}) = \left(\frac{\chi - \chi_0}{\chi \chi_0} \right) \langle M_n(\mathbf{r}) \rangle. \quad (169)$$

Above the Curie point in the absence of an applied field, co-operative effects alone produce a magnetization distribution such that

$$\langle M(\mathbf{r}) \rangle = \chi_0 H_{\text{mol}}(\mathbf{r}) = \left(\frac{\chi - \chi_0}{\chi} \right) Z^{-1} \sum_{\mathbf{r}_n} M(\mathbf{r} + \mathbf{r}_n), \quad (170)$$

i.e. by Taylor expansion of the sum to second order,

$$(\nabla^2 - \kappa_1^2) \langle M(\mathbf{r}) \rangle = 0. \quad (171)$$

This is familiar as a wave equation whose spherically symmetrical solution is just $r^{-1} e^{-\kappa_1 r}$; and κ_1^2 is exactly given by (120).

The method outlined here was exploited at an early date by de Gennes and Herpin (1957) and by Elliott and Marshall (1958). Since then Nagai (1963) has

discussed an Ising case, while Mori and Kawasaki (1962 b), Tahir-Kheli and Callen (1964 a, b), Morita (1968), Suzuki (1968) and Oguchi and Ono (1966) have published other discussions. The last-mentioned authors give a critical comparison of the various approximations. de Gennes (1963) has suggested a numerical test which indicates that some of these calculations should be accurate to about 20%, and this is in fact roughly the extent of the absolute agreement with experiment.

3.7. Field dependence of the correlations

By means of a thermodynamic argument that we do not reproduce here, Villain (1963, 1964) has shown that (121) and (166) are still the asymptotic critical correlation functions in the presence of an applied field. The predictions are extremely

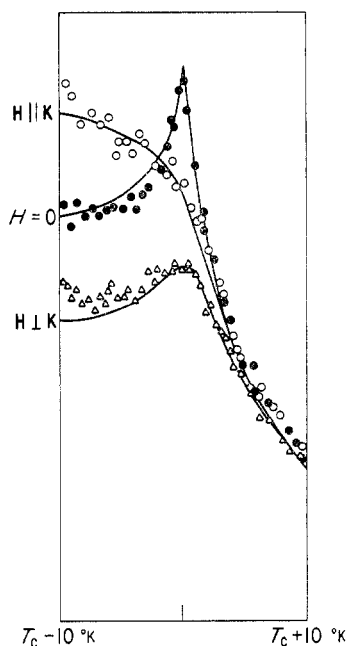


Figure 7. Experimental points and theoretical curves showing the marked influence of a small applied field of 300 Oe on the critical scattering by iron of 4.7 Å neutrons through 1.09°. (Based on Villain 1963.)

well verified for iron (figure 7). Critical scattering is exceedingly sensitive to the imposition of a field, and from the standpoint of microscopic theory this aspect has been insufficiently exploited. There is reason to believe that small terms in the Hamiltonian of an actual material, such as biquadratic exchange, might have a disproportionate effect on the field dependence which it would be interesting to study.

3.8. Other cases

We shall not occupy further space with $\chi(\mathbf{K})$. The principal omissions from this survey are the susceptibilities in the low-temperature, spin-wave region. It can be shown from spin-wave theory that χ^{xx} at low temperatures has the same

form as (160); on the other hand χ^{zz} is quite different from anything we have given, and is discussed at length by Mori and Kawasaki (1962 a) (see also Kawasaki 1967 b, Kanamori and Itoh 1968).

4. Properties of the spectral-shape function

We now go on to study the other function with a dominating influence on $R(\mathbf{K}, \omega)$:

$$F^{\alpha\beta}(\mathbf{K}, \omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega t} F^{\alpha\beta}(\mathbf{K}, t) dt, \quad (172)$$

in which

$$F^{\alpha\beta}(\mathbf{K}, t) = \frac{\{S^{\alpha}(-\mathbf{K}, 0), S^{\beta}(\mathbf{K}, t)\}}{\{S^{\alpha}(-\mathbf{K}, 0), S^{\beta}(\mathbf{K}, 0)\}} = \frac{g^2 \mu^2}{N\chi^{\alpha\beta}(\mathbf{K})} R^{\alpha\beta}(\mathbf{K}, t). \quad (173)$$

Owing to the great difficulty of deriving $R^{\alpha\beta}$ explicitly from statistical mechanics, we are reduced to discussing the properties of F in general terms.

4.1. The moments of $F(\mathbf{K}, \omega)$

Fortunately the moments of F ,

$$F^{(n)}(\mathbf{K}) = \int_{-\infty}^{\infty} F(\mathbf{K}, \omega) \omega^n d\omega, \quad (174)$$

are subject to the theorem:

$$\begin{aligned} F^{(n)}(\mathbf{K}) &= \left[\int_{-\infty}^{\infty} F(\mathbf{K}, \omega) \omega^n e^{i\omega t} d\omega \right]_{t=0} \\ &= \left[\left(\frac{1}{i} \frac{\partial}{\partial t} \right)^n \int_{-\infty}^{\infty} F(\mathbf{K}, \omega) e^{i\omega t} d\omega \right]_{t=0} \\ &= \left[\left(\frac{1}{i} \frac{\partial}{\partial t} \right)^n F(\mathbf{K}, t) \right]_{t=0}, \end{aligned} \quad (175)$$

and this leads to a number of interesting results. To begin with, (175) shows that all the moments are finite. We discuss some of them in detail below.

In the most general case, all moments of F are non-zero; but we recall from §2.10 that for substances having $S(\mathbf{K}) = S(-\mathbf{K})$ the observable quantity is $\mathcal{F}^{\alpha\beta} = \frac{1}{2}(F^{\alpha\beta} + F^{\beta\alpha})$, which is even in ω and hence has all odd moments zero. To avoid complexities, we restrict attention to such materials; and we shall further confine ourselves to substances with exchange coupling only, for which terms with $\alpha \neq \beta$ become zero so that we can deal solely with $\mathcal{F}^{\alpha\alpha} = F^{\alpha\alpha}$. It then suffices to write

$$\begin{aligned} F^{\alpha\alpha(2m+1)} &= 0, \\ F^{\alpha\alpha(2m)} &= \left[(-)^m \left(\frac{\partial}{\partial t} \right)^{2m} F^{\alpha\alpha}(t) \right]_{t=0}. \end{aligned} \quad (176)$$

In principle, a knowledge of all the moments specifies F , since by Taylor expansion of F in (172)

$$F(\mathbf{K}, \omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega t} (1 - \frac{1}{2} F^{(2)} t^2 + \frac{1}{24} F^{(4)} t^4 - \dots) dt. \quad (177)$$

However, (177) is of little use, because the whole series is required, whereas the higher moments are increasingly difficult to calculate. They are also dominated more and more by the exact shape of the remote wings of F , which are for all practical purposes unmeasurable experimentally. On account of these difficulties we shall be particularly interested only in the first three even moments:

$$\begin{aligned} F^{\alpha\alpha(0)} &= 1 \\ F^{\alpha\alpha(2)} &= -\dot{F}^{\alpha\alpha}(\mathbf{K}, t=0) \\ F^{\alpha\alpha(4)} &= \ddot{F}^{\alpha\alpha}(\mathbf{K}, t=0). \end{aligned} \quad (178)$$

4.2. The second moment

Now by (62),

$$\dot{F}^{\alpha\alpha}(\mathbf{K}, t) = \frac{g^2 \mu^2}{N \chi^{\alpha\alpha}(\mathbf{K})} \frac{i}{\hbar} \langle [S^\alpha(-\mathbf{K}, 0), S^\alpha(\mathbf{K}, t)] \rangle \quad (179)$$

$$= \frac{g^2 \mu^2}{N \chi^{\alpha\alpha}(\mathbf{K})} \frac{i}{\hbar} \sum_{\mathbf{m}, \mathbf{n}} e^{i\mathbf{K} \cdot (\mathbf{n} - \mathbf{m})} \langle [S_{\mathbf{m}}^\alpha(0), S_{\mathbf{n}}^\alpha(t)] \rangle \quad (180)$$

$$\dot{F}^{\alpha\alpha}(\mathbf{K}, t) = \frac{g^2 \mu^2}{N \chi^{\alpha\alpha}(\mathbf{K})} \frac{i}{\hbar} \sum_{\mathbf{m}, \mathbf{n}} e^{i\mathbf{K} \cdot (\mathbf{n} - \mathbf{m})} \langle [S_{\mathbf{m}}^\alpha(0), \dot{S}_{\mathbf{n}}^\alpha(t)] \rangle. \quad (181)$$

We know that

$$\dot{\mathbf{S}}_{\mathbf{r}} = \frac{1}{i\hbar} \left[\mathbf{S}_{\mathbf{r}}, \sum_{\mathbf{r}', \mathbf{r}''} -J(\mathbf{r}' - \mathbf{r}'') \mathbf{S}_{\mathbf{r}'} \cdot \mathbf{S}_{\mathbf{r}''} \right] \quad (182)$$

$$= \frac{2}{\hbar} \sum J(\mathbf{l}) \mathbf{S}_{\mathbf{r}} \times \mathbf{S}_{\mathbf{r}+\mathbf{l}}; \quad (183)$$

$$\dot{S}_{\mathbf{n}}^\alpha = \frac{2}{\hbar} \sum_{\mathbf{l}} J(\mathbf{l}) (S_{\mathbf{n}}^\beta S_{\mathbf{n}+\mathbf{l}}^\gamma - S_{\mathbf{n}}^\gamma S_{\mathbf{n}+\mathbf{l}}^\beta). \quad (184)$$

$$\begin{aligned} F^{\alpha\alpha(2)}(\mathbf{K}) &= -\dot{F}^{\alpha\alpha}(\mathbf{K}, 0) \\ &= -\frac{g^2 \mu^2}{N \chi^{\alpha\alpha}(\mathbf{K})} \frac{2i}{\hbar^2} \sum_{\mathbf{l}, \mathbf{m}, \mathbf{n}} J(\mathbf{l}) e^{i\mathbf{K} \cdot (\mathbf{n} - \mathbf{m})} \langle [S_{\mathbf{m}}^\alpha, S_{\mathbf{n}}^\beta S_{\mathbf{n}+\mathbf{l}}^\gamma - S_{\mathbf{n}}^\gamma S_{\mathbf{n}+\mathbf{l}}^\beta] \rangle; \end{aligned} \quad (185)$$

$$F^{\alpha\alpha(2)}(\mathbf{K}) = \frac{2g^2 \mu^2}{N \hbar^2 \chi^{\alpha\alpha}(\mathbf{K})} \sum_{\mathbf{l}, \mathbf{n}} J(\mathbf{l}) (1 - e^{-i\mathbf{K} \cdot \mathbf{l}}) \langle S_{\mathbf{n}}^\beta S_{\mathbf{n}+\mathbf{l}}^\beta + S_{\mathbf{n}}^\gamma S_{\mathbf{n}+\mathbf{l}}^\gamma \rangle. \quad (186)$$

Let us next discuss this formula in the low, critical and high temperature regions.

4.2.1. $F^{xx(2)}$; *low temperatures*. At the very lowest temperatures, the quantity of interest for inelastic scattering is $F^{xx(2)}$. For a ferromagnet, the expectation value in (186) tends to S^2 , and the whole moment accordingly to

$$\frac{2g^2\mu^2 S^2}{\hbar^2} \chi^{xx}(\mathbf{K}) \{J(0) - J(-\mathbf{K})\}. \quad (187)$$

We know also that in this temperature range $\chi^{xx}(\mathbf{K})$ for a ferromagnet is given by (159); whence putting $J(-\mathbf{K}) = J(\mathbf{K})$,

$$F^{xx(2)} = \left[\frac{2S}{\hbar} \{J(0) - J(\mathbf{K})\} \right]^2. \quad (188)$$

Now $F^{xx}(\mathbf{K}, \omega)$ is actually a spectrum consisting of two spin-wave lines, at $\pm E_{\mathbf{q}}$ respectively:

$$F^{xx}(\mathbf{K}, \omega) = \frac{1}{2} \delta(\omega + \omega_{\mathbf{q}}) + \frac{1}{2} \delta(\omega - \omega_{\mathbf{q}}), \quad (189)$$

so that from this point of view

$$F^{xx(2)} = \omega_{\mathbf{q}}^2. \quad (190)$$

Remembering that $J(\mathbf{K}) = J(\mathbf{q})$ in a ferromagnet, we correctly recover from (188)–(190) the ferromagnetic spin-wave energy

$$\hbar\omega_{\mathbf{q}} = 2S\{J(0) - J(\mathbf{q})\}. \quad (191)$$

The formalism ‘knows’ about spin waves in so far as we have injected the Heisenberg Hamiltonian at equation (182), and also in principle at equations (187)–(188) through $\chi(\mathbf{K})$.

For an antiferromagnet the expectation value in (186) tends to $S^2 e^{i\tau_0 \cdot 1}$, so that

$$F^{xx(2)}(\mathbf{K}) = \frac{2g^2\mu^2 S^2}{\hbar^2} \chi^{xx}(\mathbf{K}) \{J_{\text{diff}}(0) - J_{\text{diff}}(-\mathbf{K})\}. \quad (192)$$

Using (161), we obtain

$$F^{xx(2)}(\mathbf{K}) = \left(\frac{2S}{\hbar} \right)^2 \{J_{\text{diff}}(0) - J_{\text{diff}}(\mathbf{K})\} \{J_{\text{diff}}(0) - J(\mathbf{K})\}; \quad (193)$$

$$\hbar\omega_{\mathbf{q}} = 2S[\{J_{\text{diff}}(0) - J_{\text{diff}}(\mathbf{K})\} \{J_{\text{diff}}(0) - J(\mathbf{K})\}]^{1/2}, \quad (194)$$

again the correct formula for spin-wave dispersion.

4.2.2. *Spin waves*. The spin-wave dispersion law is given by the poles of F^{xx} (or R^{xx} , or g^{xx}) at low temperatures. We shall not be taking up spin-wave theory in this article, except to remark upon a recent Green function analysis of the Heisenberg ferromagnet. This has suggested that over the normal temperature range of validity of spin-wave theory the neutron-scattering cross section for spin waves should diminish with increasing temperature faster, by a factor $\langle S^z \rangle / S$, than is ordinarily predicted by taking into account their dynamical interactions. The extra factor invites a simple physical verbalization: that as the spins become increasingly disoriented the neutron ‘sees’ them as having in common only their average value. The corresponding phenomenon in metallic iron and nickel has

been observed by Stringfellow (1968). He finds, however, that in iron the diminution of effective spin with temperature is more like

$$\langle S^z \rangle \{1 - (0.30 \pm 0.08) T/T_C\}$$

(figure 8). The topic has been amplified in short reviews by the present authors (Lowde 1965, Marshall and Murray 1968).

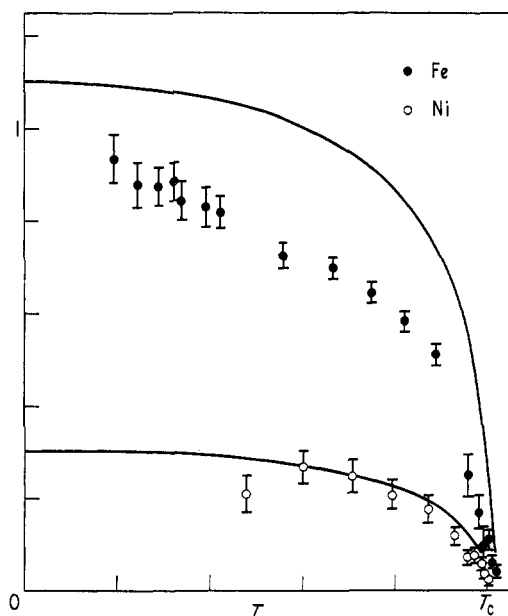


Figure 8. The effective spin component amplitude S_{eff} taking part in a spin wave in iron and nickel, measured by means of the neutron spin-wave scattering cross section, to which it is proportional. Full curves give the mean spin $\langle S^z \rangle$ for iron and nickel under the circumstances of the experiment, for which applied fields of approximately 3000 and 2000 Oe were present. (After Stringfellow 1968.)

4.2.3. $F^{\alpha\alpha(2)}$; general temperatures. The mean energy of a pair of atoms separated by $\mathbf{l}$ can be written as $-2J(\mathbf{l})\langle \mathbf{S}_{\mathbf{n}} \cdot \mathbf{S}_{\mathbf{n}+\mathbf{l}} \rangle$. Per atom, the magnetic coupling energy is half this, say

$$\mathcal{E}(\mathbf{l}) = -J(\mathbf{l})\langle \mathbf{S}_{\mathbf{n}} \cdot \mathbf{S}_{\mathbf{n}+\mathbf{l}} \rangle. \quad (195)$$

Now above T_C , (186) gives

$$F^{\alpha\alpha(2)}(\mathbf{K}) = \frac{2g^2\mu^2}{N\hbar^2\chi^{\alpha\alpha}(\mathbf{K})} \sum_{\mathbf{l}, \mathbf{n}} J(\mathbf{l}) (1 - e^{-i\mathbf{K} \cdot \mathbf{l}}) \frac{2}{3} \langle \mathbf{S}_{\mathbf{n}} \cdot \mathbf{S}_{\mathbf{n}+\mathbf{l}} \rangle, \quad (196)$$

$$= -\frac{2g^2\mu^2}{\hbar^2\chi^{\alpha\alpha}(\mathbf{K})} \frac{2}{3} \sum_{\mathbf{l}} (1 - e^{-i\mathbf{K} \cdot \mathbf{l}}) \mathcal{E}(\mathbf{l}), \quad (197)$$

$$= -\frac{4g^2\mu^2}{3\hbar^2\chi^{\alpha\alpha}(\mathbf{K})} \{\mathcal{E}(0) - \mathcal{E}(-\mathbf{K})\} \quad (198)$$

using Fourier transforms defined as in equation (14).

To follow the xx moment with lowering of the temperature through T_C , the simplification $\langle S_n^y S_{n+1}^y + S_n^z S_{n+1}^z \rangle = \frac{2}{3} \langle \mathbf{S}_n \cdot \mathbf{S}_{n+1} \rangle$ becomes invalid. For a small temperature range, however, it remains highly approximate, and with further reduction of T all the way to zero it alters only in that the factor we have written $\frac{2}{3}$ increases gradually to unity. Thus the factor $-\{\mathcal{E}(0) - \mathcal{E}(-\mathbf{K})\}$ gives a rough account of the xx second moment through the transition temperature, and a semi-quantitative indication of it right down to 0°K .

An idea of the behaviour of this factor in typical circumstances is gained from the case of nearest-neighbour coupling only, when—using Oguchi's

$$\gamma_K = \sum_{n,n'} e^{i\mathbf{K} \cdot \mathbf{r}_n} \quad (199)$$

—it becomes $-(\gamma_0 - \gamma_K)\mathcal{E}$, $\mathcal{E}$ being the magnetic energy per atom in that example. The temperature dependence of $\mathcal{E}$ gives rise to a curve having the general shape drawn in figure 9, showing no singular behaviour at T_C and falling off at high temperatures as T^{-1} .

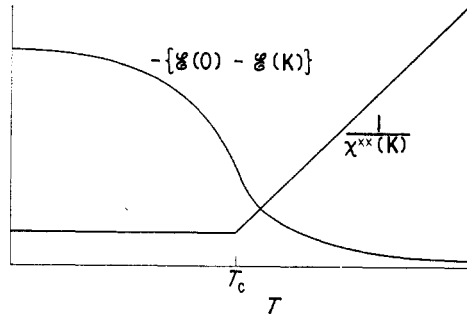


Figure 9. Temperature dependence of the two principal factors of $F^{xx(2)}$.

The inverse of $\chi^{xx}(\mathbf{K})$ is, of course, $\chi_C^{-1} r_1^2 (\kappa_1^2 + q^2 - \dots)$ and at a constant q gives the much more angular curve illustrated. $F^{xx(2)}$ itself varies as the product of the two functions in figure 9, and at sufficient remove from the critical temperature on either side it tends to be little dependent on T , becoming asymptotically constant at both temperature extremes. On the other hand, $F^{zz(2)}$ rises more sharply to the left below T_C and falls to zero at $T = 0$.

We return to this topic in §4.6.

4.3. High temperatures

4.3.1. $F^{\alpha\alpha(2)}$; high temperatures. For the remainder of the temperature range above some small multiple of T_C , the properties of $\mathcal{S}^{\alpha\alpha}(\mathbf{K}, \omega)$ are virtually constant. In (197), we know from (154) that

$$\chi^{\alpha\alpha}(\mathbf{K}) = \frac{g^2 \mu^2 S(S+1)}{3k_B T} \{1 + \beta \frac{2}{3} S(S+1) J^{\alpha\alpha}(\mathbf{K}) - \dots\}; \quad (200)$$

so that to order β only the first factor need be taken, regardless of anisotropy or

similar complications. The temperature dependence of $\mathcal{E}(\mathbf{l})$ is given by

$$\begin{aligned}\mathcal{E}(\mathbf{l}) &= -J(\mathbf{l}) \langle \mathbf{S}_{\mathbf{r}} \cdot \mathbf{S}_{\mathbf{r}+\mathbf{l}} \rangle = -J(\mathbf{l}) \frac{\text{Tr} \{ \mathbf{S}_{\mathbf{r}} \cdot \mathbf{S}_{\mathbf{r}+\mathbf{l}} e^{-\beta \mathcal{H}} \}}{\text{Tr} e^{-\beta \mathcal{H}}} \\ &= -J(\mathbf{l}) \frac{\text{Tr} \{ (\mathbf{S}_{\mathbf{r}} \cdot \mathbf{S}_{\mathbf{r}+\mathbf{l}}) - (\mathbf{S}_{\mathbf{r}} \cdot \mathbf{S}_{\mathbf{r}+\mathbf{l}}) \beta (\sum_{\mathbf{m}, \mathbf{n}} J(\mathbf{n}) \mathbf{S}_{\mathbf{m}} \cdot \mathbf{S}_{\mathbf{m}+\mathbf{n}}) + \dots \}}{\text{Tr} \{ 1 - \beta \mathcal{H} + \dots \}}\end{aligned}\quad (201)$$

which to order β is

$$-2\beta \{J(\mathbf{l})\}^2 \text{Tr} (\mathbf{S}_{\mathbf{r}} \cdot \mathbf{S}_{\mathbf{r}+\mathbf{l}})^2 \quad (202)$$

$$= -\frac{2}{3} \beta \{J(\mathbf{l})\}^2 S^2(S+1)^2. \quad (203)$$

Thus

$$F^{\alpha\alpha(2)}(\mathbf{K}) = \frac{8}{3} \frac{S(S+1)}{\hbar^2} \sum_{\mathbf{r}} (1 - e^{-i\mathbf{K} \cdot \mathbf{r}}) \{J(\mathbf{r})\}^2. \quad (204)$$

de Gennes, who derived this formula for $\langle \omega^2 \rangle$ in 1957, also observed that since the spherical average of $e^{-i\mathbf{K} \cdot \mathbf{r}}$ is $(\sin Kr)/Kr$, then for a polycrystal

$$F_{\text{poly}}^{\alpha\alpha(2)}(\mathbf{K}) = \frac{8}{3} \frac{S(S+1)}{\hbar^2} \sum_{\mathbf{r}} \left\{ 1 - \frac{\sin Kr}{Kr} \right\} \{J(\mathbf{r})\}^2, \quad (205)$$

which approaches an asymptotic value of

$$\frac{8}{3} \frac{S(S+1)}{\hbar^2} \sum_{\mathbf{r}} \{J(\mathbf{r})\}^2 \quad Kr_n > \pi \quad (206)$$

with increase of K beyond about π/r_n . Thus the periodicity of (199) in $\mathbf{K}$ space is lost for a polycrystal, the angular averages introducing an effect, outside the region of small K , completely equivalent to incoherence.

As a matter of curiosity, if we observe from (184) that at high temperatures

$$\langle \dot{\mathbf{S}}_0^2 \rangle_{T=\infty} = \frac{8}{3} \frac{S^2(S+1)^2}{\hbar^2} \sum_{\mathbf{r}} \{J(\mathbf{r})\}^2, \quad (207)$$

we see that the second moment at large K in the polycrystal takes this simple form:

$$F_{\text{poly}}^{\alpha\alpha(2)}(\infty) = \frac{\langle \dot{\mathbf{S}}_0^2 \rangle}{S(S+1)}. \quad (208)$$

Regarding the polycrystal as a sample in which the effects of coherence can be averaged out in the sense of equation (206), we find that in the high- T , large- K limit the mean square frequency $\langle \omega^2 \rangle$ transferred to the neutron is equal to the mean square of

$$\frac{\dot{\mathbf{S}}_0}{\{S(S+1)\}^{1/2}} = \omega_0 \quad (209)$$

(say), i.e. is identical with the mean square angular velocity $\langle \omega_0^2 \rangle$ in the spin system.

4.3.2. The fourth moment. By going on to calculate $\ddot{F}^{\alpha\alpha}(\mathbf{K}, 0)$, $F^{\alpha\alpha(4)}$ may be determined. Unfortunately, as equation (183) shows, each application of $\partial/\partial t$ introduces another operator $\sum_{\mathbf{i}} \mathbf{S}_{\mathbf{i}}(t) \times$, thereby doubling the number of terms, increasing the order of each in S , and raising the order of summation. As a result the calculation rapidly gets out of hand, and indeed the sixth moment has never been published.

Various authors have determined the fourth, the most recent discussion being that of Collins and Marshall (1967); we quote their result only as simplified for the case of solely nearest-neighbour coupling:

$$F^{\alpha\alpha(4)}(\mathbf{K}) = \frac{32}{9} \frac{S^2(S+1)^2}{\hbar^4} J^4(\gamma_0 - \gamma_K) \left(7\gamma_0 - 3\gamma_K - 4 - \frac{3}{2S(S+1)} \right); \quad (210)$$

at small K

$$F^{\alpha\alpha(4)}(\mathbf{K}) \simeq \frac{64}{27} \frac{S^2(S+1)^2}{\hbar^4} J^4 Z \left(Z - 1 - \frac{3}{8S(S+1)} \right) K^2 r_n^2. \quad (211)$$

(r_n is the nearest-neighbour separation.)

4.3.3. $F^{\alpha\alpha}(\mathbf{K}, \omega)$ at high temperatures. At the present state of knowledge we have to acquire some feeling for $F(\mathbf{K}, \omega)$ from a few properties of the even moments. From the definition at (39),

$$\begin{aligned} F^{\alpha\beta}(\mathbf{K}, t) &= \frac{\sum_{\mathbf{m}, \mathbf{n}} e^{i\mathbf{K} \cdot (\mathbf{n} - \mathbf{m})} \{S_{\mathbf{m}}^{\alpha}(0), S_{\mathbf{n}}^{\beta}(t)\}}{(N/g^2 \mu^2) \chi^{\alpha\beta}(\mathbf{K})} \\ &= \frac{\sum_{\mathbf{r}} e^{i\mathbf{K} \cdot \mathbf{r}} \{S_0^{\alpha}(0), S_{\mathbf{r}}^{\beta}(t)\}}{(1/g^2 \mu^2) \chi^{\alpha\beta}(\mathbf{K})}. \end{aligned} \quad (212)$$

Therefore

$$F^{\alpha\beta(n)}(\mathbf{K}) = \frac{\sum_{\mathbf{r}} e^{i\mathbf{K} \cdot \mathbf{r}} [\{(1/i)(\partial/\partial t)\}^n \{S_0^{\alpha}(0), S_{\mathbf{r}}^{\beta}(t)\}]_{t=0}}{(1/g^2 \mu^2) \chi^{\alpha\beta}(\mathbf{K})}. \quad (213)$$

In the region of high temperatures it follows from the definition (18) that

$$\{S_0^{\alpha}(0), S_{\mathbf{r}}^{\beta}(t)\} = \beta \langle S_0^{\alpha}(0) S_{\mathbf{r}}^{\beta}(t) \rangle - \beta \langle S_0^{\alpha}(0) \rangle \langle S_{\mathbf{r}}^{\beta}(0) \rangle. \quad (214)$$

Using (200) also, we find

$$F^{\alpha\beta(n \neq 0)}(\mathbf{K}) = \frac{3}{S(S+1)} \sum_{\mathbf{r}} e^{i\mathbf{K} \cdot \mathbf{r}} \left[\left(\frac{1}{i} \frac{\partial}{\partial t} \right)^n \langle S_0^{\alpha}(0) S_{\mathbf{r}}^{\beta}(t) \rangle \right]_{t=0}; \quad (215)$$

all moments become asymptotically independent of temperature.

There are two interesting limiting cases. At large K , the spherical average of $e^{i\mathbf{K} \cdot \mathbf{r}}$ tends towards independence of K , and therefore all moments for a polycrystal become independent of both K and T :

$$F_{\text{poly}}^{\alpha\beta(n)}(\mathbf{K}) \xrightarrow[T \rightarrow \infty]{K \rightarrow \infty} F_{\text{poly}}^{\alpha\beta(n)}(\infty). \quad (216)$$

Thus $F_{\text{poly}}(\mathbf{K}, \omega)$ itself tends, in the high-temperature limit, towards some definite function $F_{\text{poly}}(\infty, \omega)$. Now at high temperatures, (44) and (4) show that

$$\frac{1}{3} S(S+1) F^{\alpha\beta}(\mathbf{K}, \omega) = \frac{1}{2\pi} \sum_{\mathbf{r}} \int_{-\infty}^{\infty} e^{i(\mathbf{K} \cdot \mathbf{r} - \omega t)} \langle S_0^{\alpha}(0) S_{\mathbf{r}}^{\beta}(t) \rangle dt. \quad (217)$$

With increase of K , the spherical average $(\sin Kr)/Kr$ of $e^{i\mathbf{K} \cdot \mathbf{r}}$ tends to δ_{r0} ; thus

$$F_{\text{poly}}^{\alpha\alpha}(\infty, \omega) = \frac{1}{\frac{1}{3} S(S+1)} \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega t} \langle S_0^{\alpha}(0) S_0^{\alpha}(t) \rangle dt. \quad (218)$$

At small K the first term in the expansion of $e^{i\mathbf{K}\cdot\mathbf{r}}$ vanishes, since

$$\sum_{\mathbf{r}} \{S_0^\alpha(0), S_{\mathbf{r}}^\beta(t)\} = \int_0^\beta \langle e^{\lambda \mathcal{H}} S_0^\alpha(0) e^{-\lambda \mathcal{H}} \sum_{\mathbf{r}} S_{\mathbf{r}}^\beta(t) \rangle d\lambda - \beta \langle S_0^\alpha \rangle \sum_{\mathbf{r}} \langle S_{\mathbf{r}}^\beta \rangle, \quad (219)$$

and because

$$\sum_{\mathbf{r}} S_{\mathbf{r}}^\beta(t) = \sum_{\mathbf{r}} S_{\mathbf{r}}^\beta(0), \quad (220)$$

we see that

$$\sum_{\mathbf{r}} \{S_0^\alpha(0), S_{\mathbf{r}}^\beta(t)\} = \sum_{\mathbf{r}} \{S_0^\alpha(0), S_{\mathbf{r}}^\beta(0)\}. \quad (221)$$

Consequently

$$F^{\alpha\beta}(0, t) = F^{\alpha\beta}(0, 0); \quad (222)$$

$$F^{\alpha\beta(n \neq 0)}(0) = 0; \quad (223)$$

$$F^{\alpha\beta(n \neq 0)}(\mathbf{K}) \xrightarrow[\substack{K \rightarrow 0 \\ T \rightarrow \infty}]{-3} \frac{2S(S+1)}{\sum_{\mathbf{r}} (\mathbf{K} \cdot \mathbf{r})^2} \left[\left(\frac{1}{i} \frac{\partial}{\partial t} \right)^n \langle S_0^\alpha(0) S_{\mathbf{r}}^\beta(t) \rangle \right]_{t=0}. \quad (224)$$

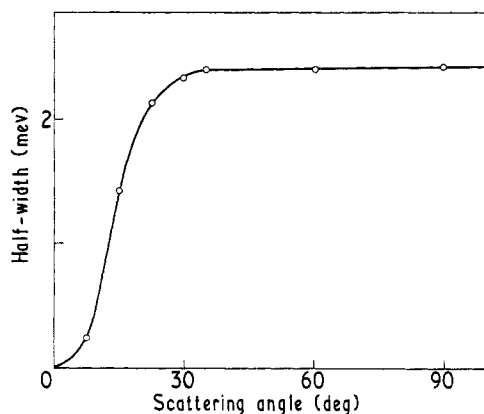


Figure 10. Half-width of the energy distribution of scattered neutrons from polycrystalline paramagnetic MnF_2 at $T = 4T_N$ as a function of scattering angle, showing the vanishing of the width at $K = 0$ and the asymptotic approach to a constant value in the limit of large momentum transfer. (Based on Cribier and Jacrot 1963.)

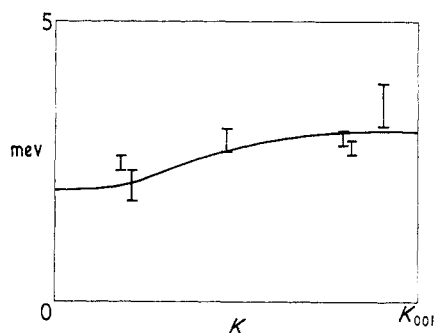


Figure 11. Experimental and theoretical values for the root-mean-square energy transfer to neutrons scattered with momentum transfer along the [001] axis in paramagnetic crystalline terbium at $T = 3T_C$. A contribution from single-ion anisotropy prevents the root-mean-square energy transfer from going to zero at $K = 0$. Theoretical effects due to the temperature being finite have been taken into account, following a calculation by Sears. (After Collins 1968.)

At high temperatures, all non-zero moments are independent of temperature and go to zero as K^2 .

These phenomena are brought out by Cribier and Jacrot's (1963) data showing the width of the scattered neutron wavelength distribution from polycrystalline paramagnetic MnF_2 as a function of K (figure 10).

4.3.4. *The effect of anisotropy.* In a brief review, Collins (1968) discusses the high temperature paramagnetic scattering of neutrons from the point of view adopted here. He has shown that the inclusion of a single-ion anisotropy term $A\sum_i S_i^z S_i^z$ in the Heisenberg Hamiltonian affects $F^{\alpha\alpha(2)}$ solely by adding

$$\frac{1}{10}\delta_{\alpha z} A^2(1 + \hat{K}_z^2)\{4S(S+1) - 3\} \quad (225)$$

to (204). (*Note added in proof:* see also Lindgard 1968.) Its effect on what is calculated and observed for terbium is shown in figure 11.

4.4. Experimentally determined fluctuation spectra

4.4.1. *Neutron-scattering experiments.* In the rather obscure situation that has been described, an ounce of observation is manifestly worth a pound of conjecture, and experiments of various kinds have helped a great deal in sorting out exactly what does occur. By way of example, we illustrate at figure 12 for the high temperature case the general trends in F that have been brought out above. The measurements there shown were made on RbMnF_3 , whose privileged place in magnetism is due to its being a simple-cubic antiferromagnet of spin $\frac{5}{2}$ having for most practical purposes nearest-neighbour coupling only and no anisotropy. Figure 12 is a selection from the data of Windsor *et al.* (1968), who have surveyed $R(\mathbf{K}, \omega)$ for this crystal at $3.5T_N$ over the whole of the $\langle 110 \rangle$ section of the Brillouin zone. The ω spectra are shown for a succession of $\mathbf{K}$'s from the origin to $(\frac{1}{2}, \frac{1}{2}, 0)$. As expected, a broad energy spectrum of fluctuations is seen at large K that sharpens up considerably as K is reduced.

We shall come back to this figure a number of times to examine detailed points.

4.4.2. *Computer-simulation experiments.* An entirely different kind of experiment, but one of no less importance, is that in which the motion of the spin system is simulated on a computer. Spins in an array of as large a size as appears to justify the expense are set up with their initial directions appropriately selected; for an experiment at $T = \infty$ a table of random numbers would be employed to fix directions, while for any finite temperature some special procedure must be used for obtaining the desired thermodynamic situation. The motion is then followed by calculating all the differentials $d\mathbf{S}$ corresponding to the motion in each of a succession of small intervals of time, and the required statistical averages are determined. This method is applicable to both classical and quantum systems, but the taking of expectation values is much simpler in the former case, and so classical spins have normally been preferred for large ensembles. Classical simulation 'works', in this field, essentially because the equation of motion of a classical magnetic moment $\mathbf{M}$ in an effective field $\mathbf{H}$ is isomorphous with the exact equation for a Heisenberg spin. The former equation is

$$\dot{\mathbf{M}} = \gamma \mathbf{M} \times \mathbf{H}, \quad (226)$$

with $\gamma = -g|e|/mc$. This is to be compared with

$$\dot{\mathbf{S}} = i\hbar^{-1} [\mathcal{H}, \mathbf{S}]. \quad (227)$$

But if we take a simple Hamiltonian $\mathcal{H} = g\mu\mathbf{S}\cdot\mathbf{H}$ with $\mathbf{H}$ some effective field—and this includes the Heisenberg Hamiltonian—(227) becomes

$$i\hbar\dot{\mathbf{S}} = g\mu[\mathbf{S}, \mathbf{S}\cdot\mathbf{H}] = ig\mu\mathbf{H}\times\mathbf{S} = i\gamma\hbar\mathbf{S}\times\mathbf{H}; \quad (228)$$

i.e.

$$\dot{\mathbf{S}} = \gamma\mathbf{S}\times\mathbf{H}. \quad (229)$$

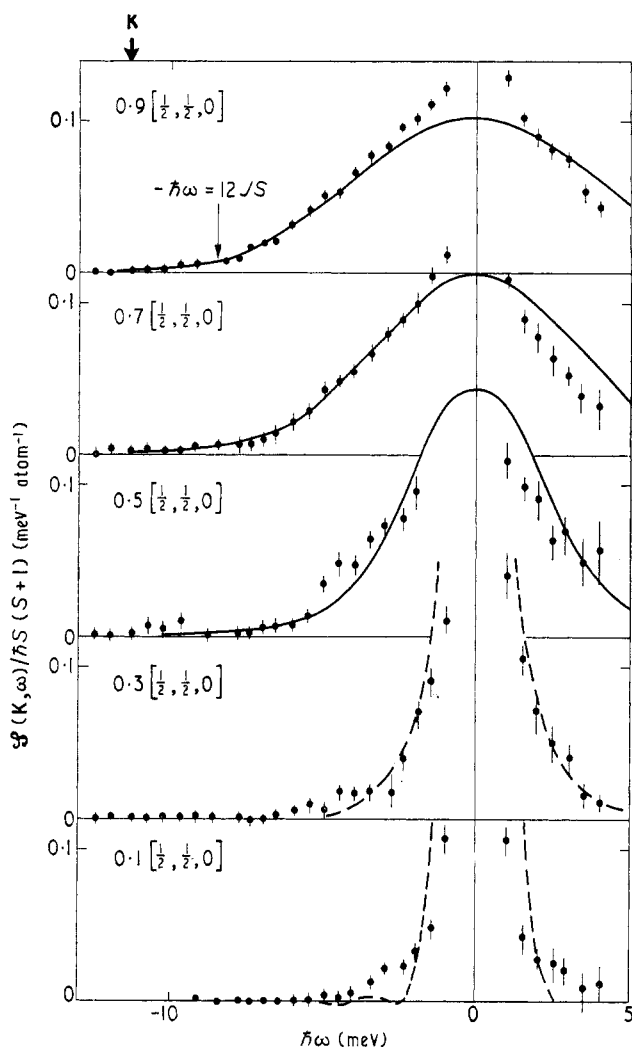


Figure 12. A neutron-scattering determination of $\mathcal{S}(\mathbf{K}, \omega)$ for RbMnF_3 at $3.5T_N$, compared absolutely with calculated distributions for $T = \infty$. Energy spectra are shown for a succession of $\mathbf{K}$'s between the origin and the point $(\frac{1}{2}, \frac{1}{2}, 0)$ on the boundary of the Brillouin zone; the widths are seen to increase with K up to roughly a limit corresponding to the maximum molecular-field-theory energy $2ZJS$. Full curves give the predictions of formula (234), and broken curves show the function determined for a computer-simulated classical spin array. (After Windsor *et al.* 1968.)

Accordingly, in the classically equivalent spin system the method of taking expectation values $\langle |S(t)| \rangle$ might be incorrect, but at least the motion of the operator is satisfactory.

Various detailed considerations relating to the computer technique are given by Blume *et al.* (1968) (for a general reference see Beeler 1966).

An interesting infinite-temperature computation is that of Windsor (1967, 1968), who performed twenty-three runs with a simple-cubic crystal of classical spins to obtain the results of figure 13. The crystal had a cubic habit, with never

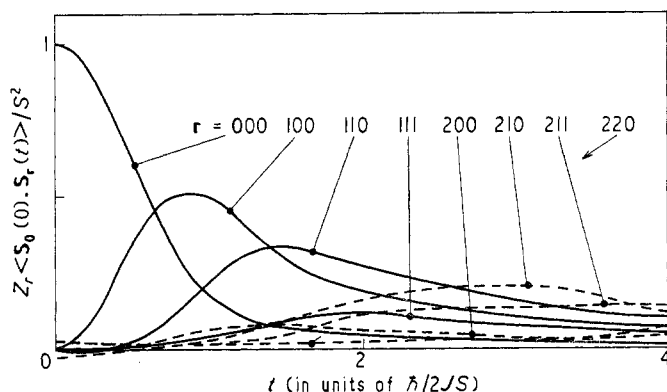


Figure 13. Time dependence of the spin-correlation function $\langle S_0(0) S_r(t) \rangle$ at $T = \infty$ for a simple-cubic paramagnet with classical spins and an energy given by the Heisenberg Hamiltonian with coupling to nearest neighbours only. The functions shown, for different interatomic vectors $\mathbf{r}$, were determined empirically from the computer-simulated motion of various arrays of a few thousand spins having initially random orientations. The statistical uncertainty is estimated to be between 1 and 3%, increasing with the multiplicity of the site type. The probable location of a unit of spin, known to be at the origin at time zero, can be seen to fall away at the origin in a time roughly given by the precession period in the root-mean-square molecular field and to rise to a maximum on the different neighbours in succession. (After Windsor 1968.)

less than ten atoms on a side; periodic boundary conditions were used to obviate edge effects. The expected spin value initially known to be S at $r = 0$ and $t = 0$ can be seen falling away in a time roughly given by

$$t_0 = \frac{\hbar}{2SJ}, \quad (230)$$

which is the oscillation period in the molecular field produced by a single parallel neighbour. (Apart from a factor $\frac{1}{3}S(S+1)/S^2$, this function is the self-correlation appearing in equation (218).) The probabilities for the spin to reappear on the different nearby sites rise from zero with the passage of time, and eventually all fall as the spin escapes to a great distance.

Transforms of the functions of figure 13 will be considered in the later sections.

4.5. $F(\mathbf{K}, \omega)$ at high T and large K

4.5.1. *The Gaussian approximation.* Turning to the case of large K , the following argument leads us to suspect that the fluctuation spectrum has something rather

like a *Gaussian* distribution. Consider a crystal of N spins $\frac{1}{2}$ at an effectively infinite temperature, so that at any moment $\frac{1}{2}N$ spins are pointing upward and $\frac{1}{2}N$ downward, and imagine that the magnetic coupling is described by a very long-range rectangular function such that each spin is coupled equally to a large number $2n$ of neighbours within a certain sphere of interaction. When a neutron scatters inelastically from any given atom at $\mathbf{r}$, the energy transfer is equal to the energy cost of turning the spin over in the molecular field $H_{\text{eff}}(\mathbf{r})$. If within the sphere of interaction there happen to be exactly n spins pointing each way, $H_{\text{eff}}(\mathbf{r}) = 0$; if there are $n+x$ spins up and $n-x$ spins down it is proportional to x . Given incoherence (which we are not), all these inelastic events would send intensity into any direction equally; the distribution of energy transfers would then be described in appropriate units by the Bernoulli probability distribution of x :

$$B(2n, n+x) = {}^{2n}C_{n+x} \left(\frac{1}{2}\right)^{n+x} \left(1 - \frac{1}{2}\right)^{n-x} \\ \xrightarrow{n \rightarrow \infty} (\pi n)^{-1/2} e^{-x^2/n}.$$

Is the Gaussian distribution appropriate in the more realistic case where there is coherent reconstruction of wave fronts and a limited range of forces? On the latter count it is clearly a worth-while first approximation, as the minimum number of neighbours ($2n = 6$) is still not small enough to destroy the conception entirely. On the other hand the picture presented is not correct, as with $2n = 6$ it would lead to three well-defined lines of energy loss or gain, whereas a continuum is observed; the point here is that the scattering from any given microcrystal is *not* incoherent, and even the polycrystal F does not really result from so simple a mechanism.

de Gennes's original test of the Gaussian approximation was to compute

$$\frac{3F_{\text{poly}}^{\alpha\alpha(2)2}}{F_{\text{poly}}^{\alpha\alpha(4)}}, \quad (231)$$

which is unity for a Gaussian distribution. In fact for a polycrystal it rises from zero as K^2 and in an oscillatory manner approaches 0.89 with increase of K beyond π/r_n (for a simple-cubic structure with $S = 1$ and nearest-neighbour coupling). If we set particular store by $F_{\text{poly}}^{\alpha\alpha(2)}$, we can express the desired approximation in terms of it as

$$F_{\text{poly}}^{\alpha\alpha}(\infty, \omega) = (2\pi F_{\text{poly}}^{\alpha\alpha(2)})^{-1/2} \exp\left(-\frac{\omega^2}{2F_{\text{poly}}^{\alpha\alpha(2)}}\right), \quad (232)$$

and this equation can be used to deduce $F^{\alpha\alpha(2)}$ from experimental data.

4.5.2. Observations on polycrystals. A number of experiments on polycrystals have shown (232) to be a good first approximation (Cribier and Jacrot 1963, Collins and Nathans 1965, Satya Murthy *et al.* 1965, 1968). From our point of view here, the most interesting study is again Windsor's (1966 a, b, 1967) on RbMnF_3 . His results for the polycrystal, as before at $3.5T_N$, are shown in figure 14. K here is chosen not to be very large (because of the fall-off of the scattering form factor), but to be sufficiently near the first zero of $\sin(Kr_n)$ to produce a frequency distribution that

should be virtually identical with the limiting case (cf. equations (217)–(218)). It is clear that the observed spectrum is non-Gaussian, and indeed close examination shows that the best-fitting Gaussian falls substantially below some of the higher points.

In the figure, comparison is made with (218) as given by the computer-simulated classical array of figure 13, and it is extremely interesting to see the close correspondence between the two sets of results. In fact, the set of measured neutron points must be distorted to some degree because the temperature was only $3.5T_N$ in the actual RbMnF_3 specimen. The exact nature of this distortion is not known, but there is reason to believe (§4.6 below) that it is not enough to diminish the significance of the good match obtained in figure 14.

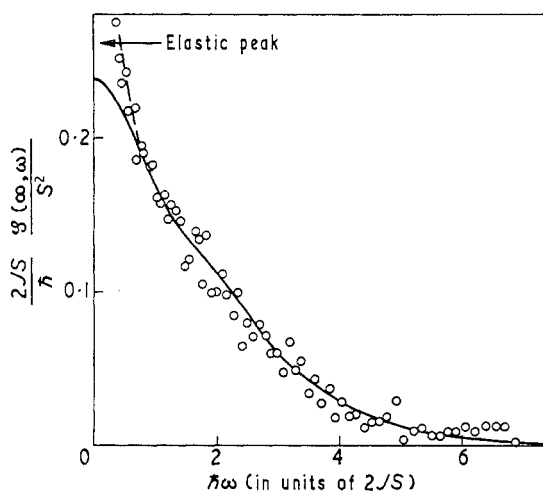


Figure 14. The asymptotic spectral-shape function attained by a polycrystalline paramagnet in the limit of large momentum transfer. A full curve, which relates to the ordinate scale, shows the Fourier transform of the 000 curve in figure 13 and gives the asymptotic form for a classical-spin magnet at infinite temperature. Open circles show the observed asymptotic distribution for polycrystalline RbMnF_3 at $3.5T_N$ as obtained by neutron scattering. (After Windsor 1967.)

The rather pointed maximum at zero frequency is qualitatively similar to, if milder in degree than, the spike at $\omega = 0$ found by Carboni and others (Rogers *et al.* 1967, Carboni and Richards 1968) for the linear antiferromagnetic chain.

4.5.3. $F_{\mathbf{x}1}^{aa(2)}$; high temperatures. The polycrystalline $F^{(2)}$ rises to its limiting value rather quickly, by $Kr_n \sim \pi$, and it is interesting to enquire whether any corresponding development occurs in the single-crystal formula. The single-crystal $F^{(2)}$ is periodic in $\mathbf{K}$ space, taking the moment small near nuclear-reflection positions. Now $\mathbf{K} \cdot \mathbf{r}_n \equiv \pi$, which makes $\gamma_K = -\gamma_0$, is the condition for an antiferromagnetic reflection in the lattices we are considering; it therefore represents $\mathbf{K}$'s that are as far as possible from the nuclear positions. As (204) and (210) show, it also gives the largest $F^{(2)}$ and $F^{(4)}$.

Calculation for a simple-cubic crystal with nearest-neighbour coupling and $S = 1$ shows that

$$\left. \begin{array}{ll} \frac{3F^{(2)2}}{F^{(4)}} = 1.30 & \text{single crystal } \mathbf{K} \cdot \mathbf{r}_n \equiv \pi \\ 0.96 & \text{polycrystal } Kr_n = \pi \\ 0.89 & \text{polycrystal } Kr_n = \infty \\ 1 & \text{Gaussian} \\ 0 & \text{Lorentzian} \end{array} \right\} \quad (233)$$

as against

Collins and Marshall (1967) point out that a useful improvement on the Gaussian approximation is the Gram-Charlier series

$$F(\mathbf{K}, \omega) = (2\pi F^{(2)})^{-1/2} e^{-(\omega^2/2F^{(2)})} \left\{ 1 + c_2 H_2\left(\frac{\omega}{\sqrt{F^{(2)}}}\right) + c_4 H_4\left(\frac{\omega}{\sqrt{F^{(2)}}}\right) + \dots \right\}, \quad (234)$$

in which the H_n are the Chebyshev-Hermite polynomials. The advantage of this series is that after the coefficients up to c_n have been determined by matching moments, the addition of a further term matched by use of a further moment does not affect the already-determined first n coefficients. Collins and Marshall show that even in the case of an exchange parameter that is a function of $\mathbf{r}$, matching of this expression (cut off at c_4) to the rigorously known second and fourth Heisenberg moments for $\mathbf{K}$ at the zone boundary gives a simple result:

$$c_2 = 0; \quad c_4 = \frac{1}{8} \left(\frac{F^{(4)}}{3F^{(2)2}} - 1 \right). \quad (235)$$

Thus if $3F^{(2)2}/F^{(4)}$ be determined, a fitting function is available that has the first three even moments correct. It is also known that when (234) is fitted to a body-centred cubic nearest-neighbour coupled Heisenberg ferromagnet, the first half-moment, defined by

$$F^{(1)} = \int_0^\infty F(\tau_0, \omega) \omega d\omega, \quad (236)$$

differs from that of a Gaussian distribution by a factor of only 1.03.

The expression (234) fits a simple cubic Ising magnet (whose spectrum can be derived exactly) with $S = \frac{5}{2}$ to better than 1% everywhere. To regard this as a check would be self-inconsistent; but in an optimistic mood one might draw an inference that the motion of Heisenberg spins is not so different from that of Ising ones as to create serious discrepancies. On the other hand, if the correct interpretation of figure 8 be that there is markedly non-Gaussian fluctuation at small ω , then (234) is of considerably diminished utility, since effects at small ω would be only poorly accommodated by a fit that relied on second and fourth moments. (*Note added in proof*: see Kwon and Gersch 1968.)

4.6. Moderately high temperatures

4.6.1. *The onset of short-range order.* Before any use can be made of the foregoing discussion it is necessary to know how high a temperature must be to be effectively infinite. As we have seen, lowering of the temperature introduces increasing

amounts of short-range order, which will eventually invalidate the arguments of §§ 4.3 and 4.5.

That pronounced effects occur is seen by extending (201) to higher order in β . By use of

$$Z = \text{Tr } e^{-\beta \mathcal{H}} = \text{Tr} \{1 - \beta \mathcal{H} + \frac{1}{2} \beta^2 \mathcal{H}^2 - \dots\} = (2S+1)^N (1 + \frac{1}{2} \beta^2 \langle \mathcal{H}^2 \rangle + \dots) \quad (237)$$

and

$$\langle \mathcal{H}^2 \rangle = \sum_{\mathbf{r}} 2J(\mathbf{r})^2 \langle \mathbf{S}_0 \cdot \mathbf{S}_{\mathbf{r}} \rangle^2 = \frac{2}{3} NS^2(S+1)^2 \sum_{\mathbf{r}} J(\mathbf{r})^2, \quad (238)$$

one finds

$$F_T^{(2)} = F_{\infty}^{(2)} \{1 - \beta \frac{2}{3} S(S+1) J(\mathbf{K}) + \dots\}. \quad (239)$$

To order β the decrease is a fraction

$$f = \frac{T_C}{T} \frac{J(\mathbf{K})}{J(0)} \quad (240)$$

(here written with $\mathbf{K}$, for a ferromagnet), and can therefore be a very appreciable correction even at temperatures of the order of $5T_C$ that one is accustomed to regarding as extremely high from the point of view of effects in the specific heat.

The detailed-balance-corrected scattering distribution is governed, through equation (43), by $R(\mathbf{K}, \omega)$. The total scattering is thus related to the zeroth moment of $R(\mathbf{K}, \omega)$; and by (41) this has the temperature dependence of $\chi(\mathbf{K})$, so that from (200) the area under the corrected scattering is

$$R_T^{(0)} = R_{\infty}^{(0)} \{1 + \beta \frac{2}{3} S(S+1) J(\mathbf{K}) - \dots\}. \quad (241)$$

To first order in β , therefore, the area goes up by the same amount that the second moment goes down.

Over the range of validity of (239) and (241) the change in the fourth and higher moments would be expected to be rather less. This is because they involve averages over quartic spin products, which will be less affected than the quadratic ones by a given amount of local order. Assuming that the behaviour of the higher moments does not introduce any very pronounced unsuspected effects, therefore, a simple understanding of the above formulae is obtained on noting that they would equally result from

$$F_T(\mathbf{K}, \omega) = \beta \frac{2}{3} S(S+1) J(\mathbf{K}) \delta(\omega) + \{1 - \beta \frac{2}{3} S(S+1) J(\mathbf{K})\} F_{\infty}(\mathbf{K}, \omega). \quad (242)$$

Here the *shape* has been changed, in that the fraction (240) has uniformly been removed and deposited at the centre: this fraction is positive near the magnetic Bragg positions, negative at the magnetic Brillouin zone boundaries, and small over the intermediate volume. Obviously, the $\delta(\omega)$ is an oversimplification; but the suggestion is, correctly, that the short-range order gradually drains off the intensity into quasi-elastic, near-Bragg peaks as the temperature is lowered. Equation (242) is a crude representation, but it gives a valuable rough estimate of the amount of area involved.

The effect to order β in the presence of a single-ion crystal-field anisotropy term in the Hamiltonian has been calculated by Sears (1967), whose result is incorporated in the predictions of figure 11.

4.6.2. Experiments with single crystals. The first extensive surveys of $\chi_T(\mathbf{K}, \omega)$ from neutron scattering were made on nickel by Cable *et al.* (1967). These authors had

considerable success in accounting for the susceptibility in terms of the theory of metals (Lowde and Windsor 1967, 1968 a, b, Allan *et al.* 1968). The phenomena in metals, however, comprise a large subject which we have avoided in the present review, and to confine our study here to the Heisenberg Hamiltonian we return to the work of Windsor *et al.* (1968), something of which has already been shown in figure 12.

The full curve in that figure makes an absolute comparison between the observed $\mathcal{S}(\mathbf{K}, \omega)$ at $3.5T_N$ and the Gram-Charlier series expression of equation (234) for infinite temperature, the latter being estimated with no disposable parameters by use of the known exchange constants, themselves measured at low temperature with neutrons (Windsor 1966 a). For $\mathbf{q}$ vectors of magnitude below about $\frac{1}{2}q_{\max}$ the broken curve shows an absolute comparison with the $\mathcal{S}$ given by the computer-simulated array of classical spins discussed in §4.4.

The picture we have foretold of a roughly Gaussian distribution at the larger K 's is confirmed, as is the fact that the infinite-temperature expression gives a convincing account of the data at representative intermediate K 's of about $\frac{3}{4}(\frac{1}{2}, \frac{1}{2}, 0)$. As predicted, short-range correlations have materially affected the $\mathcal{S}$ even at $3.5T_N$, tending to depress the centre of the distribution at low K 's and to raise it on approaching the zone boundary; this is in accordance with our interpretation of (242) for the antiferromagnetic case. Windsor *et al.* find that the areas involved in the distortions are in reasonable agreement with (242).

The maximum molecular field in a simple-cubic nearest-neighbours-only Heisenberg magnet like RbMnF_3 is $12JS$, and this magnitude (here 8.5 meV) is marked on the neutron-energy-gain side. As postulated in our rough argument of §4.5, this quantity evidently does give a workable, rule-of-thumb impression of the maximum likely energy transfer. But it is clear that at small K 's the figure $12JS$ is not significantly reached, while at $\frac{1}{2}, \frac{1}{2}, 0$ some 3% of the neutrons exceed it. At the antiferromagnetic position $\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$, the full data suggest that in the high temperature limit some 7% of the neutrons would exceed it.

Below about $\frac{1}{2}K_{\max}$, the fitting formula (234) is no longer a viable candidate, as the whole fluctuation spectrum has come under the influence of its tendency to vanish as $O(K^2)$, remarked at (224). Actually, the phenomena in this region constitute a limiting case of very great interest. To this we proceed now to devote §5.

5. Spin diffusion

5.1. Diffusion at high temperatures

5.1.1. *Fundamentals.* The distribution at large T and small K we suspect of being nearly *Lorentzian*. Firstly $F^{(4)}/F^{(2)2}$ (infinite for a Lorentzian function) diverges like K^{-2} . With a function of simple, peaky shape this tends to imply that wings of some kind are being extended from the central part with reduction of K . Secondly, for low K equation (45)—which at high temperatures effectively says

$$\frac{\langle S^z(-\mathbf{K}, 0) S^z(\mathbf{K}, t) \rangle}{\langle S^z(-\mathbf{K}, 0) S^z(\mathbf{K}, 0) \rangle} = e^{-\Gamma|t|} \quad (243)$$

—is plausible, and leads to a Lorentzian.

Equation (243) expresses the approximation of spin diffusion (Bloembergen 1949, Van Hove 1954). The idea here is that at high temperature, in the absence of any special order in the surroundings, the most elementary transition that can occur is an exchange process whereby a unit of spin is transferred from an atom to a neighbouring atom; by a succession of such steps at random the unit of spin originally at the origin arrives at a large distance $\mathbf{r}$ with a probability given by the ordinary theory of atomic diffusion. It is only a slight generalization to imagine that a diffusion equation

$$\frac{\partial}{\partial t} \langle S_0^\alpha(0) S_{\mathbf{r}}^\alpha(t) \rangle = \Lambda \nabla^2 \langle S_0^\alpha(0) S_{\mathbf{r}}^\alpha(t) \rangle \quad (244)$$

holds at large $\mathbf{r}$, with Λ the spin-diffusion constant. Of course at small $\mathbf{r}$, effects due to the crystal structure, to the exact spin dynamics and especially due to the residue of short-range order in the system will falsify (244). The approximation gives, by conventional diffusion theory,

$$\langle S_0^\alpha(0) S_{\mathbf{r}}^\alpha(t) \rangle = \frac{\mathcal{A}}{(4\pi\Lambda t)^{3/2}} e^{-r^2/4\Lambda t}, \quad (245)$$

where $\mathcal{A}$ is the (constant) integral of the correlation over all space:

$$\mathcal{A} = \frac{1}{V} \int \langle S_0^\alpha(0) S_{\mathbf{r}}^\alpha(t) \rangle d\mathbf{r} \quad (246)$$

(V is the volume per atom and $(1/V) \int d\mathbf{r} \equiv \sum_{\mathbf{r}}$). Reverting to the discrete-lattice picture to write

$$\mathcal{A} = \langle S_0^\alpha(0) \sum_{\mathbf{r}} S_{\mathbf{r}}^\alpha(t) \rangle, \quad (247)$$

we demonstrate its independence of t , and show that at high temperatures where $\langle S_0^\alpha S_{\mathbf{r}}^\alpha \rangle = \frac{1}{3} S(S+1) \delta_{\mathbf{r}0}$

$$\mathcal{A} = \frac{1}{3} S(S+1). \quad (248)$$

With lowering of the temperature, the degree of correlation in the system increases and $\mathcal{A}$ rises, ultimately to macroscopic values.

From the nature of diffusion theory it is implied that t in (245) is positive. However, we have seen at (88) that our $\langle S^\alpha(-\mathbf{K}, 0) S^\alpha(\mathbf{K}, t) \rangle$ must be even in t when $\mathbf{K} \equiv -\mathbf{K}$; for correlation theory purposes we therefore replace t by $|t|$. Equation (245) then leads to

$$\langle S^\alpha(-\mathbf{K}, 0) S^\alpha(\mathbf{K}, t) \rangle = N \sum_{\mathbf{r}} e^{i\mathbf{K} \cdot \mathbf{r}} \langle S_0^\alpha(0) S_{\mathbf{r}}^\alpha(t) \rangle \quad (249)$$

$$= \frac{N\mathcal{A}}{(4\pi\Lambda|t|)^{3/2}} \sum_{\mathbf{r}} e^{i\mathbf{K} \cdot \mathbf{r} - (r^2/4\Lambda|t|)} \quad (250)$$

which we write

$$\begin{aligned} &= \frac{N}{V} \frac{\mathcal{A}}{(4\pi\Lambda|t|)^{3/2}} \int_{-\infty}^{\infty} e^{i\mathbf{K} \cdot \mathbf{r} - (r^2/4\Lambda|t|)} d\mathbf{r} \\ &= \frac{N}{V} \frac{\mathcal{A}}{(4\pi\Lambda|t|)^{3/2}} e^{-\Lambda \mathbf{K}^2 |t|} \int_{-\infty}^{\infty} e^{-\{1/2(\Lambda|t|)^{1/2}\} \mathbf{r} - i(\Lambda|t|)^{1/2} \mathbf{K}\}^2 d\mathbf{r} \\ &= \frac{N}{V} \mathcal{A} e^{-\Lambda \mathbf{K}^2 |t|}, \end{aligned} \quad (251)$$

giving equation (243) with

$$\Gamma = \Lambda K^2, \quad (252)$$

accurate for small K .

The approximation in its simplest form (243)–(252) would give rise to an

$$F(\mathbf{K}, \omega) = \frac{1}{\pi} \frac{\Lambda K^2}{\omega^2 + \Lambda^2 K^4}. \quad (253)$$

5.1.2. *Validity of the diffusion approximation.* Going back for a moment to equation (69), (39) gives (after a field $\mathbf{H}(\mathbf{m})$ is switched off)

$$\begin{aligned} \delta \langle M(\mathbf{r}, t) \rangle &= \sum_{\mathbf{K}} e^{-i\mathbf{K} \cdot \mathbf{r}} H(\mathbf{K}) \chi(\mathbf{K}) F(\mathbf{K}, t) \\ &= \sum_{\mathbf{K}} e^{-i\mathbf{K} \cdot \mathbf{r}} H(\mathbf{K}) \chi(\mathbf{K}) \int_{-\infty}^{\infty} F(\mathbf{K}, \omega) e^{i\omega t} d\omega. \end{aligned} \quad (254)$$

If (253) be substituted in this, it is straightforward to demonstrate (by using the same contour integration as was employed at (166)) that

$$\left(\frac{\partial}{\partial t} - \Lambda \nabla^2 \right) \delta \langle M(\mathbf{r}, t) \rangle \xrightarrow[r \rightarrow \infty]{t \rightarrow \infty} 0. \quad (255)$$

In (255) we have shown that the approximation implies a diffusion equation in macroscopic terms for the magnetization over large distances and times. It is therefore equivalent in that limit to a hydrodynamic description of the magnetization, which would arrive at the same equation (255) from the macroscopic conditions

$$\frac{\partial}{\partial t} M(\mathbf{r}, t) + \text{div } \mathbf{j}(\mathbf{r}, t) = 0, \quad (256a)$$

$$\langle \mathbf{j}(\mathbf{r}, t) \rangle = -\Lambda \nabla \langle M(\mathbf{r}, t) \rangle. \quad (256b)$$

The fact that this is so shows us that the diffusion approximation is almost certainly sound for great enough times, and poor only at short times, so that its Fourier transform (253) should be reliable for small energy transfers and unreliable only for large ones. But this is satisfactory, as the experimental results are accurate over the central peak of F and much less significant in the wings. Thus where the parameters are in a favourable range the observed neutron peak can be used to give a good measurement of a phenomenological spin-diffusion constant.

It is worth emphasizing that one must be careful not to make self-inconsistent assumptions about diffusive motion. For instance, in a case where there were a number of coupling terms in the Hamiltonian, one might meet with the complication that there were several different diffusion coefficients simultaneously. As an extreme example of the havoc that could be wrought by incautious assumptions about diffusion, consider a model having the at first sight innocent distribution of Γ 's with weight $\Gamma^{1/2}$ up to some maximum value G . Then

$$f(\mathbf{K}, t) \propto \int_0^G \Gamma^{1/2} e^{-\Gamma|t|} d\Gamma,$$

which beyond times $t \gg G^{-1}$ varies as $|t|^{-3/2}$ with results completely different from those of diffusion theory.

5.1.3. *Range of validity of the approximation.* Given that the diffusion approximation is valid in the limit of high temperatures and macroscopically long wavelengths and times, the question arises to what degree is the approximation useful away from this extreme limit. We shall take up this interesting question both here for the high temperature case and in §6 for a more general range of temperatures.

At high temperatures, the approximation gains in credibility from the following observation. From (8),

$$\langle S_0^\alpha S_{\mathbf{r}}^\alpha(t) \rangle = N^{-2} \sum_{\mathbf{K}} e^{-i\mathbf{K} \cdot \mathbf{r}} \langle S^\alpha(-\mathbf{K}, 0) S^\alpha(\mathbf{K}, t) \rangle. \quad (257)$$

Now at limitingly high temperatures, this can be replaced by

$$= \beta^{-1} N^{-2} \sum_{\mathbf{K}} e^{-i\mathbf{K} \cdot \mathbf{r}} \{ S^\alpha(-\mathbf{K}, 0), S^\alpha(\mathbf{K}, t) \}, \quad (258)$$

or, with (17) and (200),

$$= \frac{1}{3} S(S+1) N^{-1} \sum_{\mathbf{K}} e^{-i\mathbf{K} \cdot \mathbf{r}} F^{\alpha\alpha}(\mathbf{K}, t). \quad (259)$$

By a Taylor expansion, therefore,

$$\langle S_0^\alpha S_{\mathbf{r}}^\alpha(t) \rangle = \langle S_0^\alpha S_{\mathbf{r}}^\alpha \rangle + \frac{1}{2} t^2 \{ \frac{1}{3} S(S+1) N^{-1} \} \sum_{\mathbf{K}} e^{-i\mathbf{K} \cdot \mathbf{r}} \ddot{F}(\mathbf{K}, 0) + \dots \quad (260)$$

and by use of (176) and (204) the last term here is

$$t^2 \frac{4}{9} \frac{S^2(S+1)^2}{N\hbar^2} \sum_{\mathbf{K}, \mathbf{m}} e^{-i\mathbf{K} \cdot \mathbf{r}} (e^{-i\mathbf{K} \cdot \mathbf{m}} - 1) J(\mathbf{m})^2. \quad (261)$$

In particular, if we consider the evolution of the expectation values relating to the spin at the origin and to the spin at a neighbouring site at $\mathbf{r}_n$, in a body-centred cubic or simple-cubic lattice with nearest-neighbour coupling only, we find on employing (207) and (209)–(210) that

$$\langle \mathbf{S}_0 \cdot \mathbf{S}_0(t) \rangle = S(S+1) (1 - \frac{1}{2} t^2 \langle \omega_0^2 \rangle + \frac{1}{7} t^4 \langle \omega_0^2 \rangle^2 - \dots); \quad (262)$$

$$\langle \mathbf{S}_0 \cdot \mathbf{S}_{\mathbf{r}_n}(t) \rangle = S(S+1) Z^{-1} (\frac{1}{2} t^2 \langle \omega_0^2 \rangle - \frac{1}{5} t^4 \langle \omega_0^2 \rangle^2 + \dots); \quad (263)$$

(the t^4 terms are approximate, within about 4% in a practical case). Thus it is confirmed that in this example, at very high temperatures, spin lost from the atom at the origin reappears in the first instance on its Z nearest neighbours at $\mathbf{r}_n$.

From the foregoing, one might be tempted to take the hint that at high temperatures the diffusion approximation was really very good, even out to quite large K 's. However, (243) cannot be correct at non-zero K , since $\dot{F}(0) = 0$. To understand what is the difficulty, let us follow the illuminating discussion of $F(t)$ given by Mori and Kawasaki (1962 a). Given that $\dot{F}(0) = 0$, then trivially:

$$F^{\alpha\alpha}(t) = 1 + \int_0^t (t-t') \frac{d^2}{dt'^2} F(t') dt', \quad (264)$$

and this directs our attention to the paramagnetic $\ddot{F}$. Now from the definition (18),

$$\frac{\partial^2}{\partial t^2} \{A(0), B(t)\} = -\{\dot{A}(0), \dot{B}(t)\}, \quad (265)$$

and we observe that $\ddot{F}$ involves the spin-velocity relaxation function.

We now argue that the decay of the spin-velocity correlation is *faster* than that of the spin correlation itself. (This is a matter which we shall be considering again in §6.) To see this one must first note that the velocity correlation is associated with the precession of each spin about the momentary effective field upon it; thus the spin motions are correlated for the length of time that the molecular fields are correlated, and this is not the same as the time for which the spins themselves are correlated. In the case of classical spins, the velocity correlation is simply the double derivative of the direct spin correlation; but of course in quantum mechanics, time differentiation is accomplished by commuting with the Hamiltonian and no such simple result follows. We have argued in §4.4 that the actual operator $S(t)$ does in fact move in a Heisenberg substance like its classical analogue, but the correlation functions under discussion require the taking of expectation values and do not follow the classical case. If we write, by means of a Taylor expansion,

$$[-\dot{F}(\mathbf{K}, t)] \simeq [-\dot{F}(\mathbf{K}, 0)] \{1 - \frac{9}{7} \langle \omega_0^2 \rangle t^2 + \dots\} \quad (266)$$

(using (204) and (211) for the small- K , high- T case), we may contrast this with the implication

$$F(\mathbf{K}, t) \simeq 1 - \Lambda K^2 t + \dots \quad (267)$$

of (243). To the extent that one can rely on these expansions being well behaved it is clear that the K^2 in (267) allows us to assume a faster decay for (266).

What happens in the classical case is revealed by Windsor's (1968) computation; figure 15(a) shows that even at $K \sim 0.4(\pi/a_0)$ the velocity correlation there is reduced to a tenth while the direct correlation is still above 80%. This result should be strongly indicative of the quantum result, certainly for the larger spin values encountered in practice. If we assume (266) to be the start of a Gaussian, a velocity correlation time

$$t_{\text{vc}} \approx \langle \omega_0^2 \rangle^{-1/2} \quad (268)$$

is obtained.

We can accordingly approximate (264) by

$$F^{\alpha\alpha}(\mathbf{K}, t) \simeq 1 + t \int_0^\infty \dot{F}^{\alpha\alpha}(\mathbf{K}, t') dt' \quad (269)$$

at times $\langle \omega_0^2 \rangle^{-1/2} \ll t \ll \Gamma^{\alpha\alpha-1}$; and this

$$\simeq e^{-G^{\alpha\alpha} t} \quad (270)$$

with

$$G^{\alpha\alpha} = - \int_0^\infty \dot{F}^{\alpha\alpha}(\mathbf{K}, t) dt. \quad (271)$$

The real part of G is the decay constant; from (81)–(82) it must be

$$\text{Re } G^{\alpha\alpha} = -\frac{1}{2} \int_{-\infty}^\infty \dot{F}^{\alpha\alpha}(\mathbf{K}, t) dt, \quad (272)$$

and we know from (224) that this can be written

$$\Gamma^{\alpha\alpha} = \Lambda K^2. \quad (273)$$

Again, as to whether all this actually occurs we may look at the classical ensemble, whose computer-determined spin correlations are illustrated in more detail in

figure 15(b). In fact the immediately striking characteristics of these curves at the small K 's are precisely those of equations (266)–(273): a horizontal commencement with $\dot{F}(0) = 0$; a transition period of duration some one or two times $\langle \omega_0^2 \rangle^{-1/2}$; and the almost abrupt onset of a surprisingly well-defined exponential decline. We also notice that this prescription is still being followed as far out in the Brillouin zone as $0.4(\pi/a_0)$.

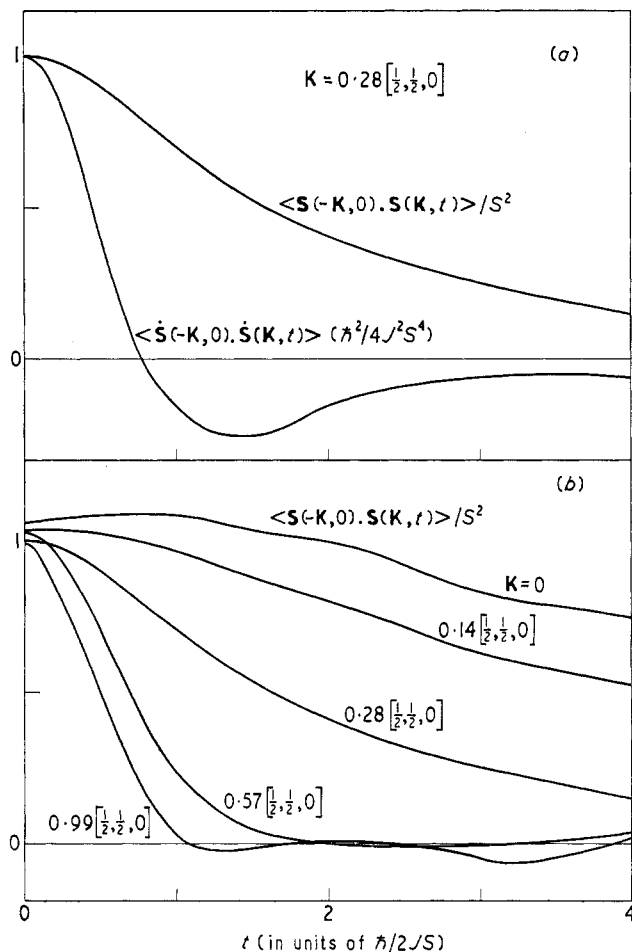


Figure 15. Spin- and spin-velocity-correlation functions for the simple-cubic classical paramagnet with Heisenberg-type nearest-neighbour couplings, as determined by computer simulation. (a) The evolution with time of the self-correlation of $\mathbf{S}(\mathbf{K})$, and that of $\dot{\mathbf{S}}(\mathbf{K})$, for a $\mathbf{K}$ of $0.28[\frac{1}{2}, \frac{1}{2}, 0]$, showing in particular the pronouncedly exponential character of the former and the relatively quick decline of the latter. (b) The self-correlation of $\mathbf{S}(\mathbf{K})$ more extensively shown as a function of $\mathbf{K}$ between 0 and $[\frac{1}{2}, \frac{1}{2}, 0]$, demonstrating that the shorter-wavelength perturbations are the more quickly lost. (Below $\mathbf{K} \sim \frac{1}{4}[\frac{1}{2}, \frac{1}{2}, 0]$ the statistical error rises dramatically from 1–2% to about 8% at $K = 0$. The function for $K = 0$ should be a horizontal straight line through unity; it suffers in practice from fluctuations and from an ultimate decline, both due to the limited size of the spin arrays studied.) (After Windsor 1968.)

5.1.4. *An expression for the diffusion constant.* The failure of the exponential approximation at short times of the order $\langle \omega_0^2 \rangle^{-1/2}$ suggests that the Lorentzian $F(\omega)$ is valid only out to some frequency $\omega_c \sim \langle \omega_0^2 \rangle^{1/2}$, beyond which the response would be expected to be greatly diminished. As an elementary approximation, following de Gennes, we take F to be a cut-off Lorentzian

$$F(\mathbf{K}, \omega) = \frac{1}{\pi} \frac{\Lambda K^2}{\omega^2 + \Lambda^2 K^4} \quad \left. \begin{array}{l} \omega < \omega_c \\ \omega > \omega_c \end{array} \right\} \quad \omega_c \gg \Lambda K^2. \quad (274)$$

$$= 0$$

The half-width at half-height is ΛK^2 , and this must be much less than ω_c for consistency. If we work out the moments of (274), an elementary calculation shows that

$$\omega_c \simeq \left(\frac{3F^{\alpha\alpha(4)}}{F^{\alpha\alpha(2)}} \right)^{1/2} \simeq \pi \langle \omega_0^2 \rangle^{1/2} \quad (275)$$

consistently with our expectation that it should be about t_{ve}^{-1} .

$$\Lambda K^2 = \frac{\pi}{2\sqrt{3}} \left(\frac{F^{\alpha\alpha(2)3}}{F^{\alpha\alpha(4)}} \right)^{1/2}. \quad (276)$$

Their ratio

$$\frac{\omega_c}{\Lambda K^2} = \frac{6}{\pi} \frac{F^{(4)}}{F^{(2)2}} \quad (277)$$

can be estimated from (204) and (211) for the case of a nearest-neighbour coupled magnet at high temperatures:

$$\frac{\omega_{\infty}}{\Lambda_{\infty} K^2} \simeq \frac{24}{K^2 r_n^2} \frac{Z-1}{Z}, \quad (278)$$

and is such a large number that (274) is clearly vindicated as a good approximation.

From the same formulae

$$\hbar \omega_{\infty} \simeq 4(Z-1)^{1/2} J \{S(S+1)\}^{1/2}. \quad (279)$$

In (276) we have also obtained a formula for the spin-diffusion constant of a nearest-neighbour magnet at high temperatures:

$$\Lambda_{\infty} = \frac{\pi r_n^2}{18\hbar} J \{S(S+1)\}^{1/2} \frac{Z}{\{Z-1-3/8S(S+1)\}^{1/2}} \quad (280)$$

$$\simeq \frac{3}{16} \frac{r_n^2}{\hbar} J \{ZS(S+1)\}^{1/2}. \quad (281)$$

5.1.5. *The range of validity again.* Turning back to (266)–(267) again with this expression for Λ , our condition that (266) must have fallen away much farther than (267) at t_{ve} amounts to

$$\Lambda K^2 \ll \frac{6}{7} \langle \omega_0^2 \rangle^{1/2}. \quad (282)$$

This works out at

$$K \ll K_{\text{max}}; \quad (283)$$

not a very exacting condition to have to satisfy.

5.1.6. *The generalized diffusion coefficient.* To take the discussion further, Benne and Martin (1965) note that if we write z as a complex number in generalization of ω ,

$$\frac{1}{K^2} \frac{\chi^{\alpha\alpha}(\mathbf{K}, z)}{\chi^{\alpha\alpha}(\mathbf{K}, 0) - \chi^{\alpha\alpha}(\mathbf{K}, z)}$$

is analytic except on the real axis. It may therefore be written in terms of an arbitrary analytic function $\Lambda(\mathbf{K}, \omega)$ as

$$\frac{1}{K^2} \frac{\chi^{\alpha\alpha}(\mathbf{K}, z)}{\chi^{\alpha\alpha}(\mathbf{K}, 0) - \chi^{\alpha\alpha}(\mathbf{K}, z)} = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\Lambda(\mathbf{K}, \omega')}{\omega'^2 - z^2} d\omega'. \quad (284)$$

From this,

$$1 - \frac{\chi^{\alpha\alpha}(\mathbf{K}, z)}{\chi^{\alpha\alpha}(\mathbf{K}, 0)} = \left\{ 1 + \frac{K^2}{\pi} \int_{-\infty}^{\infty} \frac{\Lambda(\mathbf{K}, \omega')}{\omega'^2 - z^2} d\omega' \right\}^{-1}, \quad (285)$$

and a representation for $\text{Im} \chi$ is obtained by extracting the imaginary part of both sides. As z approaches the real axis from above, the imaginary part of the integrand goes to $-\frac{1}{2}\pi\Lambda(\mathbf{K}, \omega')\delta(\omega' - \omega)/\omega'$; and relating $\text{Im} \chi(\mathbf{K}, \omega)/\chi(\mathbf{K}, 0)$ to $F(\mathbf{K}, \omega)$ by (95) gives at once

$$F^{\alpha\alpha}(\mathbf{K}, \omega) = \frac{\Lambda(\mathbf{K}, \omega) K^2}{\omega^2(1 - K^2 P)^2 + \Lambda(\mathbf{K}, \omega)^2 K^4}, \quad (286)$$

with P the principal part of the integral in (285).

Equation (286) confirms the diffusion-approximation limit at small K , and is valuable in two ways: it identifies the diffusion constant as $\Lambda(0, 0)$, and it draws attention to a more general analysis of non-Lorentzian ω -dependences in terms of a wavelength- and frequency-dependent diffusion parameter $\Lambda(\mathbf{K}, \omega)$ defined by (284).

The essential feature creating these complications is, of course, that diffusion of expectation values is not the only process taking place. Other types of phenomena occur to carry away the information that a spin of $-S$ has been put at the origin. For instance, in some ensembles the origin will chance to be in the middle of a region accidentally enjoying a high degree of ferromagnetic order; in such a situation, spin waves will exist and the spin at the origin will be dissipated by interaction with them. These and other effects are what produce character in $\Lambda(\mathbf{K}, \omega)$. On the other hand, all coherent information-carrying processes are limited in spatial range by interactions of some kind, so that over sufficiently large distances a diffusion mechanism must ultimately provide the only description; this is why the diffusion approximation is correct in the limit $K = 0$.

5.1.7. *An experiment.* The experiment of Windsor *et al.* (1968) on RbMnF_3 at $3.5T_N$ yielded several $F(\mathbf{K}, \omega)$'s at the smaller K 's. One is shown in figure 16, compared absolutely with a Lorentzian curve having the diffusion constant of Bennett and Martin's (1965) discussion. Not only is the general agreement excellent, but one can even detect a quite well-defined drop in the response at approximately $\hbar\omega_0$. As in figure 12, distortion due to the finite temperature is present, and manifests itself principally in the data being on the high side of the line at $|\hbar\omega| \sim 2-3$ meV and on the low side at ~ 1 meV. After making an appropriate allowance for effects of this kind, Windsor *et al.* obtain the diffusion parameters plotted against K in figure 17. The parabola (252) is quite closely followed right out to $\frac{1}{2}K_{\text{max}}$ (defining K_{max} as π/a_0).

5.1.8. *A 'lattice-liquid' picture.* It would seem to follow from figures 15 and 17 that at high temperature and with $S = \frac{5}{2}$, spin diffusion is overwhelmingly the dominant process for conveying information, even over atomic separations as little as two interatomic distances. Consistently with this, quite crude arguments based on the idea of an 'up spin' diffusing about the lattice by jumping into 'down spin' holes do give the correct diffusion constant. In anticipation of §6, where we shall be having difficulties with the diffusion concept, it will be of some interest to set out such an argument.

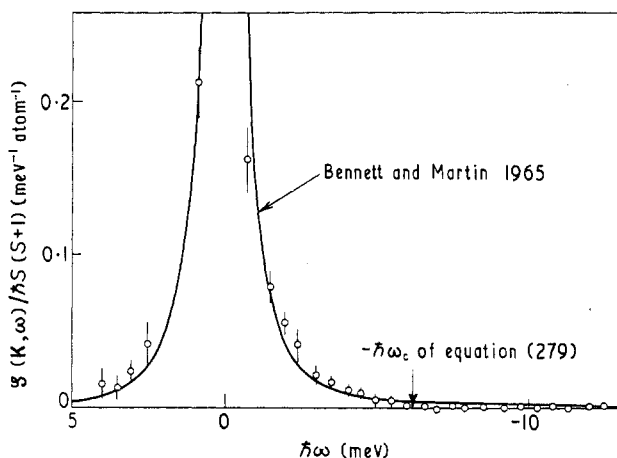


Figure 16. Spin diffusion as the dominating effect in a high temperature simple-cubic paramagnet at $K = 0.35K_{\max}$. The points show the spectral distribution function observed at this mean wave vector by means of neutron scattering from RbMnF_3 at $3.5T_N$. (Data are included for K 's between 0.27 and 0.40 times π/a_0 .) An absolute comparison is made with the Lorentzian function of spin-diffusion theory having a width given by Bennett and Martin's diffusion parameter. (After Windsor *et al.* 1968.)

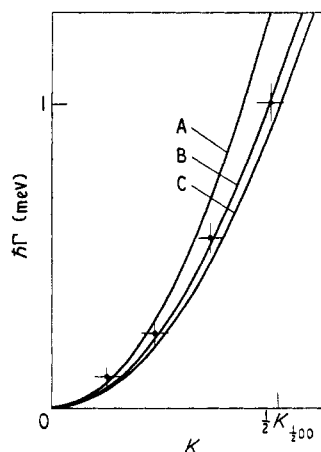


Figure 17. The half-widths of the observed Lorentzian spectral distribution functions for RbMnF_3 , plotted against K for comparison with the parabolic forms ΔK^2 corresponding to various approximate theories of Λ : A, Mori and Kawasaki 1962 a; B, Bennett and Martin 1965; C, de Gennes 1958 a. (After Windsor *et al.* 1968.)

Let us estimate the time delay t_{mn} after which $\langle \mathbf{S}_0 \cdot \mathbf{S}_{\mathbf{r}_n}(t) \rangle$ is a maximum. From (263) with just two terms on the right we obtain

$$t_{mn} \approx \frac{\hbar}{J} \frac{0.7}{\{ZS(S+1)\}^{1/2}}.$$

Actually, Windsor's classical estimate in figure 13 is about half as much again as this, and we may perhaps correct for our too brief Taylor expansion by crossing out the 0.7. If we regard t_{mn} as indicating the time taken for $1/Z$ th of the spin at the origin to leak away on average to the site $\mathbf{r}_n$, we can derive what in the language of diffusion theory would be called the 'jump frequency' ν_j of spin at high

temperatures:

$$\nu_j \approx \frac{J}{\hbar} \{ZS(S+1)\}^{1/2}. \quad (287)$$

Then by a standard result of diffusion theory,

$$\Lambda = \frac{1}{6} \nu_j r_n^2 \quad (288)$$

$$\approx \frac{r_n^2 J}{6\hbar} \{ZS(S+1)\}^{1/2} \quad (289)$$

in agreement with (281).

5.1.9. Kinematic slowing-down. According to (243) and (252), the correlation function $\langle S^z(-\mathbf{K}, 0) S^z(\mathbf{K}, t) \rangle$ is extremely long-lived at small K , that is to say as $\mathbf{K} \rightarrow \tau_{\text{nuc}}$. This effect has been named 'kinematic slowing-down' by de Gennes, in recognition of the argument that at $K = 0$ the correlation function is a constant of the motion, so that by continuity one must expect phenomena at small K to relax very slowly. The effect is not connected with any slowing of the spin motion, but rather to the fact that longer and longer wavelength fluctuations involve larger and larger aggregates of magnetism that take longer and longer to disperse at a given rate of diffusion. It is not to be confused with another type of slowing-down that will be discussed in § 6.

5.2. Diffusion at low temperatures

5.2.1. Theory. Mori and Kawasaki have evaluated (272) for the low temperature region by use of spin-wave theory expressions for S and $\dot{S}$. Partly because we are not here discussing spin-wave theory in any depth, and partly because the whole topic of diffusion at low temperatures has been insufficiently followed up, we refrain from repeating the analysis here. It is, however, extremely interesting that at low K , in the presence of limitingly small applied field and at temperatures high enough that the spin waves of wave vector $\mathbf{K}$ are excited in considerable number, Γ is again ΛK^2 , implying diffusion of $\langle S_0^z S_{\mathbf{r}}^z(t) \rangle$, and with

$$\Lambda^{zz} = \left(\frac{2Z}{3\pi} \right) r_n^2 S \left(\frac{J}{\hbar} \right). \quad (290)$$

This effect cannot be understood on the basis of non-interacting magnons. Huber (1966) has pointed out that if a wave packet be constructed out of ordinary Holstein-Primakoff magnons so as to produce a unit spin decrement δ_{r_0} on the atom at the origin, then (in a simple-cubic Heisenberg ferromagnet with nearest-neighbour coupling only) this decrement disperses as

$$|d_{\mathbf{r}}(t)|^2 = J_{x/r_n}^2 \left(\frac{2t}{t_0} \right) J_{y/r_n}^2 \left(\frac{2t}{t_0} \right) J_{z/r_n}^2 \left(\frac{2t}{t_0} \right), \quad (291)$$

where $\mathbf{r} = (x, y, z)$, the J_n 's are Bessel functions of integer order and t_0 is taken from (230). Equation (291), of course, shows no trace of diffusive behaviour. The diffusion effect of Mori and Kawasaki exists because with *interacting* spin waves there is a density fluctuation of the decrement of $S_{\mathbf{r}}^z$, dependent on the number of (finite-extent) magnons in the vicinity of the site $\mathbf{r}$ at the moment t .

This concept will come up again when we discuss critical spin motion.

5.2.2. *A computer simulation.* In the course of an extremely fruitful computer-simulation project devoted to the study of low temperature phenomena, Blume *et al.* (1968) have determined the xx correlations of classical arrays containing 4000 and more classical spins subject to Heisenberg-type coupling. At low temperatures it is necessary to employ ensembles that are carefully selected to be 'typical' of the true thermodynamic distributions of energy, total magnetization, and so on. After taking proper account of such considerations they obtained, among other things, the transverse correlation of figure 18 for a simple-cubic lattice with nearest-neighbour coupling at a temperature of $0.035T_C$.

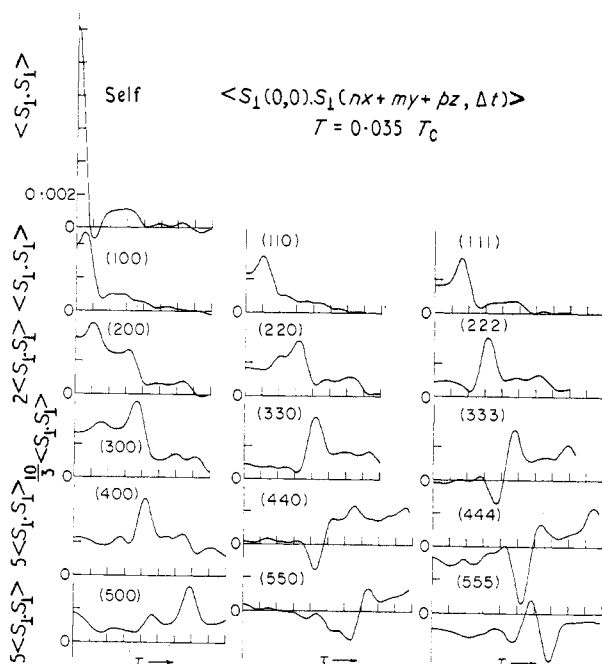


Figure 18. The evolution in time of the correlation function for the component of spin perpendicular to the magnetization at $0.035T_C$ in a simple-cubic ferromagnetic array of classical spins with Heisenberg coupling between nearest neighbours, derived by computer simulation and exhibited as a function of time for various interatomic separations $\mathbf{r}$. The instantaneous correlation as a function of $\mathbf{r}$ can be pictured from the initial intercepts. (After Blume *et al.* 1968.)

Among the extraordinary features possessed by this function is that $\langle S_0(0) S_{\mathbf{r}}(0) \rangle$ has a half-width less than an interatomic separation distance. Now in the case of a ferromagnet with a small spin, say $S = \frac{1}{2}$ or 1, $k_B T_C$ is small compared with the maximum spin-wave energy; and even at the Curie temperature in such a case, only the lower end of the spin-wave spectrum is appreciably excited. Thus at $0.035T_C$ with a small value of S , $\langle S_0 S_{\mathbf{r}} \rangle$ would have a half-width of several interatomic distances. However, for large spins the Curie temperature increases with S^2 , while the top of the spin-wave spectrum rises only with S . In the limit of large spin, therefore, all the spin-wave spectrum is excited at any non-zero temperature;

and it would seem that the classical array, which corresponds to that limit, reflects this fact by exhibiting a very short-range correlation.

Going together with this unusual localization of $\langle S_0 S_r \rangle$ is a hitherto unremarked, but none the less extremely pronounced diffusive component in the $\alpha\alpha$ motion, with propagation velocities in the ratio $1 : \frac{1}{2} : \frac{1}{3}$ in the $\langle 100 \rangle$, $\langle 110 \rangle$ and $\langle 111 \rangle$ directions. Evidently at least two types of process are competing to dissipate the disturbance originally at the origin. The jump time is seen to be about 0.6 in units of $\hbar/2ZSJ$, and by (288) this corresponds to a diffusion constant of approximately $2\frac{1}{2}$ times the Λ^{zz} of (290).

In all likelihood the perpendicular and parallel Λ 's have a common physical origin in the fluctuating magnon density, and it would be very interesting to know more about both.

6. Critical phenomena

We have seen in §3 that as the critical point is approached from either side, the magnetic correlations in the system increase in both range and degree; as a result of this fact, many of the arguments used in the foregoing begin to fail in various ways if the temperature is carried towards the region of the phase transition.

It will become apparent from what follows that our current understanding of critical spin dynamics is itself at a turning-point. In this situation we shall not set out the advanced theory of the day (for which see especially Mori 1966 a). For where the available theories have been worked out, they are inadequate; and where they are adequate, they have not been worked out. Such a discussion would in any event fall in the shadow of last year's admirable Reports on Progress in Physics by Fisher (1967) and Heller (1967). We shall try to give instead a simple account of the conceptual difficulties, of the most recently discovered facts and of what we believe may prove to be the correct basis for discussion. It is hoped that the resulting picture will set the scene for some future Report on Progress in Physics in which all will be made clear. (*Note added in proof*: see Allan and Betts 1968, Kawasaki 1968 a, b, Mori and Okamoto 1968 a, b, Oguchi and Ono 1968, Suzuki and Kubo 1968).

6.1. The question of critical diffusion

In the region of large q 's, away from magnetic Bragg reflection positions, comparatively little of note occurs on passing through the critical temperature. It has been known for some time (Riste 1962, Krasnicki *et al.* 1964, Turberfield and Okazaki 1965) that at large q 's a wide fluctuation spectrum develops continuously through the transition point without particular event. The dramatic phenomena are all in the region of the magnetic reciprocal lattice points. This fact is perhaps unsurprising; but it takes on especial force in antiferromagnets, where the very symmetry can change in the high temperature limit, so that what is a large q at $T = \infty$ (where only the nuclear symmetry need be considered) can become a small q at $T \leq T_C$, and the response function must in the interval evolve all the way from the near-Gaussian function of §4.5 to a narrow, diffusion-dominated critical peak with the various properties to be described. The small- q magnetic response in ferromagnets is also subject to a hardly less spectacular change in the same interval.

At small but non-zero q 's, we saw that the most visible difficulty in the way of obtaining a simple theory was that other processes do occur beside spin diffusion. But as the critical temperature is approached, say from above, this difficulty can only become more serious, because there will increasingly be found large regions of local order in any given ensemble in which low temperature phenomena—even spin waves—will be manifested. Throughout the paramagnetic region, nevertheless, the spectral density (286) can be discussed in terms of a generalized $\Lambda(\mathbf{K}, \omega)$. The first question must accordingly be: To what degree does $\Lambda(\mathbf{K}, \omega)$ remain comparatively featureless near the origin, so that an elementary picture can be sustained?

Clearly, as the temperature is lowered from infinity, spin diffusion must become slower, because with the range of correlation increasing according to (166) it requires more and more of an upheaval to make a given change in the magnetization at the origin. There is an interesting physical argument, due to de Gennes, suggesting that the diffusion picture remains valid and is capable of a considerable extension down the scale of temperatures. This argument appears at first sight extremely unsophisticated, but it actually embodies the essential physical content of the more advanced treatments; and indeed certain published theories have foundered due to ignoring it. Since the advanced treatments are in any case unsuccessful, we give the unsophisticated one below.

6.2. Thermodynamic slowing-down

6.2.1. *An elementary treatment.* Let us attempt to allow for the lowering of the temperature below infinity by molecular-field theory. Then the essential change introduced into the considerations at any atomic site is that a molecular field (169) has appeared. Taking the case of a simple-cubic ferromagnet with spins $\frac{1}{2}$, consider the effects of a local fluctuation in which the mean occupation-number of negative spin at the origin temporarily rises from $\frac{1}{2}$ to some greater value n_0 owing to a temporary increase of the molecular field at the origin to $\{(\chi - \chi_0)/\chi\chi_0\}g\mu(n_0 - \frac{1}{2})$. During this time the mean occupation of a neighbouring site at $\mathbf{r}_n$ is, say, n_{r_n} owing to a temporary molecular field $\{(\chi - \chi_0)/\chi\chi_0\}g\mu(n_{r_n} - \frac{1}{2})$. Then in these circumstances the mean frequency of migration of negative spin from $\mathbf{0}$ to $\mathbf{r}_n$ must be altered. If the change is slight, we can allow for it by saying that the Boltzmann occupation probabilities p_0, p_{r_n} show that the origin is now rather longer occupied than at high temperatures, in the ratio of $p_0/\frac{1}{2}$, and so the reduced rate of outflow of spin in the first few instants corresponds to that given by the $\nu_{j\infty}$ of equation (287) multiplied by $\frac{1}{2}/p_0$. The Boltzmann factor corresponding to a down spin at $\mathbf{0}$ with an up spin at $\mathbf{r}_n$ is $\exp\{-\mu\beta(H_{\text{mol}}^0 - H_{\text{mol}}^{r_n})\}$, and the modified 'jump frequency' with which a down spin leaves the origin for $\mathbf{r}_n$ proves to be

$$\nu_{0r_n} = \nu_{j\infty} Z^{-1} \left\{ 1 - 2\beta\mu^2 \frac{\chi - \chi_0}{\chi\chi_0} (n_0 - n_{r_n}) \right\}. \quad (292)$$

Relaxation of the mean occupation n_0 at the origin will therefore occur at a rate corresponding to the outflow calculated above times $n_0(1 - n_{r_n})Z$ minus the inflow from the reverse process, resulting in a jump frequency for spin of

magnitude

$$\nu_{jT} = \nu_{j\infty} \left[n_0(1-n_{r_n}) \left\{ 1 - 2(n_0-n_{r_n}) \left(\frac{\chi-\chi_0}{\chi} \right) \right\} - n_{r_n}(1-n_0) \left\{ 1 - 2(n_{r_n}-n_0) \left(\frac{\chi-\chi_0}{\chi} \right) \right\} \right] \quad (293)$$

$$= \nu_{j\infty} \frac{\chi_0}{\chi} (n_0 - n_{r_n}). \quad (294)$$

The rate is proportional to $n_0 - n_{r_n}$, so that a diffusion treatment is justified in this picture; and then appealing as at (288) to the diffusion theory result

$$\Lambda_T = \frac{\frac{1}{6}\nu_{jT}r_n^2}{n_0 - n_{r_n}}, \quad (295)$$

we obtain

$$\Lambda_T = \frac{r_n^2 J}{6\hbar} \{ZS(S+1)\}^{1/2} \frac{\chi_0}{\chi T}. \quad (296)$$

At infinite temperature, this expression reduces to as near (238) as could be expected.

As derived here, the factor χ_0/χ is justified under the condition

$$2(n_0 - n_{r_n}) \left(\frac{\chi - \chi_0}{\chi} \right) \ll 1,$$

in extremis let us say $n_0 - n_{r_n} \ll \frac{1}{2}$; and therefore the discussion, for what it is, is valid only in the situation where

$$\langle \mathbf{S}_0 \cdot \mathbf{S}_{r_n} \rangle \ll S(S+1). \quad (297)$$

In fact, numerical estimates that have been made for $\mathcal{E}(\mathbf{l})$ of equation (195) show that the Curie temperature can be approached quite closely without seriously violating (297). However, at T_C itself $\langle \mathbf{S}_0 \cdot \mathbf{S}_{r_n} \rangle / S(S+1)$ is about 0.4. Our argument might therefore be expected to apply for some way towards T_C . The principal criticism of molecular-field theory, the fact of its ignoring short-range order, does not apply to effects at small K where the diffusion picture should be sound; we are accordingly led to the idea that, certainly at small enough K , the diffusion constant Λ should begin to collapse as the temperature is lowered in a manner dominated by the collapse of $\chi^{-1}(T)$.

The same simple arguments can be extended to antiferromagnetism (de Gennes 1958 b, de Gennes and Villain 1960), and also to show that Λ , like κ_1^2 , should be expected to change by a factor 2 across the Curie temperature.

Equations (294) and (296) imply that as the temperature is lowered towards the Curie point, the spin dynamics are subjected to a slowing-down, originally called by de Gennes 'thermodynamic braking' (de Gennes 1959, de Gennes and Villain 1960) in virtue of its original derivation from the thermodynamics of irreversible processes. The phenomenon is fundamentally different from the kinematic slowing-down of §5.1. Firstly, it is associated with real changes in the spin dynamics, dependent on temperature; thus according to (245), $\langle S_0^z(0) S_r^z(t) \rangle$ actually does move more slowly in proportion to Λ . Secondly, as is particularly clear for anti-ferromagnets, where the slowing-down can be part of the process of acquiring a new symmetry group, the effect is due to the energetics of the system and not to the

continuous approach of the function studied to a constant of the motion. According to (272)–(273), the paramagnetic

$$\Gamma^{\alpha\alpha} = \frac{g^2 \mu^2}{2N} \frac{1}{\chi(\mathbf{K})} \int_{-\infty}^{\infty} \{\dot{S}^{\alpha}(-\mathbf{K}, 0), \dot{S}^{\alpha}(\mathbf{K}, t)\} dt. \quad (298)$$

Thermodynamic slowing-down is due to the denominator of (298) increasing; kinematic slowing-down to the numerator decreasing.

6.2.2. $\Lambda(\mathbf{K})$ in the elementary treatment. The temperature dependence of Λ reflects a temperature dependence of (276). We have already argued in §4.6 that the spin correlations in (213) are well behaved at T_C ; recalling that point, it is evident that the singularity of Λ^{-1} is due entirely to the temperature dependence of $\chi(\mathbf{K})$. It would follow that

$$\Lambda_T(\mathbf{K}) \simeq \Lambda_{\infty} \frac{\chi_0}{\chi(\mathbf{K})}, \quad (299)$$

giving some indication of the K dependence of our generalized $\Lambda(\mathbf{K}, \omega)$.

6.2.3. *A reservation.* Before going any farther, we must mention that (296) and (299) for Λ and $\Lambda(\mathbf{K})$ do not fit the facts. A series of experiments, some of which we shall examine in the next section, have shown that critical slowing-down does occur; indeed that so far as can be divined there is, or can be, diffusion (Jacrot *et al.* 1963); that the diffusion coefficient does then vanish at the critical temperature and scattering vector; but none of this in accordance with (296) and (299). The situation appears to be one where molecular-field theory, together with its various equivalents such as the random phase approximation in the more careful analyses, intrinsically fail to express the physics of the problem. The influence of local order, both static and dynamic, is all.

In the absence, today, of an acceptable theory we are unable to draw the firm line of argument very much farther. But perhaps for that very reason, as we approach the end of our metaphorical thread of §1 we should make sure to tie it to the sturdiest metaphorical signpost. We shall try to do this: and at the same time, in the next few pages, to describe our view of the terrain into which the necessary advances must be made.

6.2.4. *Evidence for critical diffusion.* To begin with, figure 19 shows that for $T > T_C$ in iron the width of the inelastic neutron spectrum (measured with what would nowadays be called poor resolution) increases (above the basic resolution width) in proportion to K^2 (for K 's above about $0.05 \text{ \AA}^{-1}$), thus seeming to suggest by itself that all is well with (253). Our remarks about molecular-field theory have not necessarily invalidated (298), which still depends on (269), and on the face of it a case for critical diffusion has been made out. In the first instance, therefore, it becomes a most engaging question whether (269) is still true, i.e. whether the underlying assumption that the velocity correlation is faster annulled than the direct correlation can be sustained under critical conditions.

As a preliminary to examining this matter we now study those correlations at $t = 0$.

6.3. The instantaneous spin- and spin-velocity-correlations at small K as a function of temperature above T_C

From (10), (34) and (37) it is approximately true that

$$\langle S^\alpha(-\mathbf{K}, 0) S^\alpha(\mathbf{K}, 0) \rangle \simeq \frac{S(S+1) N \chi^{\alpha\alpha}(\mathbf{K})}{3\chi_0}, \quad (300)$$

a sufficient condition being $q \ll q_{\max}$. Then by (113),

$$\langle S^\alpha(-\mathbf{K}, 0) S^\alpha(\mathbf{K}, 0) \rangle \simeq \frac{1}{3} S(S+1) N \{1 - uJ(\mathbf{K})\} \quad (301)$$

where $u = T_C/TJ(0)$; at small K it becomes

$$\simeq \frac{1}{3} S(S+1) N \frac{T}{(T - T_C) + \frac{1}{2} T_C K^2 r_n^2}. \quad (302)$$

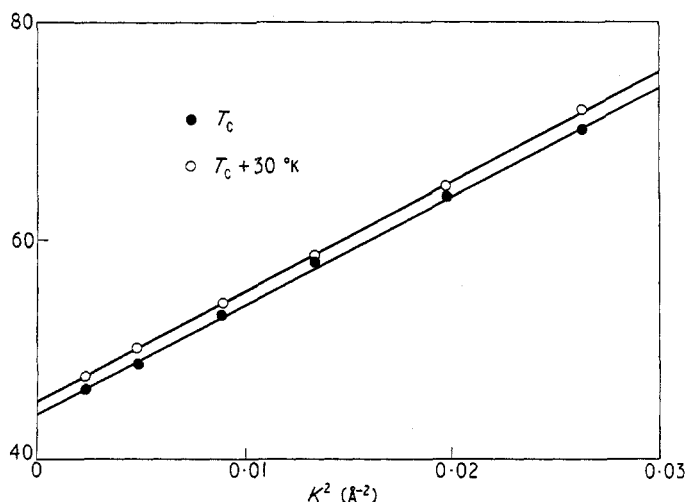


Figure 19. Energy widths (in arbitrary units) of the neutron spectrum line recorded when $4.75 \text{ \AA}$ neutrons are scattered at small angles by iron near the Curie point. The abscissa is the square of the scattering angle, here expressed in terms of the scattering vector $\mathbf{K}$. A width equal to the resolution width plus ΔK^2 is indicated (see also figure 21). (Based on Jacrot *et al.* 1963.)

It is asymptotically constant at high temperatures and diverges towards T_C in the familiar manner.

It is difficult to estimate the spin-velocity correlation; but we can obtain the closely related function

$$\beta^{-1} \{ \dot{S}^\alpha(-\mathbf{K}), \dot{S}^\alpha(\mathbf{K}) \}, \quad (303)$$

which is equal to the velocity-correlation function in the limit of high temperatures. Now in that limit, by a trivial application of (204) for small K ,

$$\beta^{-1} \{ \dot{S}^\alpha(-\mathbf{K}), \dot{S}^\alpha(\mathbf{K}) \} = \langle \dot{S}^\alpha(-\mathbf{K}) \dot{S}^\alpha(\mathbf{K}) \rangle = \frac{4N}{27} \frac{J^2}{\hbar^2} Z S^2 (S+1)^2 K^2 r_n^2. \quad (304)$$

Let us now estimate the expression (303) more generally, taking the case of a ferromagnet. We have

$$\{\dot{S}^\alpha(-\mathbf{K}), \dot{S}^\alpha(\mathbf{K})\} = \frac{N}{g^2 \mu^2} \chi(\mathbf{K}) F^{\alpha\alpha(2)}$$

which by (196)

$$= \frac{4N}{\hbar^2} \sum_{\mathbf{r}} J(\mathbf{r}) (1 - e^{-i\mathbf{K} \cdot \mathbf{r}}) \langle S_0^\alpha S_{\mathbf{r}}^\alpha \rangle, \quad (305)$$

or at small K by (257)

$$\simeq \frac{2}{\hbar^2} \sum_{\mathbf{r}} J(\mathbf{r}) \{(\mathbf{K} \cdot \mathbf{r})^2 + \dots\} \sum_{\mathbf{Q}} e^{-i\mathbf{Q} \cdot \mathbf{r}} \langle S^\alpha(-\mathbf{Q}) S^\alpha(\mathbf{Q}) \rangle \quad (306)$$

$$= \frac{2}{3\hbar^2} K^2 r_n^2 \sum_{\mathbf{Q}} J(-\mathbf{Q}) \langle S^\alpha(-\mathbf{Q}) S^\alpha(\mathbf{Q}) \rangle. \quad (307)$$

Using (301), this

$$= \frac{2S(S+1)}{9N\hbar^2} K^2 r_n^2 J(0) \frac{T}{T_C} w \quad (308)$$

where

$$w = -1 + N^{-1} \sum_{\mathbf{K}} \frac{1}{1 - uJ(\mathbf{K})}. \quad (309)$$

At the Curie point, in the case of nearest-neighbour couplings, Mori and Kawasaki give $w = 0.52, 0.39$ and 0.34 for the simple-cubic, body-centred cubic and face-centred cubic cases respectively. Accordingly at T_C ,

$$\beta^{-1} \{\dot{S}^\alpha(-\mathbf{K}), \dot{S}^\alpha(\mathbf{K})\}_{T_C} \simeq \frac{4wN}{27} \frac{J^2}{\hbar^2} Z^2 S^2 (S+1)^2 K^2 r_n^2. \quad (310)$$

While it is not clear what exactly has happened to the instantaneous velocity-correlation function, we can say that the quantity (303) has increased by wZ , that is to say by only about 3 or 4, on going from $T = \infty$ to the Curie temperature.

6.4. Facts and fallacies about critical scattering

6.4.1. *Close confinement of the critical phenomena to T_C , $\mathbf{K}_{\text{crit}}$ and $\omega = 0$.* For several years the best available discussions of critical relaxation ran rather as follows. With lowering of the temperature from infinity, $\{S, S(t)\}$ is slowed, but $\{\dot{S}, \dot{S}(t)\}$ 'obviously' is not; so the approximation justifying (269) is the better, the nearer one approaches T_C . There is accordingly diffusion; and one may use (276). Putting in $F^{(2)}$ from (310), and $F^{(4)}$ from (211), since $\{\dot{S}, \dot{S}(t)\}$ is even more 'obviously' unaffected by the lower temperature, (296) is obtained within a factor close to unity. And from the form (298) with (310), near T_C the ferromagnetic

$$\Gamma \approx K^2 r_1^2 (\kappa_1^2 + K^2); \quad (311)$$

actually at T_C , $\Gamma \approx K^4$.

Observation does not bear out these ideas. Both iron and terbium (Als-Nielsen *et al.* 1967) have been studied carefully from this point of view, and while Lorentzian behaviour is found in both cases the anticipated approach of Λ to zero with κ_1 and

K (the broken line in figure 20) is not confirmed. The implication of the data is that on reducing the temperature, what had previously been thought to be the characteristics of criticality set in much *later* than had been expected. On the other hand we know that the classical Curie-Weiss law is obeyed in Heisenberg substances (being the limiting case of the susceptibility formalism at $\chi(0, 0)$), so it would be more correct to say the phenomena of criticality are much more closely confined to the origin of $(\mathbf{K}, \omega)$ space than had been foreseen. This is confirmed by the latest analysis (Konstantinovic 1966) of the Saclay experiment on iron

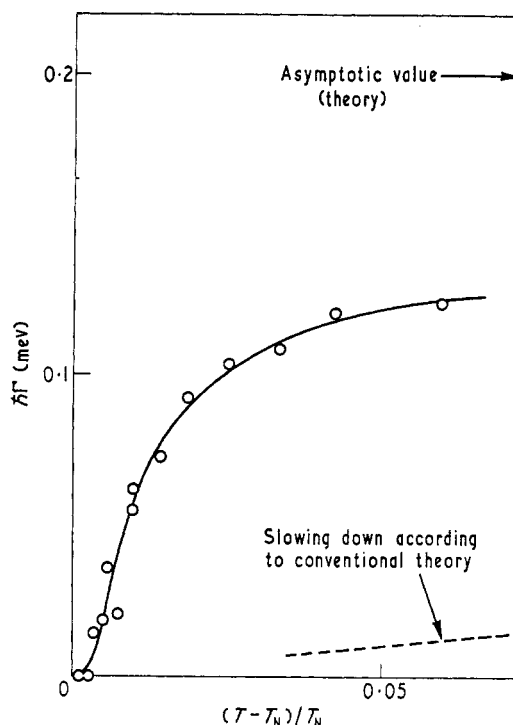


Figure 20. Critical energy widths $\hbar\Gamma$ in terbium as a function of temperature. (On descending to the critical temperature in terbium, $\chi(\mathbf{K})$ diverges for a non-zero K , leading to a spiral structure of wave vector, say, $\mathbf{Q}$. These data were taken by means of neutron scattering at $\mathbf{K} = \mathbf{Q}$.) A narrow spectral function, corresponding to severely braked spin dynamics, is found only over a much smaller range of temperatures than predicted by the theories equivalent to molecular-field theory. (After Als-Nielsen *et al.* 1967.)

(Jacrot *et al.* 1963), which on allowing carefully for instrumental effects supplements figure 19 by showing that the apparent Λ near T_C does fall off at the very smallest K 's (figure 21) (see Collins *et al.* 1968). Exactly the same conclusion was reached in a study of nickel (Lowde and Windsor 1968 b), where a neutron-scattering survey of $\text{Im}\chi(\mathbf{K}, \omega)$ showed that beyond $\hbar\omega \approx 0.05$ eV (in a substance having a maximum molecular field of $(0.36 \text{ eV})/g\mu$ on a nearest-neighbours-only, Heisenberg interpretation) the susceptibility was almost independent of temperature down to (figure 22) and through (Lowde and Windsor 1968 a) the Curie point. Iron and nickel do not have the Heisenberg Hamiltonian, but Izuyama *et al.* (1963) have

shown that the phenomena in metals must be qualitatively the same as in insulating substances, and it seems natural to believe that this is because both formalisms are constrained by deeper-lying reasons to conform to some as yet unframed law of phase transitions.

The burden of the experiments on ferromagnets is, in sum, that $F(\mathbf{K}, \omega)$ at T_C does go to an elastic $\delta(\omega)$ at $K = 0$, as indeed it must for fundamental reasons; and that at all non-trivially-zero $(\mathbf{K}, \omega)$'s the critical F is inelastic, as again one had the strongest cause to expect; but that the leading terms and coefficients in K and ω are extremely difficult to pin down because they are manifested, unexpectedly, only at exceedingly small K 's and ω 's. On the temperature scale, materially away from $K = 0$ the phenomena are compressed into unexpectedly small $T - T_C$'s.

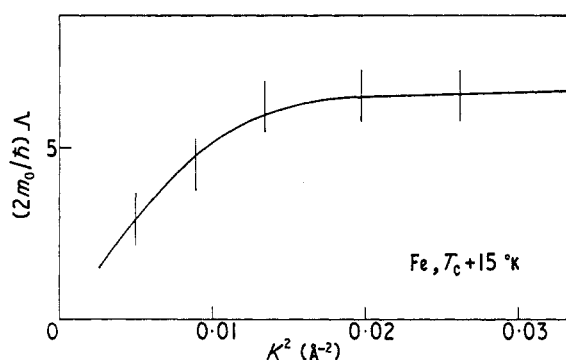


Figure 21. The diffusion parameters for iron in the range of conditions covered by figure 19, according to a detailed analysis of the Saclay experiment. (Based on Konstantinovic 1966.)

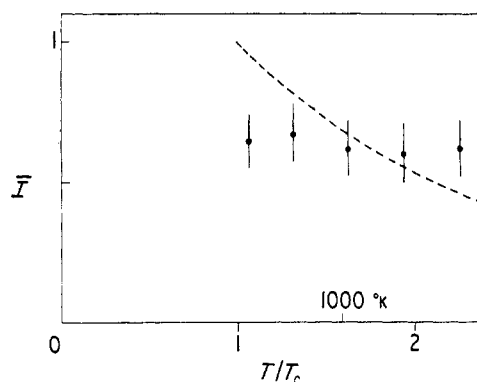


Figure 22. Values of $\bar{I}$ in the exchange enhancement $\alpha = (1 - \bar{I})^{-1}$ of the susceptibility of nickel, from a neutron-scattering determination of $R(\mathbf{K}, \omega)$ for K 's beyond $\sim 0.1 K_{\max}$ and $\hbar\omega$'s above $\sim 0.1 \times 2ZJS$, implying that outside the immediate region of the origin the critical divergence of χ is heavily suppressed. A broken curve gives the prediction of theories equivalent to molecular-field theory. (After Lowde and Windsor 1968 b.)

As often occurs in physics, the 'obvious' has proved false. Statistical mechanics is a subject in which $3N$ degrees of freedom are reduced to about half a dozen by the judicious injection of assumptions; the danger that the results will be distorted by false presupposition is always acute, and nowhere more so than in the field of critical phenomena. Only recently was a proof obtained—by Kawasaki (1967 a)—that the diffusion picture is unsound in the critical region.

6.4.2. Kawasaki's *reductio ad absurdum*. To see this, write Γ in the form (298) using for the relaxation function the time-dependent version of (306). Then

$$\Gamma_{\alpha\alpha}(\mathbf{K}) = \frac{g^2 \mu^2}{\hbar^2 N} \frac{1}{\chi_{\alpha\alpha}(\mathbf{K})} \sum_{\mathbf{r}} J(\mathbf{r}) (\mathbf{K} \cdot \mathbf{r})^2 \sum_{\mathbf{Q}} e^{-i\mathbf{Q} \cdot \mathbf{r}} \int_{-\infty}^{\infty} \langle S^{\alpha}(-\mathbf{Q}) S^{\alpha}(\mathbf{Q}, t) \rangle dt. \quad (312)$$

The diffusion assumption (243), if true, allows us to perform the integration trivially. Thereafter approximating $g^2 \mu^2 N^{-1} \langle S^\alpha(-\mathbf{Q}) S^\alpha(\mathbf{Q}) \rangle$ by $\chi(\mathbf{Q})$, which should be accurate under the conditions of a narrow, critical diffusion peak, and approximating $\Gamma(Q)$ on the right-hand side by $\Lambda_\infty \{\chi_0/\chi(\mathbf{Q})\} Q^2$ according to (299), we obtain

$$\Gamma(\mathbf{K}) = \frac{2}{\hbar^2} \frac{1}{\chi_0 \chi(\mathbf{K}) \Lambda_\infty} \sum_{\mathbf{r}} J(\mathbf{r})(\mathbf{K} \cdot \mathbf{r})^2 \sum_{\mathbf{Q}} e^{-i\mathbf{Q} \cdot \mathbf{r}} Q^{-2} \chi(\mathbf{Q})^2. \quad (313)$$

The sum over $\mathbf{Q}$ may be written

$$\sum_{\mathbf{Q}} \{(\text{small-}Q \text{ contribution}) + (\text{larger-}Q \text{ contribution})\}, \quad (314)$$

and in the critical region the small- Q contribution is

$$\sum_{\text{small } Q} \frac{\chi_c^2}{Q^2(\kappa_1^2 + Q^2)^2}. \quad (315)$$

But as the Curie temperature is approached, (315) diverges as κ_1^{-3} .

A contradiction, in other words, has resulted from inserting the diffusion approximation at (312) and using the form of χ appropriate to temperatures near T_C . An assumption is therefore in error, and one may well suspect first the assumption that $\{\dot{S}, \dot{S}(t)\}$ is little affected on approaching the transition point. $\Lambda(\mathbf{K}, \omega)$ must be more complicated than equations (292)–(311) would allow.

6.4.3. Some newly observed fluctuation spectra. That this is true, certainly below T_C , was first indicated by the sharply defined spin-wave peaks observed by Torrie two years ago (Torrie 1966) at low q in MnF_2 at $0.976T_N$. It is noteworthy that if the published distributions in this example for $T_N(1-\epsilon)$ with $\epsilon = 0.039$ and 0.024 be compared with others taken at nearly $T_N(1+2\epsilon)$ in the same investigation, it can be argued that a broad central ‘hump’ of intensity is present continuously through the Curie temperature, the spin-wave peaks apparently being just grafted on in the ordered phase.

In the following year some even more spectacular results were obtained that may well help to elucidate the mystery, when the Chalk River, Brookhaven and Harwell neutron-scattering groups all independently published data showing well-defined peaks in the scattered-neutron energy spectrum above the transition temperature that had evolved continuously as a function of temperature from the low temperature spin-wave peaks (Cowley and Martel 1968, CoF_2 ; Nathans *et al.* 1968, RbMnF_3 ; Lowde and Windsor 1968 a, nickel). Particularly striking is the spectrum obtained by Nathans *et al.* from RbMnF_3 , the development of which through T_N is shown in figure 23. In this instance it appears that unexpectedly well-defined diffusion-like and spin-wave-like peaks are present simultaneously, the phase transition appearing to have little effect on them.† Figures 20–23 together make it clear that molecular-field ideas have failed absolutely to account for the phenomena away from $\mathbf{K} = \mathbf{K}_{\text{crit}}$.

† Heller (1967) has analysed Torrie’s experiment, and suggests that the reason no central peak was found is that $\chi_{\parallel}$ was not observed, while $\chi_{\perp}$ does not diverge.

As Lowde and Windsor remark, it appears that on approaching the transition point from either above or below, the relaxation functions controlling the magnetic response strongly anticipate the behaviour that will be required of them on the other side. Thus the paramagnetic phase builds up a multi-peaked response capable of joining on to the spin-wave peaks in the transverse function below T_C , and the ordered phase develops a central component capable of going over to the

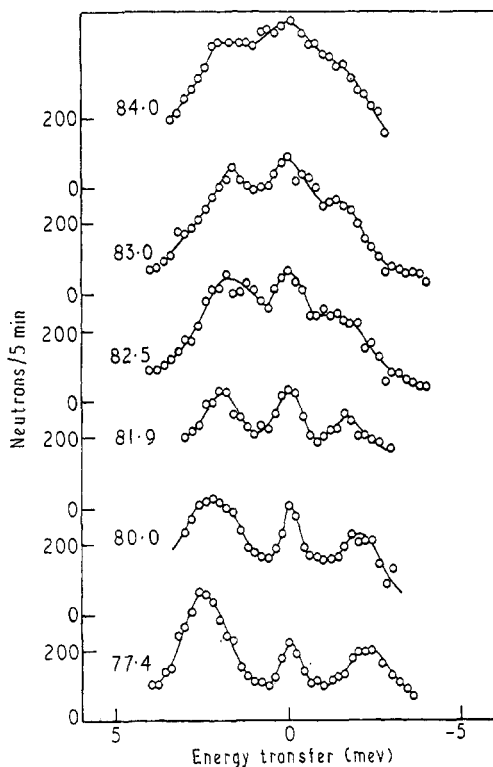


Figure 23. Neutron scattered-energy spectra from RbMnF_3 at $\mathbf{K} = 1.15[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$, taken at a series of temperatures straddling the Néel point of 82.65°K . The numbers on the curves give T in $^\circ\text{K}$. (After Nathans *et al.* 1968.)

diffusive peak above T_C . The situation as regards nickel is documented in some detail in figure 24, which reproduces the observed development with temperature of $\frac{1}{8}\sum_{\alpha}\mathcal{S}^{\alpha\alpha}$ for that substance. It is remarkable to see with what little change the correlation function evolves from $0.9 T_C$ to $1.1 T_C$.

An incidental effect of this new realization, of course, is that doubt is cast in retrospect on the interpretation of all the older experiments such as those of figures 4 and 19 taken with poor or no resolution.

6.5. Interpretation of the observed critical relaxation

Using a new theory that is too elaborate to be reproduced here, Kawasaki (1967) has demonstrated that the central peak may be diffusion-like, but that the diffusion constant, where meaningful, could vanish as some function like $\chi^{-1/4}$ as

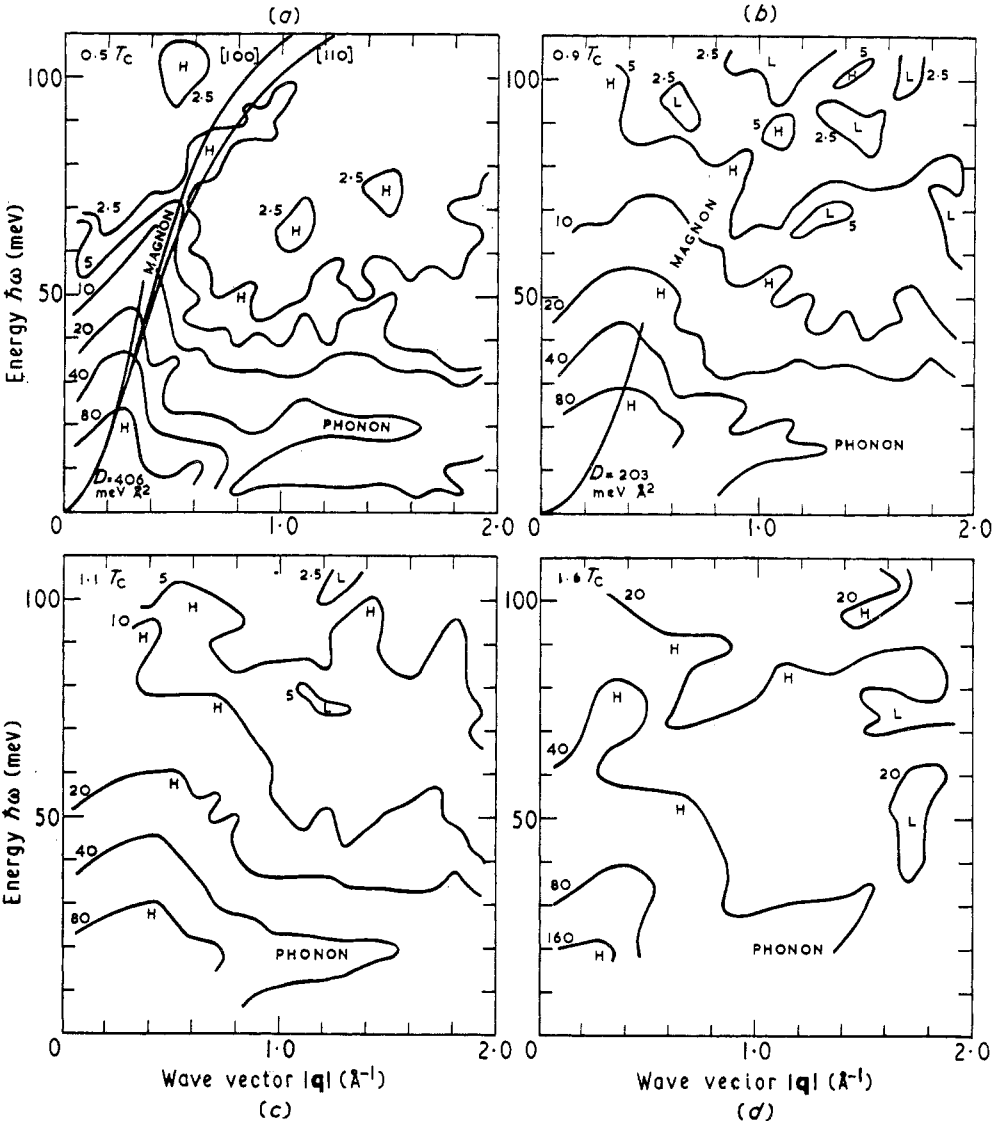


Figure 24.

$T \rightarrow T_C$. Whether this result contains the essential explanation of figures 20–21 is still unclear, and further investigation is much to be desired.

The clearest way of describing the situation is to consider a generalized, complex frequency z and to examine the poles of $R(\mathbf{K}, z)$ to which attention was drawn in § 2.12. At the lowest temperatures we know that R^{xx} is dominated by two spin-wave peaks centred at $\pm \hbar\omega_{sw}$ and corresponding to the poles at the left of figure 25; as

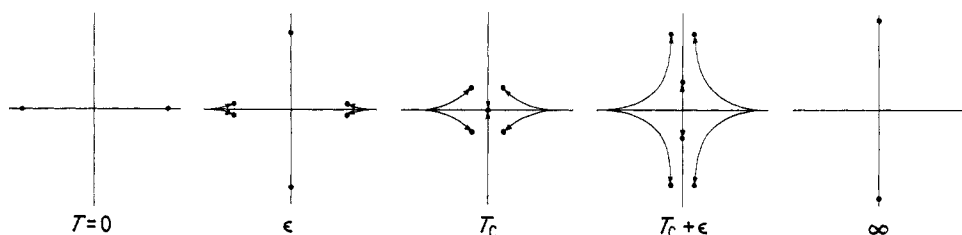


Figure 25. One conjectured mode of development with temperature of the singularities of $R(\mathbf{K}, \omega)$ in the complex plane of ω . (After Marshall 1966.)

the temperature is raised, these poles undoubtedly move off in the manner illustrated. Some distance above the transition temperature, where in fact

$$R^{xx} = R^{yy} = R^{zz}$$

we also know from (253) that (certainly at small enough K) R^{xx} has poles corresponding to spin diffusion at $\pm i\Lambda K^2$. The conventional theory of some years ago implied an evolution from the former condition to the latter without having very much to say about what took place in between.

Experiment has now revealed that, at least with a suitable choice of parameters, the transition can be exhibited as shown in the figure with all three peaks in existence together. That this is so was, indeed, conjectured four years ago (Marshall 1966) from considering the analogy to the classical fluid case. A classical fluid, at small K ,

Figure 24 (*opposite*). Contour diagrams of $\frac{1}{3}\sum_{\alpha} \mathcal{S}^{\alpha\alpha}(\mathbf{K}, \omega)$ derived from the neutron scattering of a single crystal of nickel 60, the data at all $\mathbf{K}$ with constant K having been collapsed onto a single variable K since insufficient anisotropy was found to warrant more special treatment. The contour spacing is logarithmic (H, high; L, low). In (a) the spin-wave ridge is very pronounced, and is close to the lines predicted from a non-interacting one-electron theory calculation by Thompson and Myers (1967). It is due to $\mathcal{S}^{xx} + \mathcal{S}^{yy}$, while the conspicuous bank of intensity represented by the roughly horizontal contour lines to the right of the spin-wave region is believed to be from $\mathcal{S}^{zz}$. At $1.6T_C$ in (d) the contours are in reasonable agreement with a tight-binding calculation for the paramagnetic phase by Allan *et al.* (1968) employing a random phase approximation enhancement factor of 3. In (b) renormalization and broadening of the spin-wave energy are apparent, while the outstanding feature of (b) and (c) taken together is their close similarity. In the region of temperatures just below T_C , renormalization does not take the spin-wave energies in the observed range below about half of their low temperature values, while just above T_C the 'paramagnon' phenomena of Doniach and Engelsberg (1966) and Berk and Schrieffer (1966) appropriate to 'critical' random phase approximation enhancement do not put in an appearance far enough away from the origin to be observed. A gentle transition between (b) and (c) is preferred. (From Komura, Lowde and Windsor 1968.)

may be described by the usual equations of hydrodynamics. In the absence of any dissipative effects (i.e. viscosity or thermal conductivity) the relaxation function (equivalent to R^{xx}) has two δ -function peaks at $\pm c_s K$, where c_s is the velocity of sound. The introduction of viscosity or thermal conductivity damps these sound waves, and simultaneously a diffusive mode also appears in the relaxation function. In other words: in the complex z plane, as soon as the sound-wave poles move off the real axis, a new pole appears on the imaginary axis. In suitable cases, therefore, the relaxation function measured by experiment as a function of real ω can have three maxima, one at $\omega = 0$ and two symmetrically placed on either side at the appropriate renormalized frequencies.

In the magnetic system the same phenomenon might be supposed to take place as the spin waves are damped by raising the temperature above zero. Indeed, we know it does; for as pointed out in §5.2 the interaction between magnons that damps $\langle S^x S^x \rangle$ produces diffusive motion in $\langle S^z S^z \rangle$. In the spin system we have the extra complication of discriminating, below T_C , between transverse and parallel motions; moreover spin diffusion in the ordered state, certainly in the transverse component, is not properly understood. However, as the Curie temperature is approached from below we know that R^{xx} and R^{zz} must evolve towards one another, since they are identical at T_C . Clearly just below T_C , where the z axis of alignment is only rather slightly 'preferred' by spins to the other axes, experiment and theory conspire to argue that the spin motion consists of precession around cones that occasionally flop from one axis to another. The most unexpected feature that has emerged is the strikingly good definition of the precession frequency. In crude terms: while the z or y axis is being 'preferred' as the cone axis, S^x is spin-wave-like; when the cone flops over to the x axis, S^x becomes z -like in behaviour and for a few moments suffers from diffusion. These flopping-cone motions are the 'sloppy spin waves' of Tomita (1966, 1968) (see also Kawasaki 1968 b).

When appropriate weighting is given to the poles for xx and zz responses, the most general situation as manifested in a substance of appropriate constants would seem on this view to be qualitatively as shown in figure 26.

Some remarks bearing upon the fact that on approaching T_C from above the magnetic response 'anticipates' the behaviour it must adopt on the other side of the transition point have already appeared in the literature (Kawasaki 1963, 1967 b, 1968 a, b, Tomita 1965, 1966, 1968, Mannari 1966, Brout 1967, Dalton and Hubbard 1967, Mori 1967, Resibois and de Leener 1967, Mori and Okamoto 1968 a, b). The fundamental phenomenon brought to light by the recent experiments is an underdamped $\{S, S(t)\}$ that oscillates in the process of decaying, thus corresponding to a Fourier transform with peaks in it (Villain 1967, 1968). In physical terms the local fluctuating domains of short-range order that abound at temperatures near T_C can support spin waves within; this idea leads one further to suppose that the resolvable peaks above T_C will occur only for q 's greater than some number proportional to κ_1 . In pursuing this physical picture it should be borne in mind that the much-discussed 'domains of local short-range order' do in fact exhibit remarkably *little* order; even at T_C , the correlation between nearest neighbours is as little as $\approx \frac{1}{2}$, and falls off with distance as r^{-1} . Thus again, using elementary ideas, significant effects would be expected only at the larger q 's, which is contrary to observation. To avoid some of the pitfalls of pictorial thinking, this concept has

been put on an analytical, rather than an intuitive footing by Beeby and Hubbard (1968), who show that a better-than-random-phase approximation to the motion of the spins does in fact lead to the desired effect, with a q -dependence in line with these ideas.

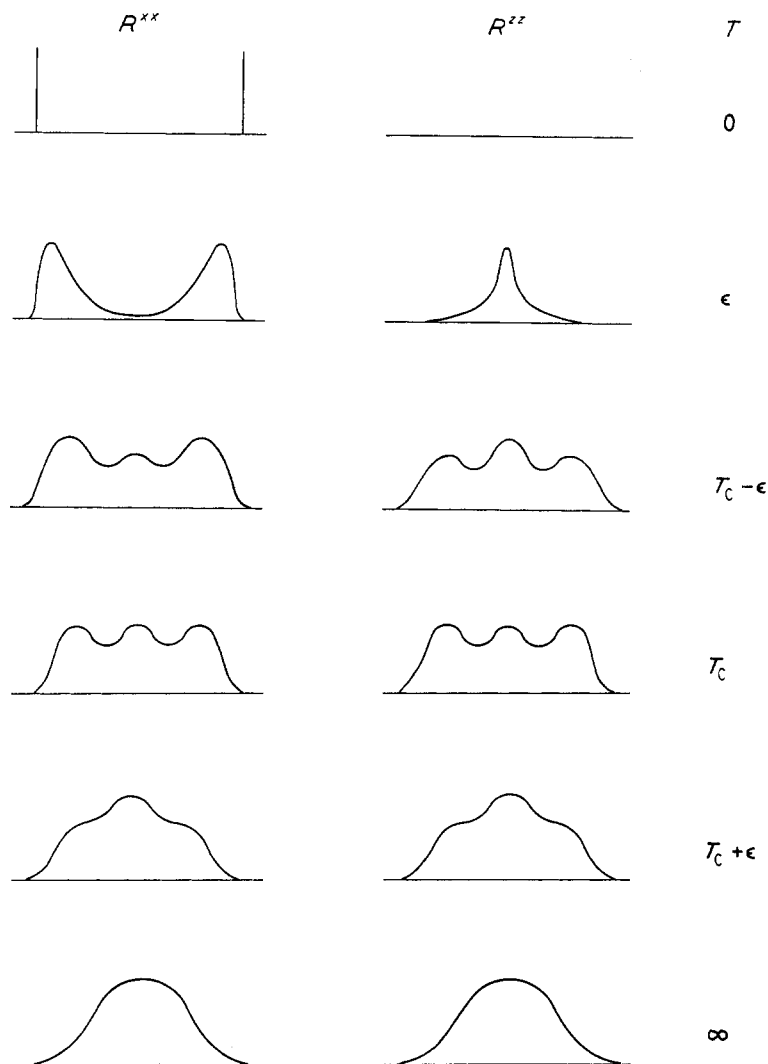


Figure 26. Conjectured form of the evolution of the ω -dependence of $R(\mathbf{K}, \omega)$ with temperature in a sufficiently general case, shown schematically to bring out in particular the different forms of R^{xx} and R^{zz} in the magnetically ordered phase, and their convergence, with increase of temperature, towards identity at T_c .

7. Concluding remarks

We began this account of magnetic correlations with a glowing description of the power of our new theoretical tools. The reader will have noticed, nevertheless, that the mean number of equations per page has been declining steadily for some

sections, and indeed is now sufficiently near zero that some account must be taken of the fact. Our long and, we hope, guided tour of this scientific terrain has been intended to map out the subject as it stands and to carry us up to the battlefield of critical spin dynamics. Just enough territory has been covered to allow us to see clearly the distant heights that must be taken.

In the face of the theoretical onslaught that has been directed against it, the problem of critical dynamics can only be said to have put up a magnificent resistance. Although the position appears now to be much clarified, our fundamental description of the situation in magnetism has degenerated badly towards verbalization, and more rigorous analysis is much to be desired. This may well flow, in the outcome, from Mori's (1965) theory. Mori finds the equation of motion for a statistical-mechanical operator $A(t)$:

$$\frac{d}{dt}A(t) = i\omega A(t) - \int_0^t \phi(t-t') A(t') dt' + f(t). \quad (316)$$

The first term is the oscillatory part, the second the damping part and the third the statistical fluctuation ignored in the preceding terms. He shows that

$$\phi(t) = \frac{\{f(0), f(t)^*\}}{\{A, A^*\}}. \quad (317)$$

In the straightforward case where A is $S^+(\mathbf{K}, t)$, it is possible to identify the thermodynamic 'random force' $f(t)$ with $\dot{S}^+(\mathbf{K}, t)$. Then if S^+ is much more slowly varying than $\dot{S}^+$, as we have seen is the case at high temperatures at small K , we obtain from this general theory the Γ of equation (272). In a more general case we are, at present, in difficulties; but we have in (316) a formalism giving the correct description.†

Whether any of the foregoing lines of thought will lead through to a worthwhile solution of the general problem is not apparent. But with a topic of such fascination it is a pleasure to be able to conclude this Report in the certainty that the end is not in sight.

References

- ALLAN, G., LOMER, W. M., LOWDE, R. D., and WINDSOR, C. G., 1968, *Phys. Rev. Lett.*, **20**, 933-6.
 ALLAN, G. A. T., and BETTS, D. D., 1968, *Can. J. Phys.*, **46**, 799-802.
 ALS-NIELSEN, J., DIETRICH, O. W., MARSHALL, W., and LINDGARD, P. A., 1967, *Solid St. Commun.*, **5**, 607-11.
 BALLY, D., *et al.*, 1967, *J. Phys. Chem. Solids*, **28**, 1947-55.
 ——— 1968 a, *Phys. Lett.*, **26A**, 396-7.
 ——— 1968 b, *Proc. Symp. on Neutron Inelastic Scattering, Copenhagen 1968* (Vienna: International Atomic Energy Agency), Paper 55.
 BEEBY, J., and HUBBARD, J., 1968, *Phys. Lett.*, **26A**, 376-7.
 BEELER, J. R., 1966, *Physics of Many-Particle Systems*, Vol. 1, Ed. E. Meeron (New York: Gordon and Breach), pp. 1-128.
 BENNETT, H. S., and MARTIN, P. C., 1965, *Phys. Rev.*, **138**, A608-17.
 BERK, N. F., and SCHRIEFFER, J. R., 1966, *Phys. Rev. Lett.*, **17**, 433.

† Note added in proof. Mori (1967) and Kawasaki (1968 a) have recently extended the discussion of equation (316).

- BLOEMBERGEN, N., 1949, *Physica*, **15**, 386.
- BLUME, M., VINEYARD, G. H., and WATSON, R. E., 1968, *J. Appl. Phys.*, **39**, 969.
- BROUT, R., 1967, *Phys. Lett.*, **24A**, 117-18.
- CABLE, J. W., LOWDE, R. D., WINDSOR, C. G., and WOODS, A. D. B., 1967, *J. Appl. Phys.*, **38**, 1247-8.
- CALLEN, H. B., 1961, *Fluctuation, Relaxation and Resonance in Magnetic Systems*, Ed. D. Ter Haar (London: Oliver and Boyd), pp. 15-22.
- CALLEN, H. B., and WELTON, T. R., 1951, *Phys. Rev.*, **83**, 34-40.
- CARBONI, F., and RICHARDS, P. M., 1968, *J. Appl. Phys.*, **39**, 967-8.
- COLLINS, M. F., 1968, *J. Appl. Phys.*, **39**, 533-7.
- COLLINS, M. F., and MARSHALL, W., 1967, *Proc. Phys. Soc.*, **92**, 390-9.
- COLLINS, M. F., and NATHANS, R., 1965, *J. Appl. Phys.*, **36**, 1092-3.
- COLLINS, M. F., *et al.*, 1968, *Phys. Rev. Lett.*, **21**, 99-102.
- COOPER, M. J., and NATHANS, R., 1966, *J. Appl. Phys.*, **37**, 1041-7.
- COWLEY, R. A., and MARTEL, P., 1968, *J. Appl. Phys.*, **39**, 1116-18.
- CRIBIER, D., and JACROT, B., 1963, *Inelastic Scattering of Neutrons in Solids and Liquids*, Vol. 2 (Vienna: International Atomic Energy Agency), pp. 309-15.
- DALTON, N. W., and HUBBARD, J., 1967, *A.E.R.E. Rep.*, No. PR/TP 13, p. 20.
- DONIACH, S., and ENGELSBERG, S., 1966, *Phys. Rev. Lett.*, **17**, 750.
- ELLIOTT, R. J., and MARSHALL, W., 1958, *Rev. Mod. Phys.*, **30**, 75-89.
- FISHER, M. E., 1967, *Rep. Prog. Phys.*, **30**, 615-730 (London: Institute of Physics and Physical Society).
- DE GENNES, P. G., 1958 a, *J. Phys. Chem. Solids*, **4**, 223-6.
- 1958 b, *J. Phys. Chem. Solids*, **6**, 43-5.
- 1959, *The Magnetic Scattering of Neutrons*, Centre d'Etudes Nucléaires de Saclay Rep. No. 925 (Paris: Presses Universitaires de France), pp. 1-64.
- 1963, *Magnetism*, Vol. 3, Eds G. T. Rado and H. Suhl (New York: Academic Press), pp. 115-47.
- DE GENNES, P. G., and HERPIN, A., 1957, *C. R. Acad. Sci., Paris*, **243**, 1611-13.
- DE GENNES, P. G., and VILLAIN, J., 1960, *J. Phys. Chem. Solids*, **13**, 10-27.
- HAMAGUCHI, Y., 1966, *Inelastic Scattering of Neutrons* (JAERI Rep. No. 1113). (Tokai-mura: Japan Atomic Energy Research Institute), pp. 122-6.
- HELLER, P., 1967, *Rep. Prog. Phys.*, **30**, 731-826 (London: Institute of Physics and Physical Society).
- HUBER, D. L., 1966, *Phys. Rev.*, **146**, 387-9.
- IZUYAMA, T., KIM, D. J., and KUBO, R., 1963, *J. Phys. Soc. Japan*, **18**, 1025-42.
- JACROT, B., KONSTANTINOVIC, J., PARETTE, G., and CRIBIER, D., 1963, *Inelastic Scattering of Neutrons in Solids and Liquids*, Vol. 2 (Vienna: International Atomic Energy Agency), pp. 317-26.
- KADANOFF, L. P., and MARTIN, P. C., 1963, *Ann. Phys., N.Y.*, **24**, 419-69.
- KANAMORI, J., and ITOH, Y., 1968, *J. Appl. Phys.*, **39**, 1358-9.
- KAWASAKI, K., 1963, *Prog. Theor. Phys.*, **29**, 156-8.
- 1967a, *J. Phys. Chem. Solids*, **28**, 1277-83.
- 1967 b, *Prog. Theor. Phys.*, **38**, 1052-61.
- 1968 a, *Proc. 2nd Conf. on Inelastic Scattering of Neutrons in Solids and Liquids*, Tokai 1967 (JAERI Rep. No. 1157) (Tokai-mura: Japan Atomic Energy Research Institute), pp. 97-102.
- 1968 b, *Proc. 2nd Conf. on Inelastic Scattering of Neutrons in Solids and Liquids*, Tokai 1967 (JAERI Rep. No. 1157) (Tokai-mura: Japan Atomic Energy Research Institute), pp. 103-6.
- KOCINSKI, J., 1968, *Acta Phys. Polon.*, **33**, 13-26.
- KOMURA, S., LOWDE, R. D., and WINDSOR, C. G., 1968, *Proc. Symp. on Neutron Inelastic Scattering*, Copenhagen 1968 (Vienna: International Atomic Energy Agency), Paper 110.
- KONSTANTINOVIC, J., 1966, *Bull. Inst. Boris Kidric*, **17**, 329-64.
- KRASNICKI, S., *et al.*, 1964, *J. Phys., Paris*, **25**, 634-41.
- KUBO, R., 1959, *Lectures in Theoretical Physics*, Vol. 1 (New York: Interscience), pp. 120-203.

- KUBO, R., 1966 a, *Rep. Prog. Phys.*, **29**, 255–84 (London: Institute of Physics and Physical Society).
- 1966 b, *Many-body Theory* (1965 Summer Lectures at the Institute Of Theoretical Physics, Tokyo), Vol. 1, Ed. R. Kubo (New York: W. A. Benjamin), pp. 1–16.
- KWON, T. H., and GERSCH, M. A., 1968, *Phys. Rev.*, **167**, 458.
- LANDAU, L., 1937, *Phys. Z. SowjUn.*, **12**, 124.
- LINDGARD, P. A., 1968, *Proc. Symp. on Neutron Inelastic Scattering, Copenhagen 1968* (Vienna: International Atomic Energy Agency), Paper 19.
- LOWDE, R. D., 1958, *Rev. Mod. Phys.*, **30**, 69–74.
- 1965, *J. Appl. Phys.*, **36**, 884–92.
- LOWDE, R. D., and WINDSOR, C. G., 1967, *Phys. Rev. Lett.*, **18**, 1136–8.
- 1968 a, *Solid St. Commun.*, **6**, 189–92.
- 1968 b, *J. Appl. Phys.*, **39**, 449–50.
- MANNARI, I., 1966, *Inelastic Scattering of Neutrons* (*JAEI Rep. No. 1113*), (Tokai-mura: Japan Atomic Energy Research Institute), pp. 120–1.
- MARSHALL, W., 1966, *Critical Phenomena, Natn. Bur. Stand. Misc. Publ. No. 273* (Washington: U.S. Govt Printing Office), p. 135.
- MARSHALL, W., and MURRAY, G., 1968, *J. Appl. Phys.*, **39**, 380–2.
- MORI, H., 1965, *Prog. Theor. Phys.*, **33**, 423–55.
- 1966 a, *Many-body Theory* (1965 Summer Lectures at the Institute of Theoretical Physics, Tokyo), Vol. 1, Ed. R. Kubo (New York: W. A. Benjamin), p. 17.
- 1966 b, *Proc. Symp. on Inelastic Scattering of Neutrons by Condensed Systems, Brookhaven Nat. Lab. Rep. BNL 940(C-45)*, Ed. L. M. Corliss (Virginia: National Bureau of Standards), pp. 49–56.
- 1967, *Bussei Kenkyu*, **9** (2), B53–B58.
- MORI, H., and KAWASAKI, K., 1962 a, *Prog. Theor. Phys.*, **27**, 529–70.
- 1962 b, *Prog. Theor. Phys.*, **28**, 690–710.
- MORI, H., and OKAMOTO, H., 1968 a, *Phys. Lett.*, **26A**, 249–50.
- 1968 b, *Proc. 2nd Conf. on Inelastic Scattering of Neutrons in Solids and Liquids, Tokai 1967* (*JAEI Rep. No. 1157*) (Tokai-mura: Japan Atomic Energy Research Institute), pp. 89–96.
- MORITA, T., 1968, *Proc. 2nd Conf. on Inelastic Scattering of Neutrons in Solids and Liquids, Tokai 1967* (*JAEI Rep. No. 1157*) (Tokai-mura: Japan Atomic Energy Research Institute), pp. 111–13.
- MÜNSTER, A., 1965, *Fluctuation Phenomena in Solids*, Ed. R. E. Burgess (New York: Academic Press), pp. 180–266.
- NAGAI, O., 1963, *J. Phys. Soc. Japan*, **18**, 74–8.
- NATHANS, R., MENZINGER, F., and PICKART, S. J., 1968, *J. Appl. Phys.*, **39**, 1237–8.
- NYQUIST, H., 1928, *Phys. Rev.*, **32**, 110.
- OGUCHI, T., and ONO, I., 1966, *J. Phys. Soc. Japan*, **21**, 2178–93.
- 1968, *J. Appl. Phys.*, **39**, 1353–4.
- PASSELL, L., BLINOWSKI, K., BRUN, T., and NIELSON, P., 1965, *Phys. Rev.*, **139A**, 1866–76.
- RESIBOIS, P., and DE LEENER, M., 1967, *Phys. Lett.*, **25A**, 65–6.
- RISTE, T., 1962, *J. Phys. Soc. Japan*, **17** (Suppl. B-III), 60–2.
- ROGERS, R. N., CARBONI, F., and RICHARDS, P. M., 1967, *Phys. Rev. Lett.*, **19**, 1016–18.
- SATYA MURTHY, N. S., et al., 1965, *Inelastic Scattering of Neutrons*, Vol. 1 (Vienna: International Atomic Energy Agency), pp. 433–42.
- 1968, *Phys. Lett.*, **26A**, 108–9.
- SEARS, V. F., 1967, *Can. J. Phys.*, **45**, 2923–9.
- SHIRANE, G., MINKIEWICZ, V. J., and NATHANS, R., 1968, *J. Appl. Phys.*, **39**, 383–90.
- SPOONER, S., and AVERBACH, B. L., 1966, *Phys. Rev.*, **142**, 291–9.
- STRINGFELLOW, M. W., 1968, *J. Phys. C. (Proc. Phys. Soc.)*, [2], **1**, 950–65.
- SUZUKI, M., 1968, *Proc. 2nd Conf. on Inelastic Scattering of Neutrons in Solids and Liquids, Tokai 1967* (*JAEI Rep. No. 1157*) (Tokai-mura: Japan Atomic Energy Research Institute), pp. 107–10.
- SUZUKI, M., and KUBO, R., 1968, *J. Phys. Soc. Japan*, **24**, 51–60.

- TAHIR-KHELI, R. A., and CALLEN, H. B., 1964 a, *Phys. Rev.*, **135**, A679-84.
— 1964 b, *J. Appl. Phys.*, **35**, 946-7.
- THOMPSON, E. D., and MYERS, J. J., 1967, *Phys. Rev.*, **153**, 574-83.
- TOMITA, K., 1965, *Proc. Int. Conf. on Magnetism, Nottingham, 1964* (London: Institute of Physics and Physical Society), pp. 103-4.
- 1966, *Inelastic Scattering of Neutrons* (*J.AERI Rep. No. 1113*) (Tokai-mura: Japan Atomic Energy Research Institute), pp. 106-9.
- 1968, *Proc. 2nd Conf. on Inelastic Scattering of Neutrons in Solids and Liquids, Tokai 1967* (*J.AERI Rep. No. 1157*) (Tokai-mura: Japan Atomic Energy Research Institute), pp. 80-4.
- TORRIE, B. H., 1966, *Proc. Phys. Soc.*, **89**, 77-85.
- TURBERFIELD, K. C., and OKAZAKI, A., 1965, *Proc. Phys. Soc.*, **85**, 743-58.
- VAN HOVE, L., 1954, *Phys. Rev.*, **95**, 249-62, 1374-84.
- VAN VLECK, J. H., 1939, *Phys. Rev.*, **53**, 924-30.
- VILLAIN, J., 1963, *J. Phys., Paris*, **24**, 622-32.
- 1964, *J. Phys., Paris*, **25**, 618-26.
- 1967, *Phys. Status Solidi*, **19**, 767-76.
- 1968, *J. Phys., Paris*, **29**, 321-8.
- WINDSOR, C. G., 1966 a, *Proc. Phys. Soc.*, **87**, 501-4.
- 1966 b, *Proc. Phys. Soc.*, **89**, 825-31.
- 1967, *Proc. Phys. Soc.*, **91**, 353-5.
- 1968, *Proc. Symp. on Neutron Inelastic Scattering, Copenhagen 1968* (Vienna: International Atomic Energy Agency), Paper 112.
- WINDSOR, C. G., BRIGGS, G. A., and KESTIGIAN, M., 1968, *J. Phys. C. (Proc. Phys. Soc.)*, **[2], 1**, 940-9.